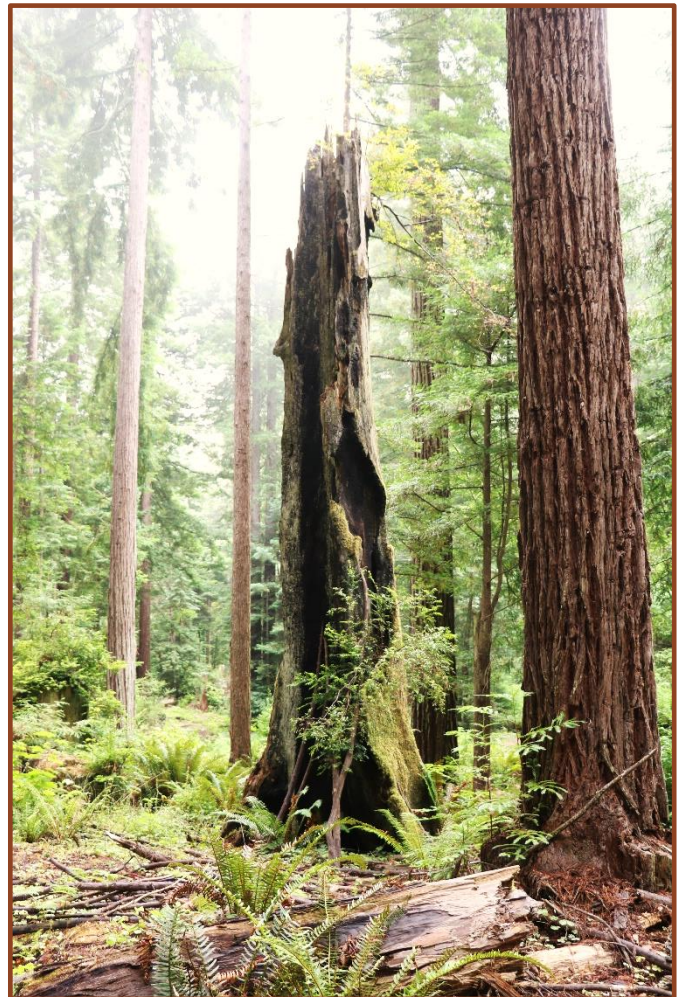
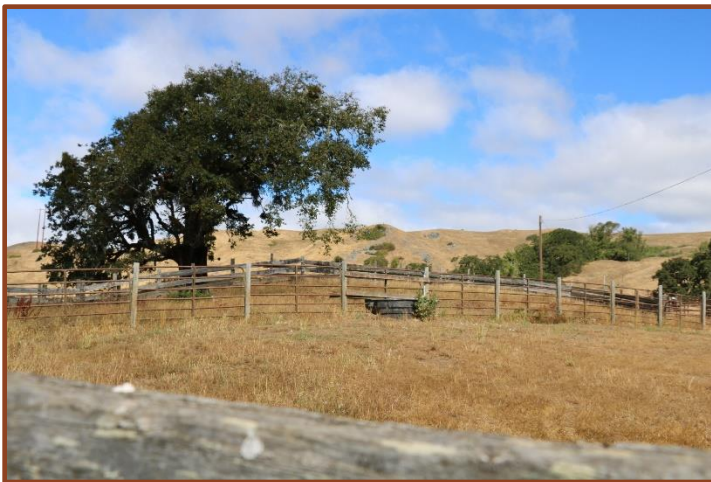


CALIFORNIA'S FORESTS AND RANGELANDS STRATEGY REPORT: 2020 UPDATE



FRAP
Fire and Resource Assessment Program
California Department of Forestry and Fire Protection

California's Forests and Rangelands Strategy Report: 2020 Update

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FIRE AND RESOURCE ASSESSMENT PROGRAM

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EXECUTIVE SUMMARY

The USDA Forest Service State and Private Forestry Program and the 2018 Farm Bill require states to periodically update their Statewide Forest Resource Strategy based on recently completed state assessments of forest resources (Forest Action Plans). This report provides an update to the original 2010 California Forest Action Plan Strategies Report, with its long-term, comprehensive, and coordinated framework for investing state, federal and stakeholder resources to address the management and landscape priorities identified in the most recent assessment.

Forests and rangelands cover about 80% of California and support abundant resources that greatly enhance the quality of life enjoyed by all residents. California forests comprise a wide diversity of habitats containing numerous species of conifer and hardwood trees. A high proportion of the forest tree species are endemic to the state.

There is a growing demand for both renewable products and environmental services provided by California's forests and rangelands. These services include: timber; livestock grazing; biomass, wind, and solar energy; recreation; clean air; clean water; carbon sequestration; wildlife habitat; and many other services. Conservation and management of our natural resources is challenged by conflicting demands, development pressure, changing climate, ongoing natural threats, the legacy of historical management practices, decaying infrastructure, varying community capacity, and limited financial incentives for landowners to maintain and enhance the broader range of forest and rangeland environmental services that benefit all Californians.

A major challenge is how to maintain or enhance the wide array of valuable natural resources in the face of often conflicting management objectives. Forest and rangeland policies must strike a balance between promoting the commercial goods and services that are produced by these lands, while implementing measures that protect and enhance the foundational ecosystems. Sustainable use of these lands requires a broad set of strategies that promotes investments to maintain, restore, and enhance the health and productivity of our forest and rangelands.

The strategies in this report stem from the findings contained in the 2017 Forests and Rangelands Assessment Report (Assessment). The 2020 strategies build upon the strategies developed for the 2010 edition of the Assessment.

California's forests and rangelands face a variety of threats, and trends indicate that these are increasing in number, extent, and severity. There are several cross-cutting issues from the Assessment:

- California forest and rangelands, and the resources they provide, will continue to increase in value for generations to come. Policies that maintain and promote the health of all resources are essential to their long-term health and productivity.

- The effects anticipated from climate changes have recently been manifesting across several forested regions, with the proximate cause being the three-year drought of record from 2012 to 2015. This has resulted in the deaths of more than a hundred million large forest trees, mostly in the southern Sierra Nevada. The high level of forest mortality has set into motion several large state and local efforts to deal with the impacts, including removing thousands of hazard trees, intensifying work to protect communities from future wildfire, such as through increased wildland fuels mitigation and the disposal of the tremendous amount of dead and dying woody material.
- Societal demands on forest and rangeland resources are increasing, especially for the ecosystem services they provide. Emerging markets, such as for solar and wind power, are placing new demands on these lands. Multiple management objectives for wildlands have created a complex framework for decision making, with increased demands for science-based approaches (e.g., the Desert Renewable Energy Conservation Plan (DRECP)).
- In addition to the regions of high forest mortality, a significant portion of forests, rangelands, urban forests, and the infrastructure required to meet demands from these lands, is in a degraded or undesirable condition.
- Opportunities exist to improve the quality of and quantity of benefits from these lands. There are management options leading to desired future conditions to sequester more carbon, improve water quality, foster more vibrant rural economies, and make natural landscapes more resilient to threats such as wildfire. Reaching desired future conditions will require surmounting numerous political, social, and economic challenges.
- Among California's great strengths are its human capital, and cutting-edge innovative spirit. The potential to reach desired future conditions across forests and rangelands will depend in large part on taking advantage of and augmenting existing institutions, collaborative efforts and groups, initiatives, strategies, and success stories.

California has a long history of a commitment to protecting, managing and investing in its natural resources. This has resulted in number of plans, strategies, and programs that influence and guide management on forest and rangelands. Many federal, state, and local agencies, tribes, as well as nongovernmental organizations, landowners, and other stakeholders are involved.

A recent, major step forward in such commitments is the recent signing of a Shared Stewardship Agreement between the State of California and the USDA Forest Service. This memorandum of understanding (MOU) "...establishes a joint framework to enhance science-based forest and rangeland stewardship in California." Among other items, the MOU commits the State and Forest Service, by 2025, to each implementing vegetation treatments on 500,000 acres per year on the forestlands for which they have responsibility.

The State Board of Forestry and Fire Protection (BOF), under State Law PRC 4789, uses the information from the Assessment to develop and update forest and rangeland policies on private lands. The most recent Policy Statement of the BOF was adopted in 2010. In addition, this updated Strategy Report incorporates key elements from existing statewide management plans, including, but not limited to, the following plans:

- California Forest Carbon Plan (2018)
- Strategic Fire Plan for California (2018)
- California Water Plan (Update 2018)
- California Wildlife Action Plan and Companion Plans (2015-2016)
- California's 2017 Climate Change Scoping Plan (2017)
- California's Climate Adaptation Strategy: Safeguarding California Plan: 2018 Update (2018)

This report lists strategies that address the priority issues that were identified through indicators and key findings in the 2017 Forests and Rangelands Assessment Report and related 2020 addenda, including a set of priority landscapes that were developed for CAL FIRE forest health grant programs. The 2017 Assessment is organized into twelve chapters, each focusing on specific aspects of and challenges confronting the management of forests and rangelands. (Note: The Assessment also address the three broad national themes that were put forth in the Redesign framework of the U.S. Forest Service for the first Forest Action Plans¹). The 12 headings below reiterate the 12 parallel chapters in this strategy and in the companion assessment report and list the strategies discussed in this new document that fall logically under each of these topics. This report describes multiple specific actions that will serve to implement each of the strategies.

Sustainable Forests (National Theme 1: Conserve Working Forest and Range Landscapes):

- Strategy 1.1 Maintain and improve the capacity of the wood products industry statewide.
- Strategy 1.2 Increase the capacity to provide incentives to non-industrial forest landowners to increase active sustainable management.

Sustainable Rangelands (National Theme 1: Conserve Working Forest and Range Landscapes):

- Strategy 2.1 Maintain and improve the capacity of range industries statewide.
- Strategy 2.2 Increase the capacity to provide incentives to range landowners.

Urban Forestry (National Theme 3: Enhance Public Benefits from Trees, Forests, and Rangelands):

- Strategy 3.1 Promote urban tree planting to improve air quality and energy conservation.
- Strategy 3.2 Maintain urban tree canopy to conserve energy and improve air quality.

Wildfire (National Theme 2: Protect Forests and Rangelands from Harm):

- Strategy 4.1 Reduce the occurrence of damaging wildfires and reduce life, property and natural resource losses through the implementation of effective and efficient fire prevention programs and activities.

¹ The three original national themes have been modified here to reflect California's approach to addressing both forests and rangelands through natural resources programs (modifications in **bold** typeface):

Theme 1: Conserve Working Forest **and Range** Landscapes

Theme 2: Protect Forests **and Rangelands** from Harm

Theme 3: Enhance Public Benefits from Trees, Forests, **and Rangelands**

- Strategy 4.2 Protect life and property from wildfire through efficient and effective fire protection planning and suppression, financial management, and firefighter/public safety strategies.
- Strategy 4.3 Reduce the impacts of wildfire on ecosystem health, public safety and private property through appropriate scientific research, education and training.
- Strategy 4.4 Address post-fire responsibilities for natural resource recovery including watershed protection, reforestation, and ecosystem restoration.

Forest Pests (National Theme 2: Protect Forests and Rangelands from Harm):

- Strategy 5.1 Restore forest lands impacted by current and historical forest pest outbreaks, air pollution and invasive species.
- Strategy 5.2 Reduce/prevent forest pest outbreaks and control their spread to maintain ecosystem health, preserve ecosystem services and avoid public safety hazards associated with large scale tree mortality events.
- Strategy: 5.3 Prevent the introduction and spread of new exotic pests and invasive plant species.
- Strategy: 5.4 Rapidly control or contain outbreaks of exotic forest pests and invasive plant species.
- Strategy: 5.5 Monitor forestland to quickly identify new, and evaluate current, outbreaks of exotic forest pests and invasive plant species to protect the most vulnerable and valued forest and range land assets.

Population Growth and Development Threat (National Theme 1: Conserve Working Forest and Range Landscapes):

- Strategy 6.1 Reduce urban sprawl by promoting redevelopment and infilling of available land within the urban matrix, strengthening planning at the local level, capacity building, and improving access to tools and data sources.
- Strategy 6.2 Aid in efforts to reduce development sprawl in rural communities.
- Strategy 6.3 Support comprehensive planning at state and regional scales that is coordinated with wildlife habitat conservation efforts.
- Strategy 6.4 Support local and regional community efforts in preserving scenic landscapes.

Climate Change (National Theme 1: Conserve Working Forest and Range Landscapes):

- Strategy 7.1. Protect and enhance the capacity of California's forests to sequester carbon through reducing risk of loss from disturbance, protecting existing forest land, and expanding forest area through tree planting.
- Strategy 7.2. Support Adaptation Needs for Forests by Assessing Climate Vulnerabilities, Improving Institutional Capacity, and Promoting a Priority Research Agenda.
- Strategy 7.3. Support Actions that Maintain, Enhance, and Protect Ecosystem Functions to Promote Biodiversity and Increase Resilience to Climate Change.

Non-Metro Regional Economy (National Theme 3: Enhance Public Benefits from Trees, Forests, and Rangelands):

- Strategy 8.1 Economic-, employment-, and housing-related measures
- Strategy 8.2 Resident-related measures
- Strategy 8.3 Politics- and governance-related measures

Water Resources (National Theme 3: Enhance Public Benefits from Trees, Forests, and Rangelands):

- Strategy 9.1 Promote Watershed Protection and Watershed Restoration.
- Strategy 9.2 Improve Water Quality through Implementation of Best Management Practices and Monitoring in High Priority Watersheds.

Wildlife Habitat (National Theme 3: Enhance Public Benefits from Trees, Forests, and Rangelands):

- Strategy 10.1 Reduce the loss and modification of habitat that supports wildlife, and maintains California's unique biodiversity.
- Strategy 10.2 Develop policies and incentives to facilitate better integration of wildlife conservation considerations into local and regional planning and land-use decision making.
- Strategy 10.3 Sustain healthy forest ecosystems to maintain California's unique biodiversity.

Reducing Community Wildfire Risk (National Theme 2: Protect Forest and Rangelands from Harm):

- Strategy 11.1 Promote formation of Local Fire Safe Councils for priority communities.
- Strategy 11.2 Promote participation in the National Firewise/USA program.
- Strategy 11.3 Establish a statewide comparative database of community wildfire planning.
- Strategy 11.4 Reduce wildfire risk to communities.

Renewable Energy (National Theme 3: Enhance Public Benefits from Trees, Forests, and Rangelands):

- Strategy 12.1 Continue to Develop, Implement, and/or Expand Initiatives that Reduce Demand for Large Renewable Energy Projects on Forests and Rangelands
- Strategy 12.2 Promote Continued Research to Improve Guidelines, Standards, and Best Practices to Minimize the Environmental Impact of Large Renewable Energy Projects on Forests and Rangelands
- Strategy 12.3 Support expansion of transmission infrastructure for emerging renewable energy generation from sources such as biomass, wind, hydro and solar in a way that minimizes environmental impact on forests and rangelands.
- Strategy 12.4 Support Development of Biomass Energy Projects that can Facilitate Projects to Reduce Wildfire Risk and Improve Forest Health

Collaboration

The strategies and actions that appear in this report were developed by CAL FIRE staff with the input from numerous stakeholders who participated in meetings and workshops, and who reviewed draft versions of the assessment and strategies reports. For many topics, the proposed strategies have been built on existing state or federal plans. The guidance provided in the existing plans was further refined by the priority issues that were identified in the 2017 Forests and Rangelands Assessment.

Synthesis

California covers more than 100 million acres and is home to almost 40 million people, with complex ecological and social systems, a history of forest and range land use, and a world-class economy. In this context, several over-arching themes and goals are found in the 2017 Forests and Rangelands Strategy Report. Collectively, the strategies contribute to the national State and Private Forestry goals for conserving working forests, protecting forests from harm, and enhancing public benefits from trees and forests. The following section briefly discusses the cross-cutting nature of the proposed strategies.

Strategies for promoting sustainable working forests and those for reducing development impacts are both intended to meet the larger national theme/goal of conserving working forests and rangelands. California has a complex mix of laws, agencies, policies, and programs that focus on disclosing, evaluating, avoiding, and mitigating the impacts of development – many dealing explicitly with development impacts on forests and rangelands.

A number of strategies address improving range and forest ecosystem health and resilience in California, which contribute to national goals of protecting forests from harm. Underlying most strategies is the need for monitoring, accurate problem analysis, effective program design and delivery, feedback on results, and making appropriate alterations (core principles of adaptive management). Implicit in the former is the understanding that many California ecosystems have been altered from elements such as fire suppression, land management, and spread of invasive species. Changing climatic patterns and other factors are likely to increase the vulnerability of some ecosystems and to add to uncertainty in making management decisions. In such cases, accurate monitoring, information, assessment, planning and research become even more important.

An overarching theme in the 2017 Assessment and the strategies described herein is the impact of wildfire and other natural disturbances on forested and range landscapes. Fire and other disturbances are an inherent and often necessary part of the ecosystem dynamics, including impacts to wildlife and fish. A number of ecosystems and ecosystem components are dependent on wildfire or other disturbances. In these cases, it may be necessary to reintroduce wildfire, or find ways to mimic its effects through management. At the same time, fire and other natural factors can be significant threats to life, public health, natural resources, and other property. Thus, in the case of wildfire, a variety of approaches have evolved to include management programs that address reduction of wildfire risk and attendant smoke and particulate matter, as well as use of wildfire as a tool to enhance ecosystems.

Similarly, many insects, diseases and other pests are endemic to forests and rangelands. They can play a key role in ecosystem function. Conversely, pests introduced from outside of California are having an increasingly significant impact in many areas. In the case of forest pests and invasive species, strategies and approaches focus on education, prevention (such as through forest stand thinning), control of existing outbreaks or spread, and restoration of impacted areas. Issues with both endemic and invasive insects, diseases, and plant species

remain a very significant challenge to maintaining and improving forest and rangeland resilience.

Because many people live in or near forests and rangelands, there is a set of strategies focused on community protection from wildfire and other risks. Collectively these strategies emphasize expanded pre-fire planning and fire prevention efforts that include: increasing awareness of the need to establish and maintain defensible space around buildings; adopting and implementing fire safe building standards for new construction; retrofitting existing homes to make them more fire resistant; strategic placement of vegetation treatments to remove hazardous fuels; and greater collaboration in the development of community-based wildfire protection plans.

The benefits of California's forests are found in urban settings as well as natural wildlands. Urban forests provide shade, improve air quality, contribute to carbon sequestration, filter stormwater runoff, add aesthetic value, increase property values, and contribute to mental health. Strategies were developed to expand urban tree planting and maintain the many benefits derived from urban forests.

People living in or near forest and rangelands also place numerous demands on these lands. Strategies to conserve, manage, and connect people to the concept of ecosystem services can enhance quality of life by providing open space, scenic vistas, outdoor recreation and educational opportunities, watershed value, and wildlife habitat close to population centers most in need of these services. Strategies can also engage local populations in being advocates for and stewards of green infrastructure and its associated values. Further, helping urban residents to understand urban forest ecosystems also can lead them to better understand and appreciate wildland forest ecosystems.

An additional set of strategies is organized around maintaining and enhancing the public benefits and broad range of environmental services that are provided from California's forests and rangelands. These strategies include watershed protection and restoration actions in priority watershed areas. These actions address water supply and water quality issues in upper watersheds that support a range of downstream beneficial uses. Strategies and actions were also developed to protect wildlife, fish, and related riverine and riparian habitat needs.

While still important in many local areas, the relative economic importance of the forest products and range-livestock industries statewide in California has declined. Because of this, emerging markets that can support investment in forest and range resources, as well as the related economic and social infrastructure, are critical for local and regional economies in particular. Ecosystem services, such as wildlife habitat and clean water, have grown in recognition of their economic value, but the markets to capture the value of these services have been limited. Recently, market mechanisms for forest health treatments tied to protection of water quality expanded. For several decades now, there has been increased interest in the capacity of forests to generate biomass as biofuel and electrical energy sources, and, more recently, in the role of forests to maintain and enhance carbon sequestration. These three areas represent emerging markets that offer a large public benefit and potential financial incentives to landowners. There is additional interest in solar, wind, and other sources of renewable energy, infrastructure for which could potentially be sited on forest or rangeland areas. Strategies are proposed for supporting the development of carbon markets, forest biomass, and other types of renewable energy. In addition, strategies are needed to evaluate potential environmental impacts from these emerging markets to ensure that they are developed in a sustainable manner.

Climate change is another over-arching issue that will have an increasing effect on the policy choices and management of forests and rangelands. Strategies for climate change include actions to protect and enhance the capacity of forests to sequester carbon both through reforestation (expanding forested areas) and through actions that are intended to reduce the risk of loss through human caused and natural disturbances. Recognizing that some degree of climate change is certain to occur, additional strategies have been proposed to support adaptation needs in forests and rangelands, to help these areas become more resilient to climate change.

Many state, federal and local laws already define a complex framework of goals, programs, and funding sources that apply to forests and rangelands in the state. Strategies already exist at many different levels, from governmental agencies to tribes, communities and community groups, nongovernmental organizations, private landowners, and other stakeholders. The challenge is to take advantage of and incorporate this rich fabric of interest and involvement.

Historically, California has shown a strong commitment to investment in natural resources through a variety of funding mechanisms. Environmental bond fund ballot initiatives or referendums in particular have had a substantial influence on priorities for protection and enhancement of natural resources on forests and rangelands. California voters in the past two decades have been supportive of programs and ballot measures that support conservation, restoration, open space, and improved environmental quality on urban and wildland landscapes alike. Using state monies for improved environmental justice for disadvantaged communities has increasingly become a priority.

In many instances, strong programs of cooperation and coordination have developed between agencies and stakeholders with an interest in forest and rangeland resources. With increased uncertainty, rapid change, and limited resources, it is imperative that such cooperation continue and grow. The use of place-based and other community related approaches, such as watershed organizations, urban forestry groups, and Firesafe Councils, is well developed in California and will continue to play a critical role in developing and implementing strategies.

Progress Since Publication of the Strategies 2010 Document

California continues to move forward to strengthen sustainable management of the state's forests. Several new state programs, proclamations, and plans to support these efforts have emerged since the 2010 edition of this document and are listed with other similar initiatives in Appendix A. Some of the most relevant to forest concerns are:

- California Forest Carbon Plan
- CAL FIRE Forest Health Grant Program
- CAL FIRE Forest Health Research Grant Program
- 2018 Strategic Fire Plan for California
- CAL FIRE Fire Prevention Grant Program
- California Forest Management Task Force (FMTF)
- State Wildlife Action Plan Update
- Timber Regulation and Forest Restoration Fund and Program
- Tree Mortality State of Emergency Proclamation (Governor Jerry Brown, 2015)
- Wildfire State of Emergency Proclamation (and subsequent CAL FIRE 45-day Report) (Governor Gavin Newsom, 2019)
- Agreement for Shared Stewardship of California's Forest and Rangelands between the State of California and the USDA Forest Service Pacific Southwest Region (2020)

Another area of substantial change since the Strategies 2010 document is the sources and level of funding available for CAL FIRE programs related to the state's wildland and urban forests. Funds generated from the State's auction of greenhouse gas emission allowances under Assembly Bill 32 have resulted in a flow of many millions of dollars of California Climate Investments monies to a wide range of forest health treatments, conservation easements, and urban greening projects for example. This includes forest health grants to the Forest Service for use on National Forest System lands. The creation of the Timber Regulation and Forest Restoration fund under Assembly Bill 1492 in 2012 provides a new funding source for forest restoration on nonfederal lands and for enhancement of timber harvesting regulation programs across multiple state agencies. Unfortunately, the significant economic disruptions due to the COVID-19 pandemic are creating great uncertainty as to the funding that will be available through these programs going forward.

Table ES-1 at the end of this executive summary provides a high-level summary of the strategies developed in this report, along with highlights of recent accomplishments/outcomes, and program resource needs or concerns.

Recommendations

Key recommendations for implementing strategies on forest (including urban forests) and rangelands include:

1. Maintain and Enhance the Resilience and Health of Forest and Rangeland Ecosystems

– At the heart of any set of strategies for forest and rangeland conservation must be understanding, maintaining, and enhancing the resilience and health of forest and rangeland ecosystems in California. Resilience refers to the ability of an ecosystem to respond positively to or recover quickly from the effects of disturbance. If forests and rangelands are not in good health and are unable to respond to disturbance, they will be less able to produce a wide range of goods and services.

2. Maintain and Enhance Urban Forests and Support Urban Forestry Programs -Urban forests are increasingly recognized as vital to improving air quality, reducing heat island effects, and promoting the general health and high quality of life of residents. Recognizing and supporting successful urban forestry programs is essential to the National Priority of connecting people with trees and the out-of-doors. Making efforts to improve urban forests and to initiate forestry programs where nonexistent, particularly in economically challenged settings and areas of high summer temperatures and low air quality, is a core goal of this work.

3. Invest in Forests and Rangelands – There is growing public demand for managing forests and rangelands to support a broad range of environmental services. There is also pressure in some areas to convert forests and rangelands to other uses. The loss of forests and rangelands diminishes their value as open space, wildlife habitat, and the many other resources that they provide. Investments in infrastructure and support to landowners (private and public) are needed to manage lands to maximize the benefits from these environmental services. Sustaining long-term investments in forest and range resources would benefit from diversifying funding sources and reducing reliance on bond funded programs. Promoting emerging markets for forest biomass, carbon sequestration, water quality, and other ecosystem services will provide additional incentives for landowners to provide these public benefits.

4. Promote a Collaborative Approach – Protecting, enhancing, and restoring forest and rangeland ecosystems requires a commitment to a science-based understanding of the threats

and risks to forests. This can be done most effectively through a collaborative approach that recognizes the wide range of forest uses and differing management objectives. While management objectives differ across ownership and administrative boundaries, a more cohesive approach is needed across all forests and rangelands. A collaborative approach among stakeholders is essential to further refine the preliminary priority landscapes that were identified in the 2010 Forests and Rangelands Assessment and for the CAL FIRE Forest Health Grant Program.

5. Prioritize Strategies Based on Co-benefits – With limited resources, the implementation of strategies for addressing priority management issues brought forth in 2010 and 2017 Forests and Rangelands Assessments needs to consider the co-benefits associated with any one strategy. For example, the strategic placement of fuel reduction projects to protect communities will likely have additional co-benefits for watershed protection.

6. Reduce Wildfire Risk to Communities and Structures – Continue to promote programs that assist with implementing fuel reduction efforts in and around at-risk communities, and home-hardening projects to make structures less flammable and more wildfire resistant.

7. Integrate Policies, Planning, and Organizations – Promoting more cohesive policies on forests and rangelands will require a more integrated management of private and public lands. Ecosystem health, wildfire management, water resource management, and many other important public benefits derived from forest and rangelands can only be achieved through a program that integrates policies and actions across larger landscape ecosystem units.

8. Research, Information Needs, and Decision Support for Adaptive Management – Understanding the threats to forests and rangelands, and the effectiveness of management actions requires a commitment to research, data collection and monitoring, and assessment. The 2010 and 2017 Forests and Rangelands Assessments made use of the best available data and assessment methods, but there are many data gaps and analysis limitations (see data limitation section for additional information). In addition, there are many knowledge gaps that would benefit from a sustained research agenda to better understand forest ecosystem processes, response to natural and human caused disturbances, and effectiveness of management actions. The limitations in data collection, monitoring, and research reduce the capacity to support and inform important policy choices.

9. Outreach to the Public and Landowners – Public understanding and support are essential for implementing strategies related to management of natural resources on forests, rangelands, and urban landscapes. In addition, expanded outreach to landowners is needed to encourage participation in vegetation management, reduction of fuel hazards, related programs that benefit overall forest health, and programs for urban tree planting and maintenance. There is also a need to expand the capacity of existing State and Private Forestry programs, related governmental agencies, councils, resource conservation districts and other regional entities that provide education and outreach to both landowners and the public.

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
National Theme/Priority 1: Conserve Working Forest and Range Landscapes			
<i>California Topic 1: Sustainable Working Forests</i>			
1.1. Maintain and improve the capacity of the wood products industry statewide.	47	Under Senate Bill 859 (2016), the California Natural Resources Agency and the Wood Products Working Group in 2017 prepared the report, Recommendations to Expand Wood Products Markets in California: Investing in Communities and California's Climate Resilient Future . The report identified three overarching recommendations and designated lead state agencies for each: Remove barriers to market and create pathways for success; promote innovation; and invest in human capital. Establishment, per Executive Order in 2018, of the Joint Institute for Wood Products Innovation , under Board of Forestry and Fire Protection (BOF) , and their completion of the 2020 reports, Recommendations to Expand Wood and Biomass Utilization in California and extensive Literature Review and Evaluation of Research Gaps to Support Wood Products Innovation . The Governor's Forest Management Task Force is working to address this issue through recommendations developed by its Rural Economic Development/Wood Utilization Working Group.	More funding for R&D on innovative wood products and biomass utilization. Public funding and relevant tax-exempt finance structures for incentives to leverage private capital to support infrastructure development and market demand for innovative wood and biomass products. Funding for grants and other incentives to support workforce development and biomass supply chain development.
1.2. Increase the capacity to provide incentives to non-industrial forest landowners to increase active sustainable management.	4	Use of California Climate Investment (CCI) funds to support Forest Stewardship and CA Forest Improvement Programs and the Forest Health Grant Program at CAL FIRE . The CAL FIRE Forest Health Grant Program provided \$142.6 million total in grants for FYs 2018-19 and 2019-20. Funding for forest restoration via Timber Regulation and Forest Restoration Fund (TRFRF) was legislatively established in 2012. CAL FIRE's Wildfire Resilience Grants offer an opportunity for CAL FIRE to work	Future of CCI funds, at least in the short term, is uncertain due to economic effects of COVID-19 pandemic. No funds have been allocated for current FY. TRFRF monies also may be facing a drop due to the COVID-19 economy. Another funding source—e.g., federal funding—may needed to sustain these

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		with cooperating entities through block grant agreements to help deliver expanded incentives, assistance and benefits to private nonindustrial landowners. In 2020, CAL FIRE granted over \$2.2 million in Prop 68 funds that will result in forest improvements to over 1300 acres in Napa, Sonoma, Lake, Mendocino and Plumas counties. More flexible “exemptions” from Timber Harvesting Plan requirements have been promulgated in recent years by Board of Forestry and Fire Protection, paired with multi-agency monitoring to evaluate their effectiveness and environmental outcomes. Completion of the 2019 Draft Natural and Working Lands Climate Change Implementation Plan that identifies the potential roles for forestlands in climate change mitigation and adaptation. A landmark accomplishment occurred for the Forest Legacy program (FLP) in 2014 with the completion of the 50,000-acre working forest conservation easement on Usal Forest, in collaboration with other state agencies and The Conservation Fund. In fiscal year 2018/19 the FLP invested \$9.327 million to complete four projects totaling 8,950 acres, including old growth redwood forest protection and public access enhancement.	important programs. A large, immediate surge of additional funding likely will be needed to help landowners restore their forests following the record fires of 2020.
<i>California Topic 2: Sustainable Rangelands</i>			
2.1. Maintain and improve the capacity of range industries statewide.	36	Ongoing activity of the BOF Range Management Advisory Committee , including completion of a new 2020 Strategic Plan . Research identifies that rangelands can have significant potential for increased carbon sequestration. Completion of the 2019 Draft Natural and Working Lands Climate Change Implementation Plan that identifies the potential roles for rangelands in climate change	Continued or enhanced funding of incentives through NRCS programs. Funding to re-establish state subventions paid to support Williamson Act zoning protections for rangelands. More funding is needed from state, federal, and
2.2. Increase the capacity to provide incentives to range landowners.	4		

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		mitigation and adaptation. Funded under Proposition 68, a new program at the CA Department of Conservation (DOC) in November 2020 issued \$2 million in grants for watershed restoration and conservation projects on agricultural lands in Marin, Sonoma, Kings and Ventura counties. The Sustainable Agricultural Lands Conservation (SALC) Program , operated by the Strategic Growth Council with assistance from the DOC fights climate change by protecting our productive farms through grants for planning and conservation easements, using CCI monies. To date, sixty-two easement projects have protected 90,700 acres.	private sources for conservation easements as interest outstrips funding at a time of large generational turnover in family ownership of ranches. As noted above, CCI fund allocations are suspended due to COVID-19 economic impacts; thus, an alternative funding source may be needed to continue work under the SALC Program. Infrastructure investment needs have been identified for new slaughterhouse construction and for maintaining and upgrading infrastructure for watering livestock
<i>California Topic 6: Population Growth and Development Impacts</i>			
6.1 Reduce urban sprawl by promoting redevelopment and infilling of available land within the urban matrix, strengthening planning at the local level, capacity building, and improving access to tools and data sources.	7	Major state legislation passed in 2008 (Senate Bill 375) is continuing implementation, building on the existing framework of regional planning to link the regional allocation of housing needs and regional transportation planning to reduce greenhouse gas emissions from motor vehicles. The state Strategic Growth Council was established to provide policies and grant funding to achieve these ends.	CCI funds for Strategic Growth Council grant programs have been frozen due to the COVID-19 economic impacts. A new source of funding may be needed to replace CCI monies to continue addressing this strategy.
6.2 Aid in efforts to reduce development sprawl in rural communities.	3	CAL FIRE and BOF Programs provide inputs to counties on their General Plan updates. Thanks to the availability of CCI funds, increases in conservation easements on farm, range, and forest lands have been achieved through multiple state programs, often in collaboration with local lands	Funding source needed to re-establish state subventions paid to support Williamson Act zoning protections for agricultural and rangelands. Increased funding for the Strategic Agricultural Lands

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		trusts. The number of land trust organizations in California has grown substantially in recent years.	Conservation Program. Need a new funding source to replace CCI monies, which have been frozen due to the COVID-19-driven economic downturn.
6.3 Support comprehensive planning at state and regional scales that is coordinated with wildlife habitat conservation efforts.	6	Completion of 2015 update to the State Wildlife Action Plan and companion plans , including a companion plan for forests. Programs at CAL FIRE and the BOF to assist or review local land use planning processes have grown in recent years.	No new resource needs identified.
6.4 Support local and regional community efforts in preserving scenic landscapes.	4	See Strategies 6.1 and 6.2.	See Strategies 6.1 and 6.2.
National Theme/Priority 2: Protect Forests and Rangelands from Harm			
California Topic 4: Wildfire			
4.1. Reduce the occurrence of damaging wildfires and reduce life, property and natural resource losses through the implementation of effective and efficient fire prevention programs and activities.	13	CAL FIRE and BOF completion of the 2018 Strategic Fire Plan for California . CAL FIRE units regularly update their Fire Plans consistent with the Strategic Fire Plan and taking into account local Community Wildfire Protection Plans. The 2018 California Forest Carbon Plan sets State and Federal goals for conducting fuels treatments on a total of one million acres per year. The 2020 Shared Stewardship Agreement between Forest Service and State of CA further reinforces this commitment. CAL FIRE has expanded its communications program and their educational and outreach efforts. CAL FIRE has secured funding and contracts to replace its aging Huey helicopters with state-of-the-art Sikorsky Firehawks. The BOF has completed a Programmatic Environmental Impact Report to facilitate compliance with the CA Environmental Protection Act (CEQA) for fuels management projects. The Governor's Forest Management Task Force provides	Many activities related to these strategies are funded with CCI monies, which are now threatened by the COVID-driven economic contraction. No CCI funds have been allocated for the current State fiscal year. The bulk of fire suppression hand crews come from the State's inmate-based Conservation Camps. Because of changes in the State's corrections laws and early inmate releases due to COVID-19, Conservation Camps have been closed and inmate numbers at still-open camps have declined. The State needs to find an approach and
4.2. Protect life and property from wildfire through efficient and effective fire protection planning and suppression, financial management, and firefighter/public safety strategies.	33		
4.3. Reduce the impacts of wildfire on ecosystem health, public safety, and private property through appropriate scientific research, education, and training.	12		

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		leadership, policy coordination, and actions to address forest health and management needs throughout the state. The CAL FIRE Forest Health Grant Program and Fire Prevention Grant Program have targeted \$219 million dollars of CCI monies to restoring resilient forest conditions, increasing the use of prescribed fire, reducing fire risks, and supporting research needs over State FYs 2018-18 and 2019-20. CAL FIRE in 2020 executed a research MOU with UC Berkeley, establishing the California Initiative for Research on Fire and Forests, to enhance collaboration with top-level UC fire researchers. The Regional Forest and Fire Capacity Program was established in 2019 at the CA Department of Conservation uses CCI monies to provide grants to increase regional capacity to prioritize, develop, and implement projects that improve forest health and fire resilience, facilitate greenhouse gas emissions reductions, and increase carbon sequestration in forests throughout California. \$20 million was granted in 2019.	funding to replace this lost hand crew capacity. Potentially use new fuel reduction crews to augment to existing crews to meet the current need for hand crew work.
4.4. Address post-fire responsibilities for natural resource recovery including watershed protection, reforestation, and ecosystem restoration.	9	State agencies have developed and instituted standard procedures for assigning multi-agency post-fire emergency watershed response teams to identify needs for watershed restoration. CAL FIRE has renovated its L.A. Moran Reforestation Center , including its seedbank facilities, and restarted reforestation seedling production.	The COVID-driven economic contraction may result in a drop in revenues to the Timber Regulation and Forest Restoration Fund, which is supported by a 1% tax on retail lumber and wood product sales. This fund supports the operation of the L.A. Moran Reforestation Center as well as forest restoration grant programs. An alternative funding source may be needed

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Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
			to keep these programs working at a time when there is a record need for post-fire restoration.
<i>California Topic 5: Forest Pests</i>			
5.1 Restore forest lands impacted by current and historical forest pest outbreaks, air pollution and invasive species.	13	The state mounted a significant response to the record drought and tree mortality event of 2011-2015+, including a governor-level Tree Mortality Task Force, which later broadened into the Forest Management Task Force. Addressing public safety, fire risk, and forest health have been critical components of this ongoing work. Participation of state and federal agencies, local governments, NGOs, Fire Safe Councils electrical utilities, and others has been significant. As described above, CAL FIRE's expanded Forest Health Grant Program and revitalization of the L.A. Moran Reforestation Center both contribute to greatly enhanced restoration capacity.	CCI monies have been an important source of funding in this area. These are now threatened due to the COVID-driven economic downturn, with no funds having been allocated during the current State fiscal year. An alternative funding source may be needed. Federal funding for forest pest treatments and restoration, primarily delivered through NRCS, need to be maintained or have increased funding to meet landowner needs.
5.2 Reduce/prevent forest pest outbreaks and control their spread to maintain ecosystem health, preserve ecosystem services and avoid public safety hazards associated with large scale tree mortality events.	8		
5.3 Prevent the introduction and spread of new exotic pests and invasive plant species.	13	The " buy it where you burn it " firewood educational program has been widely adopted and very successful. CA Department of Food and Agriculture inspection stations have become increasingly attuned to looking for firewood on vehicles entering the state.	
5.4 Rapidly control or contain outbreaks of exotic forest pests and invasive plant species.	9	State, federal, and local government, the UC system, and NGO cooperators have been very actively identifying and responding to addressing new invasive pests such as sudden oak death, gold-spotted oak borer, Mediterranean oak borer, and shot hole borers. Related research and education efforts have been substantial. The Forest Service's Region 5 Forest Health Detection Program provided	Research and funds to develop and implement enhanced forest mortality detection and monitoring using remotely-sensed data sources such as LiDAR imagery and NAIP orthophotos.
5.5 Monitor forestland to quickly identify new, and evaluate current, outbreaks of exotic forest pests and invasive plant species to protect the most vulnerable and	4		

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
valued forest and range land assets.		a robust and essential response to the forest mortality monitoring needs that arose during the 2011-2015 drought.	
<i>California Topic 11: Reducing Community Wildfire Risk</i>			
11.1 Promote formation of Local Fire Safe Councils for priority communities.	2	The 2018 Strategic Fire Plan recognizes these needs and establishes state goals and objectives to address them. CAL FIRE and the BOF have the primary responsibilities for implementing the Strategic Fire Plan, including plans for each of CAL FIRE's 21 local units and 6 contract counties.	Resources and incentives to enhance ignition resistance of structures/homes exposed to wildfire hazards.
11.2 Promote participation in the National Firewise/USA program.	1		
11.3 Establish a statewide comparative database of community wildfire planning.	2	Ongoing collaboration between Region 5 Forest Service and CAL FIRE on a statewide risk database and framework for mapping wildfire risk continues, and is expected to develop statewide risk assessment by 2022. Analysis supports operational needs and planning to reduce areas at highest risk.	Federal and State funding is contributing to a consistent risk assessment framework; additional support for data and information delivery, as well as ongoing maintenance is needed.
11.4 Reduce wildfire risk to communities.	4	The 2018 Strategic Fire Plan recognizes this need and establishes state goals and objectives to address it. CAL FIRE unit Fire Plans bring this down to the local level. Over just the 2018-19 and 2019-20 State FYs, CAL FIRE's Forest Health Grant Program and Fire Prevention Grant Program have provided \$219 million for forest health and fuels reduction projects that contribute to community protection. Programs at CAL FIRE and the BOF to assist or review local land use planning processes have grown in recent years. CAL FIRE recently established new teams with the sole responsibility of implementing fuels reduction projects. The intent is that these staff will not get dispatched for fire suppression work.	CCI monies have been an important source of funding in this area. These are now threatened due to the COVID-driven economic downturn, with no funds having been allocated during the current State fiscal year. An alternative funding source may be needed.

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
National Theme/Priority 3: Enhance Public Benefits from Trees, Forests, and Rangelands			
California Topic 3: Urban Forestry			
3.1. Promote urban tree planting to improve air quality and energy conservation.	13	CAL FIRE’s Urban and Community Forestry Program has increased funding for grants, including for tree planting and inventories, due to the availability of CCI funds and of bond funds through Proposition 68. This program awarded \$35.5 million in grants over the 2018-19 and 2019-20 FYs. Relatively new urban forestry and urban greening grant programs have been started at the Natural Resources Agency and the Strategic Growth Council. CAL FIRE and collaborators, including the Forest Service, are now conducting regular, periodic surveys of urban tree canopy to provide better information on status and trends.	CCI monies have been an important source of funding in this area. These are now threatened due to the COVID-driven economic downturn, with no funds having been allocated during the current State fiscal year. An alternative funding source may be needed.
3.2. Maintain urban tree canopy to conserve energy and improve air quality.	5		
California Topic 7: Climate Change			
7.1. Protect and enhance the capacity of California’s forests to sequester carbon through reducing risk of loss from disturbance, protecting existing forest land, and expanding forest area through tree planting.	18	The 2018 California Forest Carbon Plan evaluates forests, climate change, and carbon flux in detail, as well as setting State and Federal goals for conducting forest health and fuels treatments on a total of one million acres per year. The 2020 Shared Stewardship Agreement between Forest Service and State of CA further reinforces this commitment. The Forest Carbon Plan also highlights the need for more frequent data collection through the Forest Service Forest Inventory and Analysis program to aid adaptive management under the current circumstances of relatively rapid forest change. Preparation to shift the FIA 10-year measurement cycle to a 5-year cycle began in 2020 with full implementation of the annual program funded through 2021. Completion of the 2019 Draft Natural and Working Lands Climate Change Implementation Plan identifies the potential roles for forestlands in	The current primary funding source for these activities, CCI funds, has been significantly affected by the COVID-19 economic contraction. No CCI funds have been allocated to these programs for the current fiscal year. A new funding source, such as federal funds, may be needed to allow continuation of these important programs.
7.2. Support Adaptation Needs for Forests by Assessing Climate Vulnerabilities, Improving Institutional Capacity, and Promoting a Priority Research Agenda.	12		
7.3. Support Actions that Maintain, Enhance, and Protect Ecosystem Functions to Promote Biodiversity and Increase Resilience to Climate Change.	5		

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		climate change mitigation and adaptation. CAL FIRE is collaborating with the Forest Service, the Oregon Department of Forestry, the Washington Department of Natural Resources, the British Columbia Ministry of Forests, Land, Natural Resource Operations, and Rural Development, and academia to explore forest management and wood utilization scenarios for climate mitigation through the Pacific Coast Carbon Initiative. The Safeguarding California Plan: 2018 Update – California’s Climate Adaptation Strategy identifies current and needed actions state government should take to build climate change resiliency for forests and other sectors. Under the requirements of Assembly Bill 1504 (2010), the Board of Forestry and Fire Protection in 2016 began producing the California Forest Ecosystem and Harvested Wood Product Carbon Inventory Annual Reports, working in close collaboration with the Forest Service’s Pacific Northwest Research Station. The Board also has completed a Programmatic Environmental Impact Report to facilitate compliance with the CA Environmental Protection Act (CEQA) for fuels management projects. The Governor’s Forest Management Task Force provides leadership, policy coordination, and actions to address climate-change-related forest health and management needs throughout the state. Primarily CCI-funded, the CAL FIRE Forest Health Grant Program and Fire Prevention Grant Program have delivered \$219 million in grants over the past two state fiscal years to restore resilient forest condition, treat fuels, and support related research needs. CAL FIRE has renovated its L.A. Moran Reforestation Center,	

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		including its seedbank facilities and restarted reforestation seedling production. CAL FIRE is collaborating with the Forest Service, academia, and others to address adaptation issues for seedling production, species and genotype selection, and related issues.	
California Topic 8: California's Non-Metro Regional Economy			
8.1 Economic-, employment-, and housing-related measures.	30	Under Senate Bill 859 (2016), the California Natural Resources Agency and the Wood Products Working Group in 2017 prepared the report, Recommendations to Expand Wood Products Markets in California: Investing in Communities and California's Climate Resilient Future . The report identified three overarching recommendations and designated lead state agencies for each: Remove barriers to market and create pathways for success; promote innovation; and invest in human capital. Establishment, per Executive Order in 2018, of Joint Institute for Wood Products Innovation under the BOF, which in 2020 issued recommendations for expanding wood and biomass utilization. The Governor's Forest Management Task Force is working to address these issues through recommendations developed by its Rural Economic Development/Wood Utilization Working Group.	Specific funding needs identified in these reports include: - Publicly-funded financial incentives to leverage private capital to support infrastructure development and demand for innovative wood and biomass products markets. - Grant funding and other financial incentives to support workforce development and biomass supply chain development. - Development of science and technology-based regional strategies that prioritize achievable solutions.
8.2 Resident-related measures.	10		
8.3 Politics- and governance-related measures.	4		
California Topic 9: Water Resources			
9.1 Promote Watershed Protection and Watershed Restoration.	7	These strategies are addressed by the legislative creation of the Timber Regulation and Forest Restoration Fund and Program in 2012. This has enhanced interagency collaboration and effectiveness for regulation of timber harvest on nonfederal lands and provided new funds for fisheries and forest restoration. Funds provided by	The funding sources for much of these activities, the Timber Regulation and Forest Restoration Fund and CCI monies, are at risk due to the COVID-19-driven economic contraction. No new CCI
9.2 Improve Water Quality through Implementation of Best Management Practices and	11		

Table ES-1. Summary of California Strategies and Actions, with Related Outcomes/Accomplishments and Resources Needed, Organized by the Three National Themes/Priorities.

Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
Monitoring in High Priority Watersheds.		the Natural Resources Agency have strengthened water quality monitoring on forestlands through the Surface Water Ambient Monitoring Program at the State Water Resources Control Board . The previously-described CAL FIRE Forest Health Grant Program, Fire Prevention Grant Program, and California Forestry Incentives Program also help to address watershed protection and restoration. Collaborative activities such as the Sierra Nevada Conservancy's Watershed Improvement Program and the Tahoe-Central Sierra Initiative also work to address monitoring, assessment, and restoration needs at the watershed level. The recently re-established Watershed Coordinator Program at the CA Department of Conservation (DOC) provides grants to local government or non-profits to work with local stakeholders and downstream beneficiaries to develop plans and projects to improve watershed health. DOC's Working Lands and Riparian Corridors Program provides grants (\$2 million in 2020) to protect, restore, and enhance working lands and riparian corridors through conservation easements and restoration projects on agricultural lands. Over the past ten years, California has formalized the use of multi-agency Watershed Emergency Response Teams to evaluate watersheds post fire and to recommend actions needed to protect public safety and restore healthy watershed function.	monies have been allocated to these efforts for the State 2020-21 fiscal year. Alternative funding sources, such as federal monies, may be needed to sustain these activities at a meaningful level into the future.
<i>California Topic 10: Wildlife Habitat</i>			
10.1 Reduce the loss and modification of habitat that supports wildlife, and maintains California's unique biodiversity.	8	California completed a new State Wildlife Action Plan and companion plans in 2015. The State has multiple programs for the development and purchase of conservation easements to prevent	A critical funding source for State conservation easement programs, including Forest Legacy, CCI monies, is at risk

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Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		habitat loss, including the Forest Legacy Program, and these programs have been expanded in recent years through the availability of CCI funds.	due to the COVID-19-driven economic contraction. No new CCI monies have been allocated to these efforts for the State 2020-21 fiscal year. Alternative funding sources, such as Forest Service funding for the Forest Legacy Program, may be needed to sustain these activities at a meaningful level into the future.
10.2 Develop policies and incentives to facilitate better integration of wildlife conservation considerations into local and regional planning and land-use decision making.	12	The CA Department of Fish and Wildlife's Natural Landscape Conservation Planning Program has several elements, including data and assessment tools, that work to meet this strategy.	
10.3 Sustain healthy forest ecosystems to maintain California's unique biodiversity.	9	See discussion under Strategies 1.2, 2, 4, 5, 7, 9, and 11. The Board of Forestry and Fire Protection established the Effectiveness Monitoring Committee (EMC) in 2014 to create an independent body to assess the efficacy of the Forest Practice Rules and other related laws and regulations at protecting forest resources. An advisory body to the Board, the EMC helps implement an effectiveness monitoring program that provides an active feedback loop to policymakers, managers, agencies, and the public by soliciting robust scientific research that addresses the effectiveness of these laws at meeting resource objectives and ecological performance measures related to Assembly Bill 1492.	EMC funding for research grants, which comes from the TRFRF, may be at risk due to the COVID-19 economic downturn. An alternative funding source may be needed to continue this activity.

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Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
<i>California Topic 12: Renewable Energy</i>			
12.1 Continue to Develop, Implement, and/or Expand Initiatives that Reduce Demand for Large Renewable Energy Projects on Forests and Rangelands	2	In 2016, the California Energy Commission, the California Department of Fish and Wildlife, the U.S. Bureau of Land Management (BLM), and the U.S. Fish and Wildlife Service completed the Desert Renewable Energy Conservation Plan (DRECP) that identifies areas in the desert appropriate for the utility-scale development of wind, solar, and geothermal energy projects. The comprehensive plan also provides for the long-term conservation and management of covered species and preserves the natural resources, recreational areas, and scenic values.	
12.2 Promote Continued Research to Improve Guidelines, Standards, and Best Practices to Minimize the Environmental Impact of Large Renewable Energy Projects on Forests and Rangelands	1		
12.3 Support expansion of transmission infrastructure for emerging renewable energy generation from sources such as biomass, wind, hydro and solar in a way that minimizes environmental impact on forests and rangelands.	2	Senate Bill 100 (2018) established a State policy requiring renewable energy and zero-carbon resources supply 100 percent of electric retail sales by 2045. It requires the California Energy Commission, California Public Utilities Commission, and California Air Resources Board to prepare by January 2021 a report that addresses technologies, forecasts, transmission, and the maintenance of safety, environmental and public safety protection, affordability, and system and local reliability. The 2020 recommendations of the Joint Institute for Wood Products Innovation addresses transmission connectivity for forest biomass projects.	Funds to renew and expand the CA Public Utilities Commission (CPUC) incentive program for pipeline biogas interconnection and reserve a portion for forest biomass projects. The CPUC should also adopt a similar program to incentivize interconnection for forest BioMAT projects to the electricity grid.
12.4 Support Development of Biomass Energy Projects that can Facilitate Projects to Reduce Wildfire Risk and Improve Forest Health	4	Under Senate Bill 859 (2016), the California Natural Resources Agency and the Wood Products Working Group in 2017 prepared the report, Recommendations to Expand Wood Products Markets in California: Investing in Communities and California's Climate Resilient Future . The report identified three overarching recommendations and designated lead state agencies for each: Remove	Specific funding needs identified in these reports include: Publicly-funded financial incentives to leverage private capital to support infrastructure development and demand for innovative

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Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
		barriers to market and create pathways for success; promote innovation; and invest in human capital. Establishment, per Executive Order in 2018, of Joint Institute for Wood Products Innovation under the BOF, which in 2020 issued recommendations and a literature review and research gap evaluation for expanding wood and biomass utilization. The Governor’s Forest Management Task Force is working to address these issues through recommendations developed by its Rural Economic Development/Wood Utilization Working Group.	wood and biomass products markets. • Grant funding and other financial incentive to support workforce development and biomass supply chain development. • Development of science and technology-based regional strategies that prioritize achievable solutions. • Funding to explore efficient and effective permitting for bioenergy projects. • Funding to explore and recommend options for state procurement of bioenergy (including electricity, heating, combined heat and power [CHP], and low-carbon transportation fuels) generated from forest waste, mill residues, and wood processing waste as well as other vegetation removed for wildfire mitigation.

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Strategies	Number of Actions/Sub-Actions	Highlighted Outcomes/Accomplishments	Resources Needed
List of Acronyms used in this table: • BioMAT—Biomass Market Adjusting Tariff • BLM—Bureau of Land Management • BOF—California Board of Forestry and Fire Protection • CAL FIRE—California Department of Forestry and Fire Protection • CCI—California Climate Investments • CEQA—California Environmental Quality Act • CPUC—California Public Utilities Commission • DOC—California Department of Conservation • DRECP—Desert Renewable Energy Conservation Program • EMC—Effectiveness Monitoring Committee • LiDAR—Light Detection and Ranging • NAIP—National Agriculture Imagery Program • NGO—nongovernmental organization • NRCS—Natural Resources Conservation Service • SALC—Strategic Agricultural Lands Conservation • TRFRF—Timber Regulation and Forest restoration Fund • UC—University of California			

INTRODUCTION

Under the Forestry Title of the 2008 and 2014 Farm Bills, the U.S. Forest Service State and Private Forestry Redesign requires states to develop a Statewide Forest Resource Strategy based on their completed state assessments of forest resources (SAFR). California's 2020 Forests and Rangelands Strategy Report provides a long-term, comprehensive, coordinated plan for investing state, federal, and partner resources to address the management and landscape priorities identified in the 2010 and 2017 State Forest and Rangeland Assessments. The strategy report incorporates existing statewide forest and resource management plans and provides the basis for future program, agency, and partner coordination. This plan updates strategies that were first developed as part of the 2010 Forests and Rangelands Assessment and presented in the publication, California's Forest and Rangelands: 2010 Strategy Report (Fire and Resources Assessment Program, California Department of Forestry and Fire Protection).

In 2008, the U.S. Forest Service implemented a "Redesigned" State and Private Forestry (S&PF) program. The S&PF Redesign effort was conceived in response to the combined impacts of increasing pressures on our nation's forests and decreasing S&PF resources and funds. Significant threats to forests, such as insect and disease infestations, destructive wildland fires, and the loss of critical forested landscapes to development; coupled with the pressure placed on some local economies by the increasingly global nature of the forest products industry, pointed to the need for more progressive strategies to conserve our nation's forest resources.

The 2020 California Forests and Rangelands Strategy Report does the following:

- Provides long-term strategies that address issues and priority landscapes identified in the 2017 Forests and Rangelands Assessment and its 2020 addendums;
- Identifies how federal, state, tribal, private and other resources could be invested and aligned to address issues in priority landscapes that cross ownerships;
- Identifies how State and Private Forestry program areas, along with stakeholders and key partners can contribute to long-term goals and strategies;
- Provides indicators that can be used to evaluate the effectiveness of strategies over time and;
- Describes how California's strategy goals tier to the national goals of the federal State & Private Forestry Redesign, Montreal Process, and California Board of Forestry and Fire Protection policy objectives.

Relationship to Board of Forestry (Policy Statement)

The Board of Forestry and Fire Protection (BOF or Board) has responsibility for developing and implementing forest and rangeland policies in California. By statute (PRC 4789) the Board is required to develop a policy statement following the periodic assessment of forest and rangeland resources conducted by the California Department of Forestry and Fire Protection's (CAL FIRE) Fire and Resource Assessment Program (FRAP). The BOF adopted the 2010 Forest and Rangelands Strategy Report in lieu of crafting a new policy statement. As such, the revisions presented in this updated strategy report are consistent with the BOF's Policy Statement. At their discretion, the Board may elect to prepare a new policy statement in response to the 2017 Assessment and this updated Strategy Report.

To provide a transition between existing forest and range policies and proposed strategies, the 2017 Forests and Rangelands Strategies Report is structured to provide a crosswalk between strategy goals developed under Redesign and existing policy goals that followed the Montreal Process.

Other California, Regional, and National Plans

In addition to the Board of Forestry and Fire Protection's Policy Statement, there are many existing planning efforts, statewide or broader, that have established strategies for managing forests and rangelands. Existing plans provided the foundation for developing strategies. In most cases, strategies from existing plans were incorporated directly into this report or refined based on findings from the Forests and Rangelands Assessment. The following plans were consulted.

- The 2018 Strategic Fire Plan for California
- National Fire Plan
- Board of Forestry and Fire Protection Policy Statement
- California Forest Carbon Plan
- California Water Plan
- Water Board Strategic Plan Update
- Regional Water Quality Control Board (Basin Plans)
- Delta Vision Strategic Plan
- Roadmap for Biomass Development – California Energy Commission report
- California State Wildlife Action Plan
- California Outdoor Recreation Plan
- California's 2017 Climate Change Scoping Plan
- Safeguarding California Plan: 2018 Update (climate adaptation)
- Indicators of Climate Change in California (Office of Environmental Health Hazard Assessment, California Environmental Protection Agency)
- Northwest Forest Plan
- National Fish Habitat Action Plan
- US Forest Service Strategic Framework for Responding to Climate Change

Context

California is a large and diverse state covering roughly 100 million acres. Although it is the most populous state in the nation, the population is concentrated in urban areas and roughly 80 percent of the land base still consists of forests and rangelands (Figure 1).

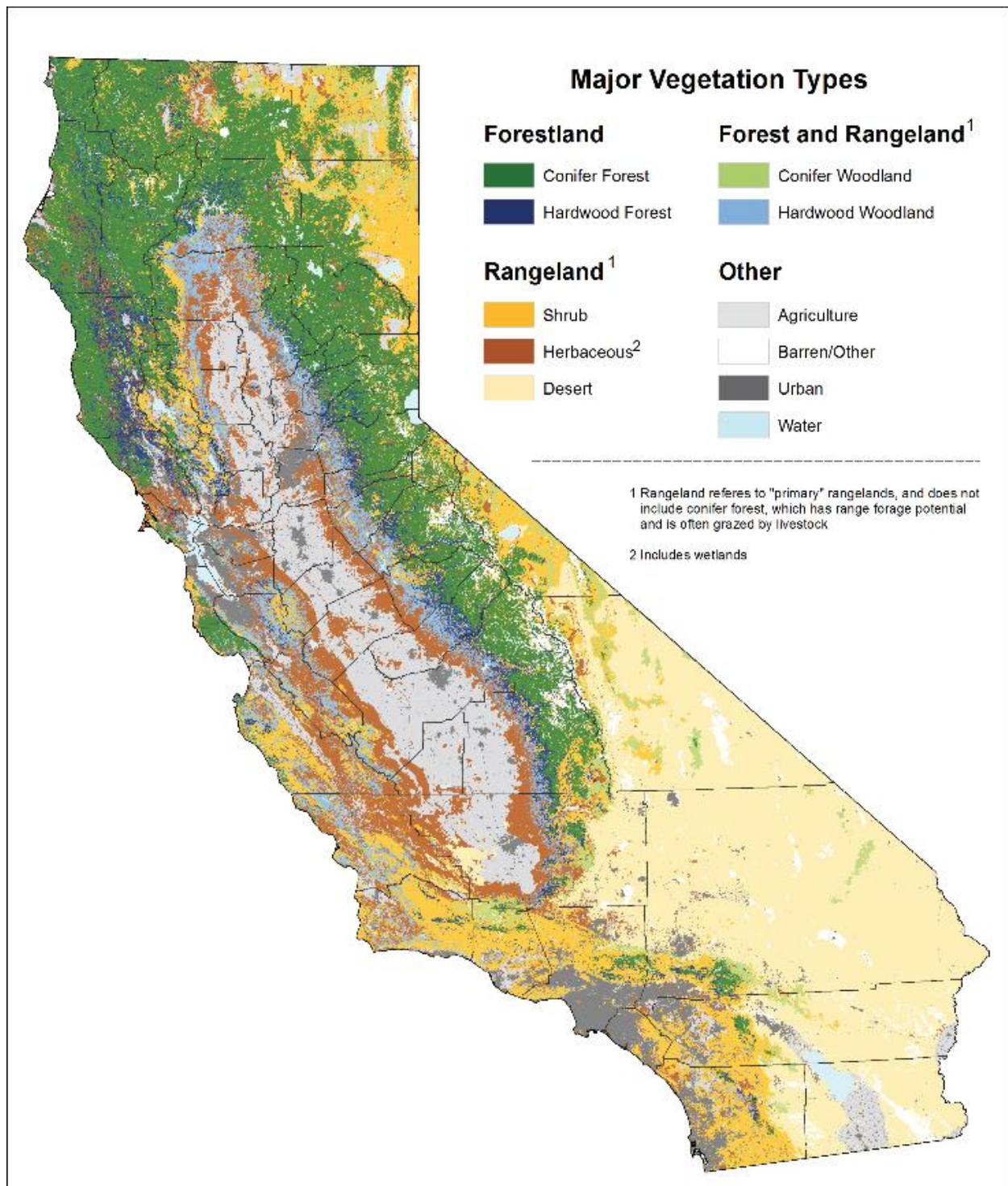


Figure 1 – Forests and rangelands in California. Source (FRAP, 2017).

Covering over 80 million acres, California's forests and rangelands provide a wide range of environmental services including: recreation, scenic vistas, wildlife and fish habitat, clean air, watershed functions (water supply and water quality), forest and agricultural products, carbon sequestration, and other services. With a population nearing 40 million (and growing) and outstanding wildlands that attract large numbers of U.S. and international visitors, there are many demands placed on our forests and rangelands. Population growth has increased concerns over water resources, water quality, preservation of open space and habitat, species extinction, and increased risk from wildfire.

California forests are comprised of both conifer forest types (~ 19 million acres) and hardwood forest types (~ 13 million) of varying age classes. Overall, forest stands are mostly dominated by medium age (80 to 120 years old) trees (FRAP, 2003; Christensen et al., 2008). However, forests on public lands tend to consist of older forest stands than those found on private lands (Christensen et al., 2008). There are general concerns that current forest stands are much denser (number of trees per acre) than they were historically; and that there are increased risks to forest health associated with this. In addition to affecting forest condition, population pressures and resulting land conversions can also reduce the total extent of forest land across the state.

Challenges

Drought and Elevated Tree Mortality

Since the initial strategy report was written in 2010, California forests have experienced significant tree mortality brought on from prolonged drought conditions. California experienced five years of drought from 2012–2016 resulting in a massive tree die off in the southern Sierra Nevada.

Year	Cumulative Totals of Dead Trees in millions (statewide)
2010	3.1
2011	1.6
2012	1.8
2013	1.3
2014	11
2015	40
2016	102
2017	129
2018	148
2019	163

In 2012, there was an increase in western pine beetle, mountain pine beetle and fivespined Ips attacks in low- to mid-elevation pine forests. In 2013, oaks in the Sierra foothills dropped their leaves early, a natural defense to limit water loss during periods of water stress and extreme heat. Gray pine mortality was also increasing in these same areas of the southern Sierra in 2013 and continuing in 2014, as well as in areas on the central coast and near Clear Lake. By March of 2015 tree mortality began to increase dramatically, first affecting ponderosa pine, then moving up slope and affecting sugar pine, incense cedar, and true firs. Drought stress allowed pests to kill trees that might otherwise have survived and resulted in

a bark beetle epidemic that threatened to spread statewide. While the rate of tree mortality has declined in recent years, the cumulative impacts are long-lasting and have heightened public awareness over forest health issues and how best to manage forests under a warming climate.

Wildfire Trends

Over the last three decades, California wildfires burned an average of 558,000 acres annually, with a marked increase in area burned since 2000 (i.e. ~700,000 ac/yr since 2000). Much of this increase has been in conifer forests [indicator (①)4.3]. Eight of the ten largest wildfires in modern state history have occurred since 2000, including several high-profile “mega-fires”. Fire exclusion in frequent-fire adapted forests has contributed to uncharacteristic patterns of high severity fire (①4.4) with potentially long-lasting effects. Fuel management activities, including prescribed and managed wildfire and mechanical fuel treatment, averaged about 261,000 acres annually over the last decade; this represents only a fraction of treatment needs and the statewide goals of treating one million acres per year across all forestlands.

Policy Challenges

Forest and range policies must strike a balance between promoting the goods and services that are produced by these lands while protecting and enhancing the underlying ecosystems. The complexity and diversity of California’s forests and rangelands creates a set of difficult policy choices. The following policy issues, framed under the Montreal Process criteria, were identified as part of the 2017 Assessment (Table 1).

A central goal of this report is to integrate proposed strategies with existing plans and strategies that are already being implemented across California’s forests and rangelands. Many state, federal and local laws already define a complex framework of goals, programs, and funding sources that apply to forests and rangelands in the state. Strategies already exist at many different levels, from governmental agencies and tribes to communities and community groups, nongovernmental organizations, private landowners, and other stakeholders. The development of strategies is also greatly influenced by public ballot initiatives. California voters in the past two decades have been supportive of programs and ballot initiatives that support conservation, restoration, open space, urban forests, and improved environmental quality. Compared to other states, California’s investment in natural resources is relatively high. The California State Legislature has invested approximately 1 billion dollars to improve forest health and reduce community wildfire risk.

Table 1. Montreal Process criteria, policy issues, options and opportunities.

Criterion 1: Policy Issues for Conservation of Biological Diversity	
Policy Issues	Options and Opportunities
Protect Forest and Rangelands from Conversion	Resume Williamson Act state government subvention payments to counties. Sustain and increase efforts to protect biologically important areas and working landscapes through easements. Continue state and local efforts to reduce sprawl.
Wildlife Habitat Connectivity	Coordinate land acquisitions and easements to connect biologically important areas to each other, connect upland and downslope habitats, and expand and connect ecologically fragmented areas into more viable habitats.
Adaptation to Climate Change	Improve overall ecosystem health (see MPC3). Support research and adaptive management approaches for changing management practices to maintain resilient ecosystems as climate changes.
Water Quality and Riparian and Fish Habitat	Support increased capacity for CA Department of Fish and Wildlife to expand effective stream science, fisheries and watershed recovery and restoration and regulatory enforcement. Encourage stakeholder partnerships to protect local and regional habitats.
Ecosystem Health	Ecological restoration projects and treatments to restore natural fire regimes should include consideration of biodiversity and wildlife habitat.
Criterion 2: Policy Issues for Maintenance of Productive Capacity	
Policy Issues	Options and Opportunities
Forest and Rangelands Conversion	Resume Williamson Act subvention payments to counties. Sustain and increase efforts to protect working landscapes through easements.
Forest and Rangeland Health	See MPC3 for detailed discussion of forest health issues. Restoring natural fire regimes would also improve rangeland health.
Climate Change	Improve overall ecosystem health (see MPC3) to maintain resiliency and reduce climate change impacts. Support research and adaptive management approaches for changing management practices to maintain resilient ecosystems as climate changes.
Landowner Assistance	Continue and enhance efforts to provide technical and financial assistance to small landowners to improve productivity, and forest and rangeland health. Expand CAL FIRE nursery operations to maintain a robust reforestation seedbank and to provide seedlings to landowners for replanting following tree mortality from pest outbreaks and wildfire.
Criterion 3: Policy Issues for Maintenance of Forest Ecosystem Health and Vitality	
Policy Issues	Options and Opportunities
Restoration of Natural Fire Regimes	Support collaborative landscape-level efforts, to meet Forest Service and State treatment goals and improve forest health across all ownerships.
Wood Processing Infrastructure Deficit	The Little Hoover Commission called for the state to develop a long-term plan for forest materials. This could lead to more specific tactics such as grants for community wood processing infrastructure. The Joint Institute for Wood Products Development was established recently under the Board of Forestry and Fire Protection.
Nonindustrial Forest Management	Reduce barriers for more active management to maintain productive and resilient forests, including regulatory cost reduction, increased marketing options, and increased availability of technical/financial assistance.
Adapting Policies and Practices for Climate Change	Support critical research needed to anticipate changes and adapt policies and practices for climate change.
Criterion 4: Policy Issues for Soil and Water Resources	
Policy Issues	Options and Opportunities
Water Supply and Quality Protection and Improvement on Livestock Grazed Rangelands	Federal and State programs providing funding and technical assistance to build and/or maintain livestock watering infrastructure according to BMPs.
Protection and Restoration of Headwater Forests	Prioritize funding for restoration of headwater forests, target fuel reduction and expanded use of prescribed fire and managed fire, restoration of forests subject to severe wildfire. Restoration of mountain meadows. Where practical, work with ranchers to develop agreements to retire or modify grazing practices in allotments with ecologically and hydrologically important mountain meadows. Utilize authorities and opportunities provided by legislation passed in recent years, such as AB 2480 (Bloom, 2016) and AB 2551 (Wood, 2018).

Montreal Process Criterion 5: Policy Issues for Forest Contribution to Global Carbon Cycles	
Policy Issues	Options and Opportunities
Forest Health	See MPC3; Forest Carbon Plan; Safeguarding California (Forestry Chapter)
Increase Carbon Storage in Forests and in Wood Products	Forest Carbon Plan
Urban Heat Islands	Maintain and enhance urban tree cover.
Criterion 6: Policy Issues for Maintenance and Enhancement of Long-term Multiple Socio-economic Benefits to Meet the Needs of Societies	
Policy Issues	Options and Opportunities
Limited Job Opportunities in Rural Communities and Disadvantaged Urban Communities	Increase education and training opportunities for rural communities. Provide incentives for businesses to locate in rural disadvantaged communities. Support Governor's designation of Opportunity Zones. http://dof.ca.gov/Forecasting/Demographics/opportunity_zones/ Utilize urban forestry and urban greening grant programs as a means to provide training, education, and jobs to residents of disadvantaged urban communities.
Expanding Renewable Energy to Meet 2030 Targets for 50% Energy from Renewable Sources	Support policies for small-scale renewable energy that contribute to the local economy. Promote urban programs that increase energy conservation.
Declining Health and Well-being	Improve access to health care. Support programs that address opioid addiction and return people back into the workforce.
Restoring Ecosystem Health	See MPC3
Maintain Urban Quality of Life	Continue to support local urban forestry programs, including tree planting and maintenance.
Criterion 7: Policy Issues for Legal, Institutional and Economic Framework for Conservation and Sustainable Management	
Policy Issue	Options and Opportunities
Research and Education	With increasing pressures on forests and rangelands, including climate change, research and education are more important than ever to develop the proper science-based policies and practices to adapt to changing conditions.
Collaboration	Continue to develop and support mechanisms to foster collaborative landscape-level initiatives, particularly for improving ecosystem health and restoring natural fire regimes.
Funding	Continue to develop stable funding sources for critical sustainability programs and initiatives.
Assessment	Agencies need to collaborate to develop and implement a more comprehensive and coordinated system to track and communicate environmental indicators.

Building on the 2017 Assessment

The California 2017 Forests and Rangelands Assessment Report consists of twelve chapters that cover a broad range of environmental and resource management issues. Each chapter consists of one or more indicators that are used to track status and trends. In some cases, indicators are evaluated against established policy and resource targets.

In addition, several of the priority landscapes that were developed in the 2010 Forest and Range Assessment report have been updated. Priority landscapes were derived by intersecting asset and threat layers. The combination of high value assets coinciding with high threats produced high priority areas.

Cross-Cutting Issues

The strategy report organizes the 2017 assessment framework by placing the assessment topics under the related three broad national themes provided by the Forest Service. Also, note that there are many cross-cutting issues that emerge. Examples include:

* **Socio-Economic:** Forests and rangelands, and urban forests, are critical to the economic, social and environmental well-being of California.

* **Emerging Markets:** Uses of forest resources are increasing and changing. Emerging markets are placing new demands on these lands, such as for renewable energy and ecosystem services. At the same time, the forest products industry and range-livestock industries face difficult challenges.

* **Forest Health:** California's forests and rangelands, and their many uses, are under a variety of threats (forest pests, development, wildfire, climate change), and trends often indicate an increasing number, extent, and severity of threats.

* **Infrastructure:** A significant portion of forests and rangelands, urban forests, and the infrastructure required to meet demands from these lands, needs improvement.

* **Uncertainty:** Among other factors, changes in the climate and shifting economic factors add considerably to the uncertainty surrounding forest and rangeland issues.

* **Management Solutions:** There are potential management solutions leading to desired future conditions that produce more carbon, improved water quality, more vibrant rural economies, and landscapes more resistant to threats.

However, reaching these desired future conditions will require meeting numerous political, social and economic challenges.

Outreach Process

CAL FIRE conducted extensive outreach throughout the development of the 2017 Forest and Range Assessment Report. Additional outreach was done in the preparation of this strategy report. Outreach was targeted at government agency partners, tribes, nongovernmental organizations (NGOs), and other public groups. Outreach consisted of meetings, workshops, webinars and use of an assessment website for posting information and soliciting public input.

Federal and State Agencies

Following the Redesign guidance, outreach was conducted with the following groups: State Forest Stewardship Coordinating Committee, State Technical Committee, the state wildlife agency, applicable federal land management agencies such as the Forest Service and Bureau of Land Management, the State Urban Forestry Council, and the Forest Legacy Program.

The Natural Resources Conservation Service (NRCS) is another key agency and meetings were held to brief staff on findings from the assessment and on the development of strategies to address both priority issues and priority landscapes. Other important federal agencies consulted, briefed, or notified included the National Oceanic and Atmospheric Administration's (NOAA) National Marine Fisheries Service, the Fish and Wildlife Service, and the National Park Service.

CAL FIRE provided briefings and solicited input on strategies from a number of state agencies including: Department of Water Resources, Department of Conservation, State Water Resources Control Board, Department of Fish and Game, Department of Parks and Recreation, California Energy Commission and California Air Resources Board. Each of these agencies has responsibilities and expertise that involve many of the resource issues in the assessment and would be important partners for implementing strategies.

Public Outreach

As part of the 2017 Forest and Range Assessment CAL FIRE held several webinars/workshops that were open to the public. Topics covered included methods, analytical framework, themes and subthemes, preliminary results, and proposed strategies. Attendance at the webinars/workshops was good and a variety of input was received, primarily in regard to data inputs, issues being addressed, and suggested strategies. Announcements were made to a mailing list of approximately 150 stakeholders representing state, federal, private, non-profit, academic and tribal interests. The announcements were then circulated beyond this to additional mailing lists belonging to other agencies. These mailing lists were also used to broadcast the information posted on the CAL FIRE website about the assessment.

Forest Service Partnership

The State and the Forest Service have long been significant partners for the protection and enhancement on forest and rangelands of all ownership classes across California. This partnership includes an ongoing flow of funding, expertise, data and information, and staff support in both directions. Recent salient points of partnership and collaboration include the signing of two Master Stewardship Agreements and an Agreement for Shared Stewardship

Entered into in August, 2020 the Agreement for Shared Stewardship of California's Forest and Rangelands defines a new level of collaboration and commitment between the State and the Forest Service. This MOU:

...establishes a joint framework to enhance science-based forest and rangeland stewardship.... The U.S. Forest Service and the State...commit to maintain and restore healthy forests and rangelands that reduce public safety risks, protect natural and built infrastructure, and enhance ecological habitat and diversity. The Parties agree to develop shared tools, coordinated processes, and innovative approaches to increase the pace, scale, and effectiveness of forest and rangeland stewardship in California.

The MOU establishes the principles of:

1. Utilize science;
2. Prioritize community safety and ecology;
3. Improve efficiency;
4. Scale up ecologically-based forestry across public, tribal and private sectors;
5. Coordinate land management; and
6. Collaborate and innovative with all stakeholders.

Per the MOU, the State and the Forest Service are committed to these actions:

1. Sustainably treat one million acres per year, with each party responsible for half of this total;
2. Develop a joint twenty-year project plan for forest and vegetation management;
3. Use sustainable vegetation treatments, including an expansion of the use of prescribed fire and the establishment of world-class monitoring and research program;
4. Expand forest management and associated infrastructure;
5. Promote ecological co-benefits;
6. Develop markets for wood products and recycle forest byproducts;
7. Improve access to sustainable recreation for California's diverse population;
8. Foster fire-adapted communities; and
9. Advance science and share monitoring and data analytics.

Although a nonbinding agreement, the MOU establishes a comprehensive, new framework for what might be described as a quantum leap in collaboration and goal setting for the signatories. Much of what the MOU seeks to do falls well within the scope of CAL FIRE's forestry assistance and fire prevention programs, and meshes well with the strategies that are developed within this 2020 Strategy Report Update.

Statewide Coordination

An important resource for statewide coordination on forest management issues in California is the [California Forest Management Task Force](#) (FMTF). First established in 2015 as the Tree Mortality Task Force in response to Governor Jerry Brown's [tree mortality state emergency proclamation](#), the body was then shifted to a broader mandate as the FMTF in 2018 in order to meet the directives of Governor Brown's [Executive Order B-52-18](#).

The California Natural Resources Agency leads the FMTF, which works to protect the environmental quality, public health, and economic benefits that healthy forests provide to California. The Task Force aims to increase the rate of forest treatments and expand state wood product markets through innovation, assistance, and investment. Advancing forest health project capacity, readiness, and completion statewide aligns with the 2018 California Forest Carbon Plan, the goal of which is to establish healthy and resilient forests that can withstand and adapt to wildfire, drought, and a changing climate.

As a central part of the FMTF, subject matter Working Groups work with regional Prioritization Groups to achieve the directives laid out in the Governor's 2018 Executive Order and the California Forest Carbon Plan. The Working Groups have developed management goals consistent with the Executive Order and are focused on coordination among the entities with funding, implementation, or regulatory authority over their management goals and will work to resolve any statewide barriers.

Resources for CAL FIRE's State and Private Forestry Programs: From Plenitude to Significant Uncertainty

CAL FIRE's State and Private Forestry Programs have seen a remarkable growth in staffing and funding, including funds for grants, over the past ten years. This funding has come from

several sources, including a now defunct fire protection parcel assessment fee applied to State Responsibility Area lands, the creation of a lumber and wood products assessment with revenues flowing into the Timber Regulation and Forest Restoration Fund (TRFRF) under Assembly Bill 1492 (2012), and the increasing dedication of California Climate Investment (CCI) monies to a broad range of forest health grants and activities. CCI monies have been substantial enough to permit the State to fund millions of dollars of forest restoration work on National Forest lands, for example.

This very positive situation has taken a stark turn towards significant funding uncertainty in the wake of the COVID-19-driven economic downturn in the United States and the rest of the world. For example, the fiscal year 2020-21 budget approved by the state legislature did not allocate any CCI funds to CAL FIRE's grant programs. While TRFRF revenues have been holding their own, these are likely to drop as the economic downturn continues and eventually has a greater effect on home construction and renovation. The significant decline in state general revenues and the fact that states cannot incur budget deficits further indicates a threat to the level of general fund revenues available for California's State and Private Forestry programs. At the federal level, there currently appears to be little consensus among the White House, Senate, and House of Representatives as to how provide another round of funding to address the economic and health care needs resulting from the pandemic. More generally, the Federal government this Fall reported a \$3.1 trillion deficit for its last fiscal year, which was \$2 trillion more than initially projected when it was approved. This uncertainty includes whether the Federal government will provide general funding to state and local governments to offset some of their revenue downfalls. How long the current economic slowdown will continue and how deep it will get remain uncertain.

In addition, support for bond funded environmental programs, typically established through ballot initiatives or referendums, tends to be cyclical and may be difficult to sustain with increasing budgetary pressure, particularly under the current economic climate. Long-term solutions should include a mix of revenue sources that in addition to CCI and bond funding include fees, where appropriate, for risk reduction, and market based incentives to promote sustainable forest management that reduces environmental risks and enhances environmental services.

With these foregoing uncertainties come similar uncertainties for the levels of State and Federal funds that will be available for CAL FIRE's State and Private Forestry programs. Currently, CAL FIRE is waiting for more resolution of these uncertainties before it attempts to identify what program funding needs may be. At one level, however, funding needs for post-fire restoration and planting are likely to be great following this year's record wildfire season, having burned well over 4 million acres of wildlands this year. Also, as this year's record wildland fire season has shown, vast levels of funding are needed to address the problematic high fuel levels on forestlands across all ownership classes in the state.

One bright light in the recent California legislative session is that the legislature has created a new funds account where revenues can be deposited from forest products receipts that the State generates from the management of federal forestlands under Good Neighbor Authority or Stewardship agreements. Once in this special fund, the monies can be allocated to CAL FIRE or other state agencies to cover States costs in implementing the source projects or for implementing new forest health treatments on other areas, for example. This revenue capture arrangement has been done previously in California between Resource Conservation Districts and the Forest Service, but never with CAL FIRE or other state agencies. This model also has been successful in other states.

Organization of the Document

The strategy topics correspond directly with the assessment topics. For each of the following assessment topics one or more strategies have been proposed. The Strategies and supporting indicators are crosswalked between the National Goals in Redesign and the Montreal Process Criteria.

- Sustainable Working Forests
- Sustainable Rangelands
- Urban Forestry
- Wildfire
- Forest Pests
- Population Growth and Development Impacts
- Climate Change
- California's Non-Metro Regional Economy
- Water Resources
- Wildlife Habitat
- Reducing Community Wildfire Risk
- Renewable Energy

National Theme 1: Conserve Working Forest and Range Landscapes

Chapter 1: Sustainable Working Forests

Indicators and key findings report on trends that vary by forest ownership group.

Forest industry in California operates under statutory and regulatory mandates requiring a demonstration of long-term sustained yield. Inventory data confirm that growth exceeds harvest and mortality on industrial and nonindustrial lands (①1.1). About 86% of industrial timberland is under some form of voluntary third-party forest sustainability certification (①1.5).

INDICATORS

- ①1.1 Growth, removals, mortality
- ①1.2 Timberland restoration
- ①1.3 Silvicultural methods
- ①1.4 Historic timber harvest
- ①1.5 Sustainability initiatives
- ①1.6 Carbon Forests

Nonindustrial forestland owners manage for a range of objectives. Based on the National Woodland Owner Survey (Butler, Hewes et al. 2016) (Butler, Hewes et al. 2016), the most common reasons for owning forestland in California are scenic beauty, residence or vacation home, protecting nature, passing the land to heirs, and investment property. Federal and state programs continue to be important sources of technical and financial assistance to small landowners for their forest management planning and activities. Active management on 11 percent of nonindustrial timberland is currently facilitated by a Nonindustrial Timber Management Plan (①1.5).

On public timberlands (primarily owned by USDA Forest Service), stand condition in terms of stocking levels (①1.2) and deviation from historic fire regimes (①4.1) are major issues. According to the Forest Service, “nearly a century of fire exclusion in California, coupled with other management decisions on both private and public land, has resulted in forests that are at an increasing risk of loss due to large scale disturbances.” The Agency further states “only an environmental restoration program of unprecedented scale can alter the direction of current trends.” Forest Service goals set in 2011 for California include treatments to improve forest resiliency and other ecological restoration activities such as restoring degraded meadows, reforestation of burned areas, increasing habitat connectivity, restoring natural fire regimes, and decreasing the impact of invasive species (USDA Forest Service 2015). To achieve their goals, the Forest Service would need to find the resources to increase ecological restoration activities from the current 200,000 acres a year to approximately 500,000 acres per year (USDA Forest Service). The federal agency has committed to achieving this annual goal through both its participation in the California Forest Carbon Plan and through its recent Shared Stewardship Agreement with the State of California.

California imports about 80% of the lumber and 90% of all wood products used in the state (California Department of Forestry and Fire Protection 2003, Laaksonen-Craig, Goldman et al. 2003, Battles, Gonzalez et al. 2014). Meeting in-state demand requires importing lumber and wood products from other states and countries that may have less restrictive rules governing forest practices. However, numerous statewide factors and trends are impeding more active timber management in California that could better contribute to our demand for wood products. These include a decline in the skilled workforce and physical infrastructure such as sawmills and biomass plants (①1.4, ①12.2), the impact of global competition on prices of lumber and wood products, elevated risk from changing climate and disturbance regimes, and regulatory compliance costs. Public sentiment against timber harvest is another factor that state and

federal agencies must address, as part of the Timber Harvesting Plan review process for private lands and the public input process for public land planning and management.

Opportunities are emerging to improve forest health and/or enhance the levels of certain ecosystem services provided by working forests. On the North Coast and in the Northern Interior, there are numerous examples of working forest acquisitions by conservation organizations, voluntary participation in carbon offset programs (①1.6), and use of conservation easements to diversify income and influence management direction. The result is a significant and increasing portion of timberland being managed under longer rotations and uneven-aged management regimes, typically with a greater emphasis placed on protecting fish and wildlife, as well as water quality and carbon sequestration (①1.3).

In other parts of the state, both public and private forests face much higher risks, as demonstrated by current widespread tree mortality in the Sierra Nevada and recent large stand-replacing wildfires. Current high stocking levels and past fire exclusion are major concerns (①1.2). The scope and scale of treatments needed at the landscape level to address these critical issues is significant and challenging, given limited wood processing infrastructure and funding. There are, however, numerous examples of collaborative landscape-level projects designed to improve forest health, reduce fire threat, and contribute to local economic development. There are programs that support these collaborative efforts at the federal level (e.g. Joint Chiefs' Landscape Restoration Partnership, National Cohesive Wildland Fire Management Strategy, Collaborative Forest Landscape Restoration Program) and state level (e.g. CAL FIRE Forest Health Initiative/California Climate Investments grant program, Sierra Nevada Conservancy Watershed Improvement Program, and Department of Conservation Regional Forest and Fire Capacity Program).

KEY FINDINGS

The Resource Base

- 🔑 Based on our modeled scenarios, in the future a portion of current timberlands will no longer be suitable for growing the tree species that are of primary commercial value in California. Statewide, there is a projected loss of between 0.6 (3%) and 1.4 million acres (8%) of timberland under the warmer/wetter and hotter/drier climate change scenarios, respectively. These former timberlands will likely transition into oak woodland, brush, or grassland, with different ecological and economic values (e.g., vegetation type conversions may increase opportunities for livestock grazing).
- 🔑 Conservation easements are an increasingly effective tool for preserving timberlands with important environmental or social values and for protecting working forests from conversion to uses incompatible with sustainable forest management (①6.3).
- 🔑 Invasive plants, many of which are not native to California, are having a major impact on both working forests and other forestlands, as well as rangelands.

①1.2 Indicator: Timberland in Need of Restoration Treatment to Reduce or Increase Stocking

- 🔑 ① Between 5.5 and 9.5 million acres are potentially in need of treatment to improve stocking, depending on whether overstocked stands are based on a 100% or 60% Stand

Density Index (SDI) rule (see 2017 Assessment document for details). Stocking is an issue on public, forest industry, and nonindustrial timberlands.

- ① 🔑 Using the 100% SDI rule, only about 4% of public timberlands is overstocked. Using the 60% rule, about 32% is overstocked.
- ① 🔑 Hardwood dominated timberlands are a natural component of the landscape, but have expanded in some areas due to past disturbances or management. Restoration treatments are more economically feasible on higher site quality lands, such as the 1.4 million acres of Sites 1, 2, and 3 land. About 41% of these acres are in the tanoak/laurel type, typically areas on the North Coast region where redwood and Douglas-fir have been displaced by hardwoods due in part to past timber management.

Management Context

- 🔑 Forest Service goals set in 2011 and reiterated in the recent Shared Stewardship Agreement include treatments to improve forest resiliency and other ecological restoration activities such as restoring degraded meadows, reforestation of burned areas, increasing habitat connectivity, restoring natural fire regimes, and decreasing the impact of invasive species (USDA Forest Service 2015). To achieve their goals, the Forest Service would need to find the resources to increase ecological restoration activities from the current 200,000 acres a year to approximately 500,000 acres per year (USDA Forest Service).
- 🔑 Regulations affecting timber harvesting on private lands have increased over time in terms of the scope of activities regulated, including impacts on threatened and endangered species, water quality and forest roads. More recently, there has been an increase in more flexible regulatory approaches, such as new or expanded Timber Harvesting Plan exemptions, intended to facilitate fuels reduction work that generates some commercial timber.

① 1.1 Indicator: Net Growth of Growing Stock on Timberland

- ① 🔑 On forest industry timberlands, the most actively managed lands, growth exceeded harvest and mortality by an average of almost 22 ft³/acre/year over the re-measurement period (2001–2006 to 2011–2016).
- ① 🔑 On nonindustrial timberlands, a portion of which are actively managed, growth exceeded harvest and mortality by an average of over 85 ft³/acre/year over the re-measurement period (2001–2006 to 2011–2016).
- ① 🔑 On Forest Service timberlands, which are managed for multiple objectives including ecosystem services, growth exceeded harvest and mortality by an average of over 33 ft³/acre/year over the re-measurement period (2001–2006 to 2011–2016). However, projecting a “business as usual” scenario into the future (Goines and Nechodom 2009) indicates this trend will not continue and the National forests could become net emitters of carbon by the end of the century, unless an alternate management path is taken.

① 1.3 Indicator: Timberland Harvested by Silvicultural Method

- ① 🔑 There has been a steady significant decline in total acres harvested over the 1997–2015 period. Harvested acres in 2015 was about half of 1997 harvest acres.
- ① 🔑 Acres harvested using uneven-aged systems have been fairly constant through time, despite the decline in total harvested acres. Thus, the percentage of acres harvested

using uneven-aged silvicultural systems has increased, from an average of 29% of harvested acres over the 1997–2001 period, to 41% over the 2011–2015 period. In the Coast District 54% of the 2011–2015 period harvest acres were uneven-aged.

- ① 🔑 Widespread tree mortality from drought, pests, and wildfire have contributed to a major increase in acres proposed for harvesting under Emergency Conditions (1052.1(b)). In 2015, acres harvested under Emergency Conditions comprised 10%, 23%, and 89% of total acres filed for harvest in the Coast, Northern, and Southern Forest Districts, respectively.

Forest Products Sector

- 🔑 California imports about 80% of the lumber and 90% of all wood products used in the state. Meeting in-state demand requires importing lumber and wood products from other states and countries that may have less restrictive rules governing forest practices.

① 1.4 Indicator: Timber Harvest from Private and Public Lands

- ① 🔑 The total volume harvested in 2015 was 72% lower than in 1955.
- ① 🔑 The contribution from public lands to total harvest volume over the 1955–1990 period averaged about 38%. Since 2000, it has averaged only about 18%.
- ① 🔑 Although this figure does not take into account total mill wood-processing capacity, the number of sawmills declined from 675 in 1956 to only 30 in 2012.
- ① 🔑 Since 2000, private harvest levels (the bulk of which are forest industry) have averaged 1,363 million board feet. Annual fluctuations have been as much as 35% over this average and 46% under.

Management Initiatives

- 🔑 Several collaborative projects involving local communities, forest managers and environmental groups exist in forested regions. The most comprehensive of these is the Sierra Nevada Conservancy “Sierra Nevada Forest and Community Initiative.” These projects are indicative of a general interest in promoting forest health and the sustainability of rural communities.

① 1.5 Indicator: Timberland Managed Under Forest Certification, or Other Sustainable Forestry Standards

- ① 🔑 Forest industry landowners with over 50,000 acres of timberland have about 3.9 million acres in mandatory “Option (a)” or Sustained Yield Plans. This represents 91% of industrial timberland and 53% of all private timberland.
- ① 🔑 About 86% of forest industry timberland and 14% of nonindustrial timberland is certified by a third-party sustainable forestry program.
- ① 🔑 A significant and increasing acreage of timberland is under modified management (Nonindustrial Timber Management Plan or Carbon Offset Project), which gives additional emphasis to ecosystem services.

①1.6 Indicator: Acres of Forestland Being Managed as Carbon Offset Projects

- ①🔑 As part of California's "cap and trade" program, as of October 2017 the Air Resources Board (ARB) had issued carbon offsets for Compliance forest projects in California on about 207 thousand acres. There are also 64 thousand acres of Early Action projects in the state. In addition to ARB projects, there are 85 thousand more acres of projects in California registered with the Climate Action Reserve (CAR).
- ①🔑 All California projects are "improved forest management", which can take different forms. The current ARB projects typically involve combinations of reduced harvest levels, longer rotations, and uneven-aged management. Some CAR projects involve intensive even-aged management with actions to maintain stocking for optimal growth, and extended rotations for culmination of carbon growth.

Landowner Assistance

- 🔑 Federal and state programs continue to be important sources of technical and financial assistance to small landowners for their forest management planning and activities.

STRATEGIES

Forestry agencies and partners can provide landowner assistance and incentives to help keep working forests working. Providing forestry assistance to landowners can improve the economics of, and encourage sustainable forest management. In urban and suburban areas, forest agencies can assist communities to develop sustainable forest management and green infrastructure programs. Assessments and strategies can identify viable and high potential working forest landscape where landowner assistance programs, such as Forest Stewardship can be targeted to yield the most benefit in terms of economic opportunities and ecosystem services. Assessment and strategies can also identify opportunities for multi-landowner, landscape scale planning and landowner aggregation for access to emerging ecosystem service markets (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS

National Goal Supported: Conserve Working Forest and Range Landscapes

Montreal Protocol Supported:

MPC-2: Maintenance of productive capacity of forest ecosystems/Productive Capacity

MPC-5: Maintenance of forest contribution to global carbon cycles/Forests and Climate

MPC-6: Maintenance and enhancement of long-term multiple socio-economic benefits to meet the needs of societies/Socio-economic Well Being

MPC-7: Legal, institutional, and economic framework for forest conservation and sustainable management/Governance

State Assessment Theme: Sustainable Working Forests

STRATEGIES OVERVIEW

State Issues

A strong policy of providing incentives for sustainable forest management complements the stringent regulatory environment found in California. The past decade has seen substantial variation in funds available for cost-share and grant programs for achieving forest restoration or vegetation treatments. Recent years have seen a very large increase in State funds available, largely through California Climate Investments (CCI) monies, with smaller amounts available through the Timber Regulation and Forest Restoration Fund. With the substantial drop in economic activity occurring in 2020 due to the corona virus pandemic, future funding from these sources is somewhat uncertain. Development of a strategy to stabilize funding levels, preferably at a high level, could be very helpful ensure the consistency and sustainability of these programs.

Cross-Cutting Issues

Priority landscapes identified forestlands where there is a high risk of catastrophic losses and restoration is needed. There are several cross-cutting issues that apply to these lands.

- Climate Change – Strategies to address underperforming stands and carbon sequestration are addressed in the climate change section. While the focus there is on improving carbon stocks and sequestration, timber and wildlife habitat may also be improved by the same practices.
- Plant, Wildlife and Fish Habitat Protection – The plant, wildlife and fish habitat strategy is highly correlated with maintaining working landscapes.
- Wildfire and Forest Pests Prevention and Restoration – Improving the resilience of forestlands to high-impact disturbance from fires and pests will have direct benefits to landowners by avoiding investment losses and will provide higher levels of ecosystem services.
- Emerging markets – New markets and products can provide potential revenue streams to support maintenance of working landscapes.

Existing Supportive Plans and Programs

- Federal Laws, Regulations and Plans: Guide management on public forestlands; also protect water quality and special status wildlife and fish species on private lands.
- Williamson Act and Timberland Production Zone (TPZ) Tax Programs: Promote long-term stewardship and reduce costs for landowners.
- Existing programs identified in the pest and other strategy sections including federal and state landowner assistance programs, university extension, and local or regional organizations.

Key Stakeholders and Partners

Forest landowners and industry, consumers, environmental and other non-governmental organizations, Registered Professional Foresters, Licensed Timber Operators, associated professions, local and tribal governments, U.S. Forest Service, Bureau of Land Management (BLM).

Strategies and Supporting Actions

Strategy:1.1. Maintain and improve the capacity of the wood products industry statewide.

Action A – Support research to identify and quantify current and long-term key drivers, barriers and opportunities, for the forest products industry in California including both supply and demand sides.

Action B – Rigorously evaluate the full costs and benefits of new legislation and regulations to avoid unreasonable additional costs to forest landowners and wood producers. In particular, the environmental and economic effects of shifting supply outside California should be quantified.

Action C – Act on the most promising results from the research in Action A. Potential examples include demand-side actions such as 1) green building standards recognition of California regulations; 2) improvements in the development and marketing of unique product lines such as high-quality redwood lumber, 3) retail-level recognition of product sources, and 4) potential for demand for mass timber products such as cross-laminated timber. Supply-side examples might include: 1) cost reduction measures related to regulation; 2) landowner cooperatives for reliably supplying logs/lumber; 3) development of ecosystem services markets to increase revenues; or 4) portable mill rental and training for areas without sawmill access.

Action D – Implement strategies from Board of Forestry and Fire Protection Policy Statement that address productive capacity (Criterion 2, Productive Capacity).

D-1. Support management that protects and enhances the multiple values of California's urban and community forests and forests in the wildland/urban interface.

D-2. Encourage county governments to support timber production through Timber Production Zoning (TPZ).

D-3. Focus part of local general plans and related project design on integration and protection of productive timberland.

D-4. Increase use of easements and land banks.

D-5. Encourage forest landowners to manage their forests in a sustainable manner that ensures long-term wood volume growth in California equals or exceeds rates of timber harvest and mortality across all ownerships, while also mitigating fuel loading and wildfire hazards.

D-6. Support continued assessments and research on the capability of California's forests to produce timber, non-wood forest products, recreation, water, fish and wildlife habitat, carbon sequestration, and other values.

D-7. Expand research and support to select appropriate genetic sources for adapting new plantings for climate change.

Action E - Implement strategies A – D, K - AA from Board of Forestry and Fire Protection Policy Statement that addresses policy options for rising consumption and statewide limitations on California commodity output, incentives for private production of ecosystem services, maintaining large landholdings in resource industries and weak economies in local communities (Criterion 6, Socio-Economic Well Being).

- E-1. Develop an economic strategy that builds on comparative advantages of California forest industries versus international producers.
- E-2. Promote more aggressive tax policies to favor development of innovative forest technologies to meet production and conservation goals.
- E-3. Foster development of markets for new products and services, certification of wood and livestock products, and market mechanisms for carbon sequestration.
- E-4. Broaden remuneration methods to landowners for non-commodity products that complement commodity production.
- E-5. By policy, recognize the overall role of private landowners in producing ecosystem services.
- E-6. Focus on long-term plans and conservation easement conditions that clarify land tenure questions and are approved as alternatives under Forest Practice Rules that reduce compliance costs to landowners.
- E-7. Examine use of systems of forest management that depend on certified, insured, and guaranteed operations rather than a permit with civil and/or criminal enforcement.
- E-8. Develop watershed approaches to permits and restoration activities that reward landowners for attaining socially desired future conditions.
- E-9. Refine trading and credit system for habitat provision, pollution reduction, and carbon sequestration.
- E-10. Recognize the continued importance of large-scale unfragmented ownerships in the working landscape that are dependent on resource based activities.
- E-11. Develop analysis of profitability limits at the industry levels and examine if state policies can be improved to assure both private and public benefits of large unfragmented holdings.
- E-12. Maintain tax policies that encourage retention of land ownerships in parcels sizes and configurations that are economic to manage.
- E-13. Identify where new regulatory approaches are possible such as the use of environmental certification or long-range plans.
- E-14. Track the levels of management that will be permitted on federal lands and how they relate to overall resource supplies and protection strategies.
- E-15. Strengthen monitoring and adaptive management approaches for individual parcels as well as larger landscapes.
- E-16. Develop strategies to limit litigation costs by focusing on topics of common agreement such as exotics, pests, fuel reduction, and restoration activities.
- E-17. At the state level, promote diversification and strengthening of resource based communities and local economies.
- E-18. Foster community capacity to procure grants to support local forest products, recreation, and ecosystem service industries.
- E-19. Continue to leverage existing local watershed groups and Fire Safe Councils.
- E-20. At the state level, develop additional support for the biomass energy industry.

E-21. Identify, make available, and guarantee biomass fuel supplies from some sections of public lands.

E-22. Create additional distributed infrastructure to diversify log and biomass markets, such as community based sawmills, portable mills, and biomass plants

Action F - Implement strategies from Board of Forestry and Fire Protection Policy Statement that address policy options for levels of regulatory oversight and policy integration as well as conflicts over forestland management practices (Criterion 7, Governance).

F-1. Conduct an analysis of the impact of overlapping mandates and review processes to determine how to create an efficient structure.

F-2. Connect policies for investment in energy and carbon sequestration to landowner incentives.

F-3. Strengthen ability to use long-term plans and forest certification to meet environmental protection objectives.

F-4. Examine use of system of environmental management that depends on certified, insured and guaranteed operations rather than a permit with civil and/or criminal enforcement.

F-5. Focus on achieving agreement on desired landscape goals and then address potential practices and conflicts.

F-6. Support collaborative landscape-level projects that involve private landowners, state and federal agencies, local communities, and other stakeholders for improving economic and environmental sustainability.

F-7 Promote more active sustainable management of National Forests and BLM forestlands to improve forest health, reduce wildfire threat, and support local economies, including through the granting of state Forest Health funds to federal land managers.

F-8. Evaluate performance-based rules structures to replace existing prescriptive standards as a means to encourage innovative approaches to resource management.

F-9. Review role of environmental certification in providing for broader acceptance of management tools.

F-10. Provide for public input into decision making and monitoring.

F-11. Strengthen skills of resource professionals regarding public involvement and values.

F-12. Continue strong support for focused management practices, such as fuel reduction and control of exotics and pests.

Strategy:1.2. Increase the capacity to provide incentives to non-industrial forest landowners to increase active sustainable management.

Action A – Research the costs and benefits to California landowners of a carbon tax on building materials including wood products. This analysis would have to be in the context of any cap and trade program, but should incorporate externalities not captured by other

measures. An accurate system would likely benefit California grown products relative to other types of building materials and imported wood products.

Action B – Consider an environmental ballot initiative or referendum that funds landowner investments in long-term stewardship.

Action C – Develop carbon protocols for avoided wildfire emissions and biomass utilization that will make fuel reduction activities for restoring forest health eligible for offsets or other carbon-related funding.

Action D – Work with other resource protection agencies to incorporate working landscapes into their habitat protection grant programs, e.g., DFW's Forest Conservation Program (Prop 84).

Chapter 2: Sustainable Rangelands

California rangelands encompass over 57 million acres of annual and perennial grasslands, oak savannas, shrublands, hardwood and conifer woodlands, and deserts (FRAP 2015). They provide a variety of important ecosystem services, such as wildlife habitat, water quality, open space, recreation, and carbon storage and sequestration (California Rangeland Conservation Coalition, California Department of Fish and Wildlife 2015). Rangelands also produce forage for grazing by domestic livestock, which can be an important component of rural economies.

INDICATORS

- ①2.1 Rangeland Conversion
- ①2.2 Beef Cattle Farm Size
- ①2.3 Federal Grazing Allotments

Well over half of the state's rangeland is utilized for the grazing of commercial livestock (Fire and Resource Assessment Program 2003). In 2015, cattle and calves ranked as the fourth highest value agricultural commodity in the state (\$3.4 billion) (California Department of Food and Agriculture 2014). Sheep and lamb inventories in California were the highest in the nation in 2013 and lamb production was second only to Texas (National Agricultural Statistics Service 2014).

Rangeland forage quantity and quality depend heavily on the amount of annual precipitation. Exotic species such as cheatgrass, medusahead, and star thistle continue to persist and spread, lowering forage quality for livestock over large areas of grasslands and oak woodlands (Eviner 2010). Climate change is projected to impact the distribution and quality of important rangeland vegetation types, with associated changes in productivity.

The recent 2011–2016 drought greatly reduced available forage in most regions, due to direct impacts on forage production as well as access to suitable water sources. Ranchers were forced to adapt their operations to minimize losses and protect rangeland resources through increased use of supplemental feed, reduced herd sizes, and early weaning of calves (Barry 2017, Larson 2017).

Rangelands also serve as the primary source for conversions to irrigated agriculture and residential/commercial development. Since European colonization in the late 1700s, nearly 16 million acres of California have been converted to development and irrigated agriculture – almost all of it from rangeland (Fire and Resource Assessment Program 2003). More recently, from 1984–2010, rates of permanent rangeland conversion (to development) have averaged about 14,000 acres per year (①2.1).

The USDA Natural Resources Conservation Service (NRCS), California Department of conservation, as well as numerous public agencies and private conservation organizations, have programs to protect working rangeland landscapes from conversion, through either acquisitions or conservation easements (①6.4). Williamson Act contracts administered by counties continue to be important to ranchers for maintaining viable livestock operations (①6.3). However, the cessation since 2009 of subvention Williamson Act payments (subsidizing reduced property tax assessments) from the state to counties has compromised the future

status of this important program. If state government subvention payments to counties are not soon resumed, the program may experience significantly declining participation by counties(?), with severe tax impacts on ranchers and farmers. Survey results suggest that the net result could be the sale of 20% of ranch acreage, much of which would be converted to other uses (Wetzel, Lacher et al. 2012).

About half of beef cattle farms are less than 50 acres; however, larger ranches produce most commercial range-fed cattle in the state (①2.2). Surveys reveal that many small operators are in the business for the lifestyle rather than for profitability (Roche, Schohr et al. 2015). Large ranching operations near public land holdings often depend on grazing leases, for example U.S. Forest Service (USFS) or Bureau of Land Management (BLM) grazing allotments, to meet annual feed requirements for their herds. However, over the last 60 years, changes in authorized Animal Unit Months (AUMs) on both USFS and BLM lands show reductions in annual use for livestock grazing, on the order of 80% and 50% respectively (①2.3). Lands owned by state and local government can also be important sources of forage for private ranches. In some cases, ranchers are paid for “targeted grazing” to manage vegetation and reduce fuel loads in parks and open space.

Trends in farm sizes, federal grazing allotments, rangeland conversion, and productivity impacts from invasive species and climate change all combine to suggest a downward future trend in statewide livestock production from rangelands. Maintaining sustainable livestock operations, especially in rural regions where it represents a significant portion of the landscape, will in part depend on continued trends that allow ranchers to diversify income from various sources. Due to uncertain and often low economic returns, many ranchers supplement their income by taking outside employment, producing for high-value niche markets, providing recreation opportunities, or earning income from sale of conservation easements (Roche, Schohr et al. 2015). Various state and federal programs and services are available that provide education, technical assistance, cost-sharing and grants, and other services.

Ranchers have numerous motivations to apply land stewardship practices that protect and enhance environmental values. Ranches are typically also the owners’ place of residence, and are often passed on to younger generations. Managing for environmental quality can translate into economic returns from tourism, recreation, hunting, or sale of easements and avoidance of regulations such as those associated with the federal Endangered Species and Clean Water Acts. Sustaining the level of ecosystem services provided by rangelands, including working ranches, is a major issue. Increased use of conservation easements (①6.4), particularly in areas most at risk of conversion, is a positive sign that the importance of these lands is being recognized. Water quality is a continuing concern (①9.1), which is being addressed primarily by education and incentive-based initiatives. There are a few specific cases where water quality impacts from grazing have led to corrective actions under the federal Clean Water Act Section 303d. Survey results show that over two-thirds of ranchers are receptive to the possibility of financial incentives for improving environmental quality (Huntsinger and Batrolome 2014), and continued funding for these efforts is important to sustain and enhance ecosystem services.

To address sustainability, in 2010 the Sustainable Rangelands Roundtable (SRR) created a set of five sustainable rangeland management criteria with 64 indicators, modeled after the Montreal Process Criteria for forest management. However, the data to support tracking the

SRR indicators are often lacking, despite the existence of several federal programs for monitoring rangeland conditions. This lack severely limits our ability to answer important questions about trends in range productivity, soil erosion, water quality, range practices and investments, habitat quality, and oak removals and regeneration.

This chapter provides a diverse set of opportunities to improve the sustainability of rangeland production and ecosystem services. These generally involve initiatives to protect rangelands from development pressures, provide incentives for management that maintains and enhances ecosystem services, and improve the profitability of ranching operations for the benefit of local economies.

KEY FINDINGS

The Resource Base

- 🔑 California rangelands encompass over 57 million acres of annual and perennial grasslands, oak savannas, shrublands, hardwood and conifer woodlands, and deserts.
- 🔑 Exotic species such as cheatgrass, medusahead, and star thistle are now spreading and lowering the forage quality for livestock over large areas of grasslands and oak woodlands.
- 🔑 Research indicates that climate change will alter the extent of rangeland vegetation types, as well as the productivity of individual types in different areas of the state.

①2.1 Indicator: Rangeland Conversion to Other Uses

- ① 🔑 Nearly 16 million acres, almost all of it formerly rangeland, has been converted to other uses since the arrival of Europeans in California. From 1850, over the course of 167 years, this represents an average historical conversion rate of about 95,000 acres per year.
- ① 🔑 From 1992–2012, conversions of rangelands to urban occurred primarily in the Southern California, San Francisco Bay Area, and Central Valley Regions.
- ① 🔑 Over that period, rangeland conversions to urban peaked in 2004–2006 at about 37,000 acres per year, dropping by 67% (12,000 acres/year) with the onset of the recession in 2008-9.
- ① 🔑 From 1984–2010, the Farmland Mapping and Monitoring Program (FMMP) recorded the net loss of rangelands to all other land uses averaged about 14,000 acres per year statewide.

Management Context

- 🔑 About 22 million acres of rangeland are privately owned, and about 35 million acres are publicly owned.
- 🔑 About 17 million acres of private rangelands are grazed by livestock. Federal BLM and Forest Service rangelands contain about 7 million and 7.6 million acres of grazing allotments, respectively.

- 🔑 The water quality issues associated with range livestock operations include erosion and sedimentation, increases in stream temperature caused by streamflow diversions and losses of riparian vegetation, and nutrient and fecal coliform inputs from livestock wastes.
- 🔑 The most recent measures of stream quality on California rangelands show about 40% of streams in good condition, but about 50% in poor or very poor condition (①9.1).

①2.2 Indicator: Rangeland Livestock Ownership

- ①🔑 The number of beef cattle farms over 2,000 acres declined from a peak of 1,521 in 1987 to 1,020 in 2012, a 33% decline. This could be due to land conversion to development, splitting into smaller farms, aggregation into larger farms, or acquisition by government agencies or conservation groups.
- ①🔑 The number of beef cattle farms less than 50 acres increased from its lowest number of 4,542 in 1997 to 5,893 in 2012, a 30% increase. This is still 20% less than the 7,343 small farms that existed in 1982.

①2.3 Indicator: Federal Grazing Allotments

- ①🔑 Long-term (50–60+ year) changes in authorized Animal Unit Months (AUMs) on both USFS and BLM lands show reductions in their annual use for livestock grazing, on order of 80% and 50% respectively.
- ①🔑 Since 2001 grazing use trends for these lands have been relatively flat.
- ①🔑 Long-term reductions in authorized AUMs have been due to changes in federal policy (e.g. the Rescissions Act of 1995, the National Forest Management Act, and additions to the National Park System), the spread of brush and timber vegetation (lack of historical fire regime), and to changes in management, e.g. to accommodate new listings of endangered species and for the protection of natural resources.

The Range Livestock Industry

- 🔑 The volatility of beef market prices, typically low returns on investments, long distances to meat processing facilities, and year-to-year uncertainties of annual forage production (e.g., due to recent widespread drought conditions), continue to make the ranching business difficult for many livestock operators.
- 🔑 Access to meat processing plants can be a constraint to livestock marketing.
- 🔑 The economics of range livestock operations, particularly on smaller properties with smaller herds suggest that most operators are in the ranching business for reasons other than profit.

Management Initiatives

- 🔑 The marginal profitability of many range livestock operations has caused ranchers and their families to seek additional sources of income to support their lifestyle. There are a variety of options available to ranchers to diversify and enhance income sources while improving sustainable management practices.
- 🔑 Targeted grazing is used as a tool to manage vegetation and reduce fuel loads in parks and open space. In some cases, fees are paid to herd managers.

- 🔑 At least 800,000 acres of California rangeland are either held in fee title or under conservation easements by land trusts and other non-governmental agencies.

Landowner Assistance

- 🔑 Resource Conservation Districts (RCDs) assess local conservation needs and develop priorities and programs to meet those needs. RCDs provide various services including education and outreach, demonstration projects, and promote best management practices. RCDs can also serve as conduits for funding to support various types of projects, for example for erosion control, fuels reduction, water conservation or quality improvements, or watershed plans.

STRATEGIES

[There is no guidance from the federal USFS Redesign Strategies documents on concerns specific to working rangelands]

GOALS: The goals of this section are to promote the long-term economic and ecological sustainability of range lands. This is done by addressing the factors that most influence the ability to keep these landscapes “working”, which is recognized to counter conversion and short-term exploitative practices.

National Goal Supported: Conserve Working Forest and Range Landscapes

Montreal Protocol Supported:

MPC-2: Maintenance of productive capacity of forest ecosystems/Productive Capacity

MPC-5: Maintenance of forest contribution to global carbon cycles/Forests and Climate

MPC-6: Maintenance and enhancement of long-term multiple socio-economic benefits to meet the needs of societies/Socio-economic Well-Being

MPC-7: Legal, institutional, and economic framework for forest conservation and sustainable management/Governance

State Assessment Theme: Sustainable Working Rangelands

Strategies Overview

State Issues

- Continued degradation of grazed rangeland forage quality due to incursion of unwanted exotic plant species, in some areas due to climate changes, atmospheric deposition of nitrogen and drought.
- Livestock infrastructure decay (e.g., watering ponds and troughs) and new permitting standards that can be cost prohibitive to landowners.
- Increasing transportation distances to marketing free range livestock and general lack of such facilities.
- Increased costs of raising livestock relative to the amount received at market
- Scheduling livestock herd grazing and rotations in regions with more fragmented rangeland ownership patterns (e.g., the south and east Bay Area).

Cross-Cutting Issues

- Climate Change – Strategies to address water conservation (e.g., for provision to range livestock) in the face of a changing climate are addressed in the climate change section.
- Plant, Wildlife and Fish Habitat Protection – The plant, wildlife and fish habitat strategy depends on preserving working landscapes.
- Wildfire – Improving the resilience of range lands to wildfire will have direct effects to landowners in avoiding investment losses.
- Emerging markets – Provide potential revenue streams to support working landscapes and reduce the costs of sustainable management.

Existing Supporting Plans and Programs

- Federal Laws, Regulations and Plans: Guide management on public range lands; also protection of water quality and special status wildlife and fish species management on private lands.
- Williamson Act Tax Program: Promotes long-term stewardship and reduces costs for range landowners.
- Existing programs identified in other strategy sections including federal and state landowner assistance programs, university extension, and local or regional organizations.
- Board of Forestry and Fire Protection's Range Management Advisory Committee.

Key Stakeholders and Partners

Range landowners and livestock industry, consumers, environmental and other non-governmental organizations, professional range managers, local and tribal governments, U.S. Forest Service, and the Bureau of Land Management

Strategies and Supporting Actions

Strategy: 2.1. Maintain and improve the capacity of range industries statewide.

Action A – Conduct research to identify and quantify current and long-term key drivers, barriers, and opportunities for the range industry in California, including both supply and demand sides.

Action B – Rigorously evaluate the full costs and benefits of new legislation and regulations to avoid unreasonable additional costs to landowners and producers. In particular, the environmental and economic effects of shifting to supply sources outside of California should be quantified.

Action C – Act on the most promising results from the research in Action A. Potential examples include improvements in the development and marketing of unique product lines such as grass-fed locally grown beef and lamb and retail-level recognition of product sources. Supply-side examples might include cost reduction measures related to regulation, or landowner cooperatives for reliably supplying livestock.

Action D – Reinstate State subventions to Counties to fund existing programs such as the Williamson Act to conserve working rangelands.

Action E – Implement strategies from Board of Forestry and Fire Protection Policy Statement that addresses productive capacity (Criterion 2, Productive Capacity).

E-1. Support livestock and other range-based enterprises through preserving high quality rangeland through the Williamson Act or other local zoning.

E-2. Focus part of local general plans and related project design on integration and protection of productive forage areas.

E-3. Increase use of conservation easements and land banks.

E-4. Improve range management techniques to enhance forage productivity.

E-5. Support continued assessments and research on the capability of California's rangelands to produce recreation, water, fish and wildlife habitat, carbon sequestration, and other natural values.

Action F - Implement strategies from Board of Forestry and Fire Protection Policy Statement that address policy options for rising consumption and statewide limitations on California commodity output, incentives for private production of ecosystem services, maintaining large landholdings in resource industries, and strengthening weak economies in local communities (Criterion 6, Socio-Economic Well-Being).

F-1. Develop an economic strategy that builds on comparative advantages of California rangeland-based industries versus international producers.

F-2. Promote more aggressive tax policies to favor development of innovative rangeland technologies to meet production and conservation goals.

F-3. Foster development of markets for new rangeland products and ecosystem services.

F-4. Broaden remuneration methods to landowners for non-commodity products, such as wildlife habitat, that complement commodity production.

F-5. By policy, recognize the overall role of private landowners in producing ecosystem services.

F-6. Focus on long-term plans and conservation easement terms that clarify land tenure questions.

F-7. Develop watershed approaches to permits and restoration activities that reward landowners for attaining socially desired future conditions.

F-8. Refine trading and credit system for habitat provision, pollution reduction, and carbon sequestration.

F-9. Recognize the continued importance of maintaining, in the working landscape, large and unfragmented ownerships that are dependent on resource based activities.

F-10. Develop analysis of profitability limits at the industry level and examine if state policies can be improved to assure both private and public benefits of large unfragmented holdings.

F-11. Maintain tax policies that encourage retention of land ownerships in parcels that offer management economies of scale.

F-12. Identify where new regulatory approaches are possible such as the use of environmental certification or water quality management plans.

F-13. Track the levels of management that will be permitted on federal lands and how they relate to overall resource supplies and protection strategies.

F-14. Strengthen monitoring and adaptive management approaches for individual parcels as well as larger landscapes.

F-15. Develop strategies to limit litigation costs by focusing on topics of common agreement such as exotics, pests, fuel reduction, and restoration activities.

F-16. At the state level, promote diversification and strengthening of rangeland-based communities and local economies.

F-17. Foster community capacity to obtain grants that support local range products, recreation, and ecosystem service industries.

F-18. Continue to leverage existing local watershed groups and rangeland coalitions.

Action G - Implement strategies from Board of Forestry and Fire Protection Policy Statement that address policy options for levels of regulatory oversight and policy integration as well as conflicts over rangeland management practices (Criterion 7, Governance).

G-1. Strengthen ability to use long term-management plans to meet environmental protection objectives.

G-2. Examine use of a system of environmental management that depends on certified, insured, and guaranteed operations.

G-3. Focus on achieving agreement on desired landscape goals and then address potential practices and conflicts.

G-4. Learn from experiences of the University of California Division of Agriculture and Natural Resources, the Natural Resources Conservation Service, The Nature Conservancy, other nongovernmental organizations, and regional parks on how better to understand rangeland management needs.

G-5. Review role of environmental certification in providing for broader acceptance of management tools.

G-6. Strengthen skills of resource professionals regarding public involvement and values.

Strategy: 2.2. Increase the capacity to provide incentives to range landowners.

Action A – Consider an environmental bond ballot initiative or referendum that funds landowner investments in long-term stewardship.

Action B – Use proceeds from federal, regional or state cap-and-trade auction sales to invest in carbon storage improvements.

Action C – Work with other resource protection agencies to incorporate working landscapes into their habitat protection grant programs.

Action D – Explore recommendations from the Range Management Advisory Committee (RMAC) Strategic Plan (2020) and Implementation of Objectives to inventory state-owned range lands and develop a coordinated strategic plan that utilizes public-private

partnerships to reduce maintenance costs to the state while providing income opportunities to local ranchers

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Chapter 3: Urban Forestry

Creating and maintaining healthy urban forest ecosystems is becoming more critical due to warming temperatures (①3.4), urban development creating more impervious surfaces (①3.2), the need to mitigate ongoing problems with air pollution (①3.3), and for the variety of other benefits provided.

INDICATORS

- ① 3.1 Tree Canopy Cover
- ① 3.2 Impervious Surfaces
- ① 3.3 Air Pollution
- ① 3.4 Urban Heat

The urban area in California encompasses 5% of the land base and is home to almost 95% of the population. Statewide, impervious surfaces in census-defined Urban Areas (UAs) increased 20% between 2000 and 2010, to about 1.9 million acres (①3.2). Roads, parking lots, buildings and other impervious surfaces in urban areas absorb solar radiation, making the environment warmer and creating Urban Heat Islands (UHI). Climate change is producing more frequent heat waves (CDC 2012) and warmer temperatures (①3.2) that are likely to expand and intensify UHI. This trend raises concerns for higher energy consumption, elevated emissions of air pollutants and greenhouse gases, compromised human health and comfort, and impaired water quality.

One of the more effective strategies to mitigate the negative impacts of UHI is to increase canopy cover by planting and maintaining urban trees. The benefits of urban forests in California include: CO₂ storage and sequestration (110 MMT/year); air pollution removal (23,700 MT/year); rainfall interception; improved water quality runoff; reduced energy use; and jobs and economic value to the State economy.

However, tree canopy in urban areas is unevenly distributed in the state. For California's 211 UAs, the average statewide Urban Tree Canopy (UTC) cover is 15% (①3.1). Only 44 UAs exceed the American Forests goal of 25% UTC.

An analysis was conducted (Schwarz, Fragkias et al. 2015), to identify areas with the highest degree of heat island effects and air pollution to locate priority landscapes for tree planting and maintenance. The uneven distribution of UTC across the state and within individual cities leads to an inequitable distribution of environmental benefits. Additional considerations for future analyses would include disadvantaged communities where the environmental and economic benefits from urban trees are often most needed.

KEY FINDINGS

Key findings and indicators are grouped below for the five topics covered in the 2017 Assessment online technical reports, which include a more detailed discussion of each topic area.

Urban Forest Benefits

-  The average annual cost associated with planting and maintaining an urban street tree is \$19.00/tree and the benefit is \$110.63/tree, so the return on investment is \$5.82 for every \$1 spent.

- 🔑 The estimated CO₂ stored in urban forests in California totaled 103 million metric tons in 2015.
- 🔑 Annually, the amount of CO₂ sequestered from urban forests is assessed at 7.2 million metric tons/year. The amount of CO₂ emissions avoided was estimated to be 1.3 million metric tons/year. Assuming a price of \$12.02/metric ton, these annual amounts are equal to about \$87 million/year for annual CO₂ sequestered, and \$16 million/year for avoided emissions.
- 🔑 Total pollution removal (biogenic volatile organic compounds and volatile organic compounds) is estimated at almost 24,000 metric tons with an annual value of \$1.1 billion.

Tree Canopy Cover

- 🔑 California urban tree canopy is estimated to occupy 791,680 acres, with an estimated 159.3 million trees and approximately 1,000 square feet of tree canopy per person.
- 🔑 There are approximately 9.1 million urban street trees, about 1 for every 4 residents.
- 🔑 Urban tree canopy is not evenly distributed, and 61.4% of urban areas have less than 10% tree canopy cover.

①3.1 Indicator: Tree Canopy Cover

- ① 🔑 Average Urban Tree Canopy (UTC) varies from 3% in Imperial County to 66% in Tuolumne County. Of California's 58 Counties, 55 have urban areas, and only 12 of these exceed the 25% UTC American Forests goal.
- ① 🔑 For all of California's 211 census-defined UAs, the average statewide UTC cover is 15%. Only 44 of these areas exceed the American Forests goal of 25% UTC.

Impervious Surfaces

- 🔑 The percentage of the urban area in each county that is covered by impervious surfaces varies significantly, from 61.1% in San Francisco to 1.8% in Sierra County.

①3.2 Indicator: Impervious Surfaces

- ① 🔑 Census-defined UAs in California have a high concentration of impervious surfaces – about 15% of UAs had at least 70% impervious surface area.
- ① 🔑 Statewide, the total area of impervious surfaces has increased almost 20% in all UAs from 2000-2010 to about 1.9 million acres, while the urban population has increased by just 10.6%.
- ① 🔑 Statewide, about 36% of the total land area within all UAs is impervious. This average varies greatly among climatic regions, from 19% in the interior areas, to 46% in the densely populated Southern California Coast area.

Air Pollution

①3.3 Indicator: Air Pollution

- ① 🔑 The San Joaquin Valley and South Coast air basins still exceed 8-hour ozone standards 80 to 90 days a year. Recent years show trends toward improvement in these regions and in the Sacramento Valley.
- ① 🔑 Recent trends in number of PM_{2.5} (particulate matter less than 2.5 microns in diameter) days over the standard vary by air basin region. The 24-hour PM_{2.5} level

standards in the San Joaquin Valley have lately exceeded over 40 days a year; in all other California air basins, it is generally less than 10 days annually.

Urban Heat

①3.4 Indicator: Days over 90 Degrees Fahrenheit (F)

- ①🔑 40% of census-defined Urban Areas (UAs) ranked high for urban heat, with more than 74 days in a year with a maximum temperature of more than 90°F.
- ①🔑 Southern California climate zones had an annual average of seven more days over 90°F in the last decade than in the 1980s.

Strategies

Urban and exurban forest cover, including agroforests can improve air quality, reduce energy consumption, and produce biomass for energy production. Assessments should identify areas where management, or restoration of the urban or exurban forest canopy will have significantly positive and measurable impact on air quality and produce substantial energy savings (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goal of this strategy is to improve air quality and reduce energy consumption through expansion, management and restoration of urban forests. This strategy also addresses goals identified at the national and state level, as noted below.

National Goal Supported: Enhancing Public Benefits from Forests

Montreal Protocol Supported:

MPC-3: Maintenance of Forest Ecosystem Health

MPC-6: Socio-economic Well-Being

State Assessment Theme: Urban Forestry for Energy Conservation and Air Quality

Strategies Overview

State Issues

- Population growth and warmer summer temperatures have increased the need for electricity in California.
- Development to keep up with the growing population has created urban heat islands that also increase the overall air temperature; 40% of the urban areas have more than 74 days a year with temperatures exceeding 90 degrees.
- Many daily activities, such as driving, lawn mowing, dry cleaning clothing, agriculture, and natural occurrences such as wind-blown dust and wildfires, pollute the air; 80% of Californians are at risk from unhealthy air.
- The American Lung Association 2016 State of Air Report found that California has some of the worst air in the Country, and Los Angeles and Bakersfield metro areas top the country's pollution list for ozone and particulate pollution.
- Communities located in the Central Valley, Inland Empire, and South Desert Region have the greatest concentration of air quality priority areas.

- Maintenance of urban forest areas that have higher risk factors, disadvantaged communities, and densely populated places.
- Restoration of urban forests impacted by wildfire, pest, and drought events.

Cross-Cutting Issues

Mitigating the effects of air pollution and energy consumption in urban areas relates to several other themes and issues presented in the assessment document. The most important are listed below:

- **Wildfire and Pest Threats** – Ignition sources are often a major factor in relation to the frequency of large wildfires. In high wildfire threat urban areas, tree maintenance and selection is imperative. With public education to plant the right tree in the right place, create or maintain defensible space, improve tree maintenance, create more space in the built environment to plant appropriate trees, reduce pest transportation and better control pests, and reduce human activities with high fire or pest spread risk, a sustainable urban forest can be achieved.
- **Water Quantity and Quality** – Managing urban water runoff including the many pollutants attached, is a challenging environmental issue. Green Infrastructure, that uses natural or engineered systems to mimic a natural filtering process, including bio-swales, pervious land surfaces, filtering vegetation, and detention ponds or cisterns all play a crucial role in urban water runoff management. Use of green infrastructures in the urban environment helps protect public safety, and increases water available for recharging of groundwater, irrigation to support vegetation and tree growth, and sustaining wildlife habitat.
- **Development** – Urban forestry is frequently an afterthought in the urban planning and development process. Effective urban forestry depends on supporting public policy that promotes the connection between community goals of environmental quality and ecosystem functions/services. Opportunities to conserve urban ecosystems with “Smart Growth” and integrating green infrastructure into urban planning solutions can help conserve natural ecosystem values, provide air quality improvement, reduce urban heat islands, improve urban water, and provide a variety of benefits to the community and wildlife.

Existing Supporting Plans and Programs

Supporting plans include: Individual urban forest plans are available for many local and county governments, but a general guide for the entire State did not exist until 2013. In 2009, The California Urban Forestry Advisory Committee (CUFAC) was established to advise the Director on the State’s Urban Forestry Program. Through CUFAC, the CAL FIRE Urban and Community Forestry Program [Strategic Plan for 2019-2024](#) was developed to provide direction and focus to program activities. This plan establishes three overarching goals, with objectives and strategies enumerated for each goal:

1. Optimize the benefits all Californians receive from urban & community forests;
2. Strengthen the urban forest industry workforce
3. Improve the health and management of the urban forest.

Existing programs that support urban forestry strategies include:

- The U.S. Forest Service National Urban and Community Forestry Program (U&CF) – Provides financial support and guidance to state urban forestry programs. The national

program provides financial and technical assistance to restore and sustain natural and human environments in urban areas.

- CAL FIRE Urban & Community Forestry Program (U&CF) –Guides urban forest activities in the State to create and maintain sustainable urban forests, per the Urban Forestry Act of 1978 ([PRC 4799.06-4799.12](#)). The goal of the U&CF program is to improve the quality of life in cities and communities by promoting a healthy ecosystem through urban forest management. The program seeks to ensure the vitality of communities by engaging people where they live, work, and play.
- Local Community Urban Forest Programs and Plans – Urban forests are generally developed and managed at a local level, as directed by local entities, and unlike rural forests are dominated by human activity. The state’s role in urban forest sustainability is to build capacity, reduce threats, and promote efficiency by identifying areas where efforts can be optimized and would maximize community benefits.
- Proposition 68 – (California Drought, Water, Parks, Climate, Coastal Protection, and Outdoor Access for All Act of 2018) authorized \$4.0 billion in bonds to be used for to finance a drought, water, parks, climate, coastal protection, and outdoor access for all program. The Urban and Community Forestry Program allocation was for \$15 million over a four-year period, which began in 2019.
- Greenhouse Gas Reduction Fund (GGRF) for California Climate Investments (CCI) – Further the goals of the California Global Warming Solutions Act of 2006 (AB 32), result in a net greenhouse gas (GHG) benefit, and provide environmental services and cost-effective solutions to the needs of urban communities and local agencies. The Urban and Community Forestry Program allocation was for \$5 million over a four-year period, which began in 2019.
- The Safeguarding California Plan: 2018 Update – California’s Climate Adaptation Strategy builds on nearly a decade of adaptation strategies to communicate current and needed actions state government should take to build climate change resiliency.

Key stakeholders and Partners

Urban forest expansion and improvement efforts are often the result of regional and local collaborations. Cooperative working relationships between government, non-profit, and community leaders are essential to program success and sustainability. Each community and citizen has a stake and contributes to the success of the urban forest. Urban forestry expansion is a “grass-roots” effort that requires continuous outreach and education. For this reason, community support, key stakeholders, and partners are an essential part of the program. The list of key stakeholders and partners that support urban forestry is extensive, including but not limited to: local and regional urban forestry nongovernmental organizations; city and county governments; community planners and developers; USDA Forest Service, Urban Ecosystems & Process Team; Los Angeles Center for Urban Natural Resources Sustainability; Urban Forest Ecosystems Institute; California ReLeaf; California Urban Forest Advisory Council; and the Western Chapter of the International Society of Arboriculture.

Strategies and Supporting Actions

Two urban forestry strategies were identified to broadly address the three goals of the CAL FIRE Urban and Community Forestry Strategic Plan for 2019-24. The first strategy is to increase tree planting efforts that will produce public benefit. The second strategy is to maintain existing tree canopy assets.

Strategy: 3.1. Promote urban tree planting to improve air quality and energy conservation.

Action A – Promote urban forestry ordinances and development standards to increase tree planting.

A-1. Promote tree planting as a condition for new developments and renovation projects.

A-2. Promote minimum tree canopy standards in public parking lots to mitigate urban heat islands.

A-3. Encourage native tree plantings, and “right place, right tree” philosophy to increase sustainability.

A-4. Encourage integration of planning, design, management, and enforcement to increase program efficiency and to minimize impacts on ecosystems and natural areas.

Action B – Retain strong cooperative working relationships with key stakeholders.

Action C – Promote the State Plan to increase benefits from urban forests.

C-1. Support the California Urban Forestry Advisory Committee findings that guide program activities.

C-2. Encourage public support and input.

Action D – Enhance cooperation and coordination among agencies, municipalities and non-profit organizations.

Action E – Increase public awareness regarding the benefits of urban forests and impacts of urban heat islands, impervious surfaces, fire hazards and ecological change.

Action F – Encourage new research related to urban forestry including on air pollution and energy conservation.

Action G – Support urban workforce development programs, such as the California Conservation Corps (CCC), to help meet urban forestry goals.

Strategy: 3.2. Maintain urban tree canopy to conserve energy and improve air quality.

Action A – Promote adoption of tree policies that protect valuable tree assets.

Action B – Promote policy and plans to better manage urban forests.

Action C – Promote efforts to increase space available for large trees in urban areas.

Action D – Increase age and species diversity in urban forests.

Action E – Promote appropriate regular maintenance schedules to protect urban tree assets.

Chapter 4: Wildfire

Altered fuel conditions, changing climate conditions, and millions of people and homes in or near wildlands are all contributing to increasing fire activity and impacts. Managing wildfire has become a complex endeavor that requires balancing fire protection with ecological needs and societal demands.

Much of California's wildland vegetation is adapted to periodic burning, but natural fire regimes have been disrupted by over a century of fire exclusion, land management and human-caused ignitions. Over the last century, 80% of California's natural landscape, excluding grasslands, has burned at a frequency that is significantly higher or lower than pre-settlement times (moderate or high departure - ④4.1). These disruptions have contributed to the current distribution of hazardous fuels in forests and to short fire intervals in many shrublands, resulting in over 25 million acres classified as Very High or Extreme Fire Threat (④4.2). Over 2.2 million housing units exist in the Wildland Urban Interface (WUI) (④11.3), placing increased demands on fire management organizations. Climate change has created additional stress on ecosystems (④9.4), altering fuel conditions and extending the fire season.

Over the last two decades, California wildfires burned an average of 708,000 acres annually, with a marked increase in area burned since 2000. Much of this increase has been in conifer forests (④4.3). Eight of the ten largest wildfires in modern state history have occurred since 2000, including several high-profile "mega-fires". Fire exclusion in frequent-fire adapted forests has contributed to uncharacteristic patterns of high severity fire (④4.4) with potentially long-lasting effects. Fuel management activities, including prescribed and managed wildfire and mechanical fuel treatment, averaged about 261,000 acres annually (④4.5) over the last decade, but this represents only a fraction of treatment needs.

Living sustainably in the fire-prone landscapes of California will require broad recognition of the inevitability of fire, which will in turn necessitate enhanced investment in and novel approaches to risk evaluation, fuel management, forest health, land use planning and community adaptation. As we move through the 21st century, fire managers and landowners in California are challenged to effectively utilize available resources and tools to create resilient landscapes, reduce loss of life and property, and stem rising management costs, while enhancing our compatibility with the fire environment in which we live.

INDICATORS

- ④4.1 Fire Return Interval Departure (FRID)
- ④4.2 Fire Threat
- ④4.3 Wildfire Activity
- ④4.4 Fire Severity
- ④4.5 Vegetation Treatments

KEY FINDINGS

④4.1 Indicator: Fire Return Interval Departure

- ④🔑 Areas dominated by conifer, hardwood, and mixed conifer-hardwood vegetation are burning less frequently than they did in the pre-settlement era, with over 75% of these vegetation types by area in high positive departure. This is most evident in the Sierra Nevada, Southern Cascades and Klamath-North Interior mountain ranges.

- ① Many shrub-dominated areas are burning more frequently than their reference regime, with 23% by area in moderate or high negative departure across the state. This problem is particularly acute in the South Coast Region, where over 48% of shrublands are in moderate or high negative departure.

①4.2 Indicator: Fire Threat

- ① Approximately 26 million acres in California (a third of the land area supporting of burnable wildland vegetation) are classified as Very High or Extreme Fire Threat.
- ① Areas of Extreme Fire Threat are concentrated in the South Coast and Mountains ecoregion (26%).
- ① The Klamath-North Interior region has a high proportion of Very High Fire Threat (64%).
- ① The Sierra-Cascades region is classified in roughly equal proportions of High (45%) and Very High (40%) Fire Threat, with areas of Very High threat concentrated in the low- to mid-elevations.
- ① Fire Threat in the North Coast region is mostly Moderate to High, owing to the relative infrequency of fire in recent decades.

①4.3 Indicator: Wildfire Activity – Trends in Burned Area

- ① Approximately 708,000 acres burned annually in California since 2000, up from an average of 343,000 acres for the period 1980–1999, and 186,000 acres annually from 1960–1979.
- ① Annual rates of burning in forest and shrub-dominated vegetation, and average size of large fires (>1000 acres) have increased significantly over the last 17 years.
- ① Fires in shrub-dominated vegetation historically burned the most area annually between 1960 and 2009. Since 2010, more forest vegetation has burned annually than shrub on average.
- ① Eight of the ten largest and most damaging wildfires in terms of both area burned and structures destroyed have occurred since 2000.

①4.4 Indicator: Fire Severity in Yellow Pine/Mixed-conifer Forests

- ① For at least 30 years the average proportion of high severity in yellow pine/mixed-conifer wildfires (23–32%) has been outside the historical range of 4–13%.
- ① High severity patches are becoming larger and less complex in yellow pine/mixed-conifer forests, leading to many negative ecological consequences, including reduced regeneration potential and snowpack retention, and increased erosion potential.

①4.5 Indicator: Fuel Treatment Area

- ① The average annual area treated by federal and state agencies over the last 10 years is about 261,000 acres with a range from ~159,000 to ~381,000 acres.
- ① Federal agencies have treated the most area annually by a large margin (93%).

- ① 🔑 CAL FIRE performed 15,755 acres of mechanical treatment and 13,941 acres of prescribed burn, meeting 105% and 70% of the targets set by the Director for FY 2016–17, respectively.
- ① 🔑 The average number of acres treated every year is small in comparison to fire protection and ecological needs, representing approximately 1.0% of the state’s area in high positive departure.

Strategies

The strategic management of wildfires is crucial to the health of our nation’s forests, the safety of our citizens and the contributions of forests to our economy. Assessments should identify areas where management can significantly reduce the risk of catastrophic wildfire while enhancing multiple associated forest values and services.

Many forest ecosystems are dependent on fire for their health and sustainability. Decades of fire suppression and a changing climate have disrupted natural fire regimes, resulting in fuel buildup, loss of biological diversity, changed species composition, and loss of some fire-dependent species. Assessments should identify areas where these effects of fire exclusion can feasibly be mitigated or countered through sound management, particularly where there are opportunities for federal, state and community partnerships. Resource strategies should identify appropriate treatment strategies for priority landscapes, including the use of fire as a management tool (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goals of this strategy are to prevent damaging wildfires, protect life and property and restore wildfire impacted areas to maintain ecosystem health, ecosystem services and public safety. This strategy also addresses goals identified at the national and state level, as noted below.

National Goal Supported: Protect Forests from Hazards

Montreal Protocol Supported:

MPC-3: Forest Health

State Assessment Theme: Wildfire

Strategies Overview

State Issues

- Fire environment and ownership complexity
- Fire risk to communities: the Wildland Urban Interface and local planning and development
- Conflicts in managing fire regime for disparate impacts and benefits (forest health vs community protection)
- Ecosystem type-conversion from uncharacteristic fire regimes
- Widespread drought-induced tree mortality and impacts on fuel hazards over time
- Wildfire Impacts on long-term forest carbon sequestration

- Wildland fire smoke is a significant human health issue, including consideration of smoke impact tradeoffs between wildfire and prescribed fire
- Understanding treatment impacts on long-term goals: hazard, carbon, and water resources
- Mitigating Utility-related wildfire risk

Cross-cutting Issues

Forest management activities used as strategies to reduce the occurrence and severity of wildfire or reduce impacts to natural resources, life, and property also address other key issues identified by the California statewide assessment.

- Forest Pest Threats – Forest management activities that prevent the introduction and spread of exotic forest pests and invasive plant species by the removal of dead, dying and diseased trees and thinning operations that reduce hazardous fuel loads. Elevated tree mortality and other drought related impacts.
- Climate Change – Activities that reduce the incidence and severity of wildfires yield additional climate benefits by protecting existing carbon stocks and producing more resilient forest stands.
- Water Quality/Supply – Activities that reduce the incidence and severity of wildfires lead to more resilient forests, resulting in better water quality and supply.
- Emerging Markets – Forest management activities that reduce the incidence and severity of fires provide feedstock for emerging biomass facilities and help protect existing biomass resources. Markets are emerging for water quality and quantity from wildland watersheds. Other potential program links (e.g. *CalRecycle*) to deal with organic wastes.

Existing Supporting Plans and Programs

State Plans and Programs:

Fire Prevention (Includes pre-fire hazard mitigation strategies)

- CA Fire Plan: The 2018 Strategic Fire Plan describes California's complex and dynamic natural and man-made environment and identifies a variety of actions to minimize the negative effects of wildland fire.
- Office of State Fire Marshal (OSFM) - The mission of the State Fire Marshal is to protect life and property through the development and application of fire prevention engineering, education and enforcement.
- CA Unit Fire Plans - Fire Plans outline the fire situation within each CAL FIRE Unit. Planning incorporates concepts of the National Cohesive Wildland Fire Management Strategy, the California Fire Plan, individual CAL FIRE Unit Fire Plans, as well as Community Wildfire Protection Plans (CWPPs).
- Wildland Fire Prevention Engineering (OSFM) - Fire Engineering takes into account the best design, construction, and engineering practices for planning fire safe communities and homes. Engineering principles also apply in the safe use of industrial and recreational equipment. The Fire Engineering staff recommends and interprets laws and regulations covering wildland fire safety and assists homeowners, landowners, decision-makers, and local government planners in building and rebuilding fire safety into California communities.
- Wildland Urban Interface Building Code Standards - The broad objective is to establish minimum standards for materials and material assemblies and provide a reasonable level of exterior wildfire exposure protection for buildings in Wildland-Urban Interface

Fire Areas. The use of ignition resistant materials and design to resist the intrusion of flame or burning embers projected by a vegetation fire (wildfire exposure) is essential to mitigating the structure losses resulting from our repeating cycle of interface fire disasters.

- Fire Protection - Fire and Emergency Response - The Department of Forestry and Fire Protection protects the people of California from fires, responds to emergencies, and protects and enhances forest, range, and watershed values while providing social, economic, and environmental benefits to rural and urban citizens.
- Fire Fighter training - It is the goal of the Department that every fire engine responding from a CAL FIRE station carries CAL FIRE firefighters, fire apparatus engineers and/or fire captains who have met, at a minimum, the extensive training requirements.
- Civil Cost Recovery Program - Wildland fires cost California taxpayers millions of dollars every year. If CAL FIRE investigation reveals a fire was caused by a violation of law or negligence, the person responsible can be charged criminally, civilly, or both.
- Cooperative Fire Program - Agreements between state, federal and local agencies are essential in responses to wildland fire emergencies. The CAL FIRE Cooperative Fire Protection Program staff is responsible for coordinating these agreements and contracts for the Department.
- Conservation Camp Program - CAL FIRE is currently authorized to operate 39 Conservation Camps statewide that house more than 4,300 inmates and wards, who staff 196 fire crews year round. These hand crews are available to respond to all types of emergencies including wildfires, floods, search and rescue. Fire crews perform more than 2.5 million hours of emergency response work each year.
- Aviation Program - In support of its ground forces, CAL FIRE has airtankers, helicopters and air-tactical planes. Replacement of its UH-1H rotary aircraft with new Sikorsky Firehawks and augmentation of its tanker fleet with newly acquired C-130 aircraft is ongoing.
- Grant and Landowner Assistance Programs – multiple programs provide direct funding, cost-share assistance, and technical advice for landowners to improve forest health and reduce wildfire hazard. Programs include Forest Health, Fire Prevention, Forest Legacy, Forest Research, and the California Forest Improvement Program, among others.
- Fire and Resource Assessment Program (FRAP) – provides extensive technical and public information for statewide fire threat, fire hazard, watersheds, socio-economic conditions, environmental indicators, and forest-related climate change. Much of this information involves Geographic Information System (GIS) analysis, tables, maps, data, and calculation tools.
- Climate and Energy Program – Provides large landscape-scale grants to collaborative groups and organizations seeking to improve forest health and reduce wildfire risk through fuel reduction, prescribed fire, pest management, reforestation, and biomass utilization.
- Fire Prevention (Resource Management Program) - Pre-fire activities such as clearing defensible space, planting and maintaining fire safe landscaping, utilizing prescribed fire, creating fuel breaks and managing forests effectively, are proven methods of reducing wildfire destruction.
- The Vegetation Management Program (Resource Management Program) - A cost share program that allows public and private landowners to participate in wildland fuel reduction projects. The primary tool used is prescribed fire, although in more recent years CAL FIRE has used the program for mechanical treatments of vegetation as well.

- The Board of Forestry and Fire Protection has recently completed the Vegetation Management Program Environmental Impact Report to help streamline compliance with the California Environmental Quality Act for fuels management projects.
- Fire Weather – CAL FIRE owns and maintains Remote Automated Weather Station equipment in compliance with NWCG Standards. Trained Unit staff perform daily data management to support daily operational strategies and tactics, resource placement and emergency response.
- Board of Forestry and Fire Protection – Plays an important role in review and comment on county updates to General Plan Safety Elements.

Federal Plans and Programs:

- USFS - The U.S. Forest Service plays several important roles in California: land manager, a provider of fire protection and prevention, private landowner assistance provider, and research.
- NRCS – Emergency Watershed Protection (EWP), Conservation Stewardship program (CSP), Environmental Quality Incentives Program (EQIP), Wildlife Habitat Incentives Program (WHIP).
- National Fire Plan Cohesive Strategy– Federal strategic plan for reducing costs and losses from wildland fire.
- Healthy Forests Restoration Act – To build-up the capacity to conduct hazardous fuels reduction projects on National Forest System lands and Bureau of Land Management lands aimed at protecting communities, watersheds, and certain other at-risk lands from catastrophic wildfire.
- FEMA Multi-Hazard Mitigation Plans Community Assistance – Assistance to communities may include grants and technical assistance directly to local governments or non-profit organizations.

Tribal Plans and Programs

- The Hoopa Valley Tribal Forestry Department is responsible for the management of over 87,000 acres of tribal timberlands. Timber operations, conducted under the tribe's Forest Management Plan, have been certified by the Forest Stewardship Council.
- The Karuk Tribe's Wildland Fire Program works to restore the use of prescribed burning as an ancestral cultural practice through its Department of Natural Resources' Wildland Fire Program.

Key Stakeholders and Partners

Key stakeholders include California citizens; land owners; CAL FIRE; state, federal and local governments and agencies; tribes, as well as non-profit organizations (e.g. fire safe councils, resource conservation districts).

Strategies and Supporting Actions

Strategy: 4.1. Reduce the occurrence of damaging wildfires and reduce life, property and natural resource losses through the implementation of effective and efficient fire prevention programs and activities.

Action A - Develop a method for the integration of fire and fuels management practices with landowner priorities and multiple jurisdictional goals within local, state and federal responsibility areas.

A-1. Continued support for fuel reduction projects using CAL FIRE Forest Health Grants and other types of funding.

A-2. Work to remove unnecessary regulatory barriers that limit hazardous fuel reduction activities, while maintaining assurance that environmental assets (e.g. air and water quality, wildlife habitat, etc.) are not exposed to risk of significant damage.

A-3. Promote the use of the Vegetation Treatment Program EIR and other programmatic documents that support and streamline regulatory processes

A-4. Assist collaborative partners by educating, improving grant capacity and other means that provide tools to achieve fuels reduction work on the landscape.

A-5. Promote forest and rangeland health, hazardous fuels reduction, and the improved utilization of all forest materials, including small logs, urban green waste, and biomass.

A-6. Increase public education and awareness in support of ecologically sensitive and economically efficient vegetation management activities, including prescribed fire, forest thinning and other fuels treatment projects.

A-7. Utilize the Forest Management Task Force (FMTF) to promote the development of multi-agency/landowner fuels reduction policies and activities at the watershed and fireshed level.

A-8. Continued support and utilization of CAL FIRE prescribed burn crews and other public and private sector resources, for fuels management activities.

A-9. Continue support for statewide multi-hazard plans with Office of Emergency Services and other governmental entities.

A-11. Continue to work with the California Air Resources Board, local Air Pollution Control Districts, and nongovernmental organizations to address concerns over use of prescribed fire and particulate matter from forest and range land management activities.

A-12. Use the 2018 Strategic Fire Plan to communicate state priorities and policies.

Strategy: 4.2. Protect life and property from wildfire through efficient and effective fire protection planning and suppression, financial management, and firefighter/public safety strategies.

Action A - Articulate and promote the concept of land use planning as it relates to fire risk and individual landowner objectives and responsibilities.

Action B - Support and participate in the collaborative development and implementation of wildland fire protection plans and other local, county and regional plans that address fire protection and landowner objectives.

B-1. Establish a working group, consisting of Board members, Departmental staff, and relevant stakeholders to develop minimum standard elements for inclusion in Unit Fire Plans.

B-2. Coordinate Unit Fire Plans with community wildfire protection plans to encourage and support one consistent approach. Develop county or regional fire plans by bringing

together community-based groups, such as fire safe councils and affected fire and land management agencies.

B-3. Create and support venues in which individual community members can be actively involved in local fire safe councils, citizen emergency response teams, Firewise and other community-based efforts to develop readiness plans and to educate landowners to mitigate the risks and effects of catastrophic wildland fire.

B-4. Collaborate with federal and local governments, other state agencies, fire service and other organizations, to develop and implement emergency response plans, including community evacuation plans.

B-5. Ensure planning efforts are consistent with the National Fire Plan, the Healthy Forest Restoration Act, the Statewide Hazard Mitigation Plan, National Forest plans, as well as local hazard mitigation plans, and other relevant statewide strategic planning documents.

B-6. Maximize available resources to strengthen planning efforts through the development of public/private partnerships.

B-7. Develop fire risk mitigation treatment decision support tools to assist in project design, implementation, prioritization and validation.

B-8. Investigate changes to laws regarding fire suppression tactics to allow more flexibility in managing wildfire to meet social and ecological objectives (similar to Federal policy).

Action C - Increase awareness, knowledge and actions implemented by individuals, local governments, and communities to reduce human loss and property damage from wildland fires, such as defensible space, fire prevention and fire safe building standards.

C-1. Educate landowners, residents and business owners about the risks and their incumbent responsibilities of living in the wildlands, including applicable regulations, prevention measures, and preplanning activities.

C-2. Facilitate activities with individuals and organizations, as appropriate, to assist individual property owners to comply with fire safe regulations.

C-3. Improve regulatory effectiveness, compliance monitoring and reporting pursuant to Public Resources Code (PRC) §4290 and §4291.

C-4. Utilize CAL FIRE staffing as available, as well as public and private organizations, to increase the number and effectiveness of defensible space inspections and promote an increasing level of compliance with defensible space laws and regulations.

C-5. Promote the consolidation of Fire Safe Regulations contained in California Code of Regulations (CCR) Title 14 with CCR, Titles 19 and 24, to achieve uniform application of building standards.

C-6. Continue to evaluate new, ignition-resistant construction technologies and materials, and promote the strengthening of California building standards.

C-7. Seek out incentives to promote the retrofit of existing structures to meet ignition-resistant building codes.

C-8. Actively enforce and seek updates as necessary to fire prevention codes and statutes, including those regulating utilities, railroads, small engines, and other categories of equipment use that contribute to fire ignition.

C-9. In a continuing effort to deter negligent behavior, actively investigate wildland fire cause and pursue appropriate civil or criminal actions, including cost recovery.

C-10. Analyze trends in fire cause and focus prevention and education efforts to modify behavior and effect change.

Action D - Support funding to correspond to statutory responsibilities and match the levels of service and performance goals established by the Board of Forestry.

Action F - Develop oversight policies and use of information and planning tools for analysis of cost containment alternatives, staffing, and accountability for state spending.

Action G - Determine the level of fire suppression resources for adequate protection of the values and assets at risk identified during the planning processes.

G-1. Maintain an aggressive wildland fire initial attack policy that places a priority on protecting lives, property and natural resources, while at the same time considers suppression strategies that incorporate values and assets at risk, as well as cost factors wherever possible.

G-2. Develop criteria for determining suppression resource allocation based on elements such as identified values and assets at risk, ignition density, vegetation type and condition, and local weather and topography.

G-3. Initiate studies and analyses to identify appropriate staffing levels and equipment needs commensurate with the current and projected emergency response environment, including a review of the National Fire Danger Rating System (NFDRS).

G-5. Initiate and maintain cooperative fire protection agreements with local, state and federal partners that value the importance of an integrated, cooperative, regional fire protection system and deliver efficient and cost effective emergency response capabilities beneficial to all stakeholders.

G-7. Ensure all firefighters are provided the appropriate training, equipment and facilities necessary to successfully and safely meet the increasingly complicated and challenging fire and emergency response environment.

G-8. Continue to evaluate and implement new technologies to improve firefighter safety, situational awareness and emergency response effectiveness.

G-9. Provide for succession planning and employee development at all levels within CAL FIRE to maintain emergency response leadership capabilities, administrative management skills, and pre-fire planning expertise.

G-10. Effectively engage and train employees across all disciplines to address both planning and emergency response utilizing a "total force" approach.

G-11. Implement defensible space strategies pursuant to PRC 4290, 4291 and the parallel Government codes for non SRA. Develop defensible space regulatory effectiveness/compliance monitoring/reporting program. Develop strategies to address hazardous fire protection situations in established neighborhoods/WUI areas that have substandard protection characteristics.

Strategy: 4.3. Reduce the impacts of wildfire on ecosystem health, public safety, and private property through appropriate scientific research, education, and training.

Action A - Identify and evaluate wildland fire hazards and the associated values and assets at risk. Facilitate the sharing of analyses and data collections across all ownerships for consistency.

A-1. Identify and provide appropriate automated tools to facilitate the collection, analysis and consistent presentation of datasets (fire reports, etc.).

A-2. Update and maintain consistent, detailed vegetation and fuels maps across all ownerships in an efficient and cost-effective manner.

A-3. Provide regular updates to the Department's Very High Fire Hazard Severity Zone maps.

A-4. Develop and validate weather and climatology information for use in predicting fire behavior and assessing fire probabilities.

A-5. Update fire history information and re-evaluate existing fire prediction models to determine composite fire threat across all ownerships.

A-6. Update existing data for values and assets at risk utilizing GIS data layers and other mapping solutions, including fire behavior-specific effects (i.e., fire risks).

A-7. Develop improved modeling of air quality impacts of wild and prescribed fire.

A-9. Use science-based approaches to evaluate, understand, and protect against the negative impacts of new and emerging threats such as climate change, insect and disease outbreaks, or land use changes on forest health and public safety, including the buildup of hazardous fuel conditions and resulting fire behavior.

A-10. Engage and participate with local stakeholder groups (i.e., fire safe councils, Farm Bureau chapters, and others) to validate and prioritize the assets at risk.

A-11. Engage with and support universities and other academic institutions performing scientific research into wildfire management, ecosystem health, climate change, and other critical topics.

Strategy: 4.4. Address post-fire responsibilities for natural resource recovery including watershed protection, reforestation, and ecosystem restoration.

Action A - Encourage rapid post-fire assessment and project implementation to minimize flooding, protect water quality, limit sediment flows, maintain soil productivity, and reduce other risks on all land ownerships impacted by wildland fire.

Action B - Work with landowners, land management agencies, and other stakeholders across the state to design burned area rehabilitation actions that encourage salvage and reforestation activities, create resilient and sustainable landscapes, and restore functioning ecosystems.

Action C - Effectively utilize available resources, including CAL FIRE hand crews, to accomplish restoration and protection activities.

Action D - Assess the effects of pre- and post-fire treatments to refine best management practices.

Action E - Assist landowners with evaluating the need to utilize features developed during a fire, taking into consideration those identified in previous planning efforts (e.g., fire lines).

Action F - Aid landowners in recently burned areas in developing and implementing vegetation treatment plans to manage the re-growth of fuels to reduce hazardous conditions.

Action G - Promote the maintenance of a native species seed bank and seedling production capacity to provide the availability of appropriate tree species for reforestation within all the state's diverse seed zones.

Action H - Use after-action reports to evaluate and implement new technologies and practices to improve future firefighting efforts.

Action I – Continued support for CAL FIRE's nursery to grow seedlings for small private landowners whose forestland is impacted by wildfires.

Chapter 5: Forest Pests

Forest pests serve an important ecological function by killing weaker trees, thinning overstocked stands, and creating special habitat elements and a food source for wildlife. They also can reduce timber growth and value, impact recreation value, and increase wildfire risk.

Episodic widespread severe pest outbreaks are part of natural cycles. However, there is worldwide concern that the frequency, severity, and size of these outbreaks is increasing, and that climate change is increasing drought and heat stress and shifting forest systems outside the range of normal variation, in part due to an expansion of the geographic and seasonal range of some important pest species (Allen, Macalady et al. 2010, Robbins 2010). Current levels of tree mortality in California, especially in the southern Sierra, add to these concerns (⑤.1). Assessment indicators support trends of increasing stress on forests due to changing fire regimes (④.1, ④.3, ④.4), changing seasonal water availability (⑨.2, ⑨.3, ⑦.2), rising temperatures (⑦.1), overstocked forest stand conditions (①.2), and increasing numbers of exotic forest pest species (⑤.2). Fortunately for California forests, invasive plant species are not a significant issue throughout most of the state.

INDICATORS

- ⑤.1 Tree Mortality
- ⑤.2 Native and Exotic Pests

Current tree mortality levels have called into question forest and fire management policies. This chapter includes an analysis of historic data that supports the role of active timber management for reducing pest-induced tree mortality, also supported by Chapter 1 (Sustainable Working Forests) (①.4). Active management to reduce tree mortality can also include restoring natural fire regimes (van Mantgem, Caprio et al. 2016). High tree mortality levels are a threat to forest health, delivery of ecosystem services, and public safety, and are particularly unacceptable on lands managed primarily for wood products or adjacent to human infrastructure. However, on other lands some level of tree mortality may be acceptable or even desirable for improving wildlife habitat. In addition, there is evidence to suggest that by selectively thinning trees not suited for current climactic conditions, forest pests can assist in creating persistent stands that are adaptable to future conditions under changing climate (Millar, Westfall et al. 2012). The challenge for land managers is ensuring mortality does not create landscape-level conditions that support more widespread pest outbreaks or wildfires. This concern underscores the importance of landscape-level collaborative planning and project implementation efforts across ownerships.

This chapter provides a list of specific opportunities to reduce pest damage, which generally involve: early detection and containment (especially for invasive exotic pests); policies and programs to limit pest spread; continued research to better understand pest species and how to manage for forest resiliency; support for active management to increase forest resilience and manage risk at the landscape level; and restoration of areas with severe pest damage to protect public safety, reduce fire risk, and restore healthy forests.

KEY FINDINGS

⑤5.1 Indicator: Area of Tree Mortality from Forest Pests and Drought

- ① Since 2002, annual detections of new forest tree mortality have ranged from just over 200,000 acres to over 4.1 million acres per year.
- ① Since 2002, severe mortality (at least 5 trees per acre) occurred on an average of 25% of all acres with detected tree mortality, and ranged between 16% of acres with detected mortality in 2008 to almost 60% in 2016.
- ① Of the total acres of detected mortality (2002–2016), 76% of the acres were federal forestlands and 22% were private.
- ① The percent of total acres detected with severe tree mortality (at least 5 trees per acre) by owner class (2002–2016) was similar for federal (34%), private (34%), and other (32%).
- ① The Sierra Cascades Bioregion accounted for 66% of detected mortality acres (2002–2016).
- ① Near-term data indicate a major recent increase in the severe mortality class acreage. Severe mortality is especially prevalent in the south Sierra since 2015.

⑤5.2 Indicator: Number of Native and Exotic Forest Pest Species Occurrences

- ① Occurrences of forest pest species, both native and exotic, have increased from 10 in 1955 to over 30 in recent years, or triple the number of species.
- ① The ratio of exotic to native pests has been increasing over time. Exotic pests were a minor component of occurrences in the 1950s and 1960s. The trend line shows they now comprise over one-third of occurrences. In 2007, half of pest species occurrences were exotic species.
- ① Native bark beetles and wood borers remain a high priority. There are elevated activity levels of fir engraver, western pine, mountain pine, *lps*, and red turpentine beetles, flatheaded fir borer, and pine borers throughout the South Coast and Sierra bioregions, and other areas of the state.
- ① Non-native forest pests such as sudden oak death, pitch canker disease, gold spotted oak borer, and invasive shot hole borers/*Fusarium* complex are currently of major concern to California forest pest management agencies, for wildland and urban forests alike. Of additional great concern is the arrival of the Mediterranean oak borer (*Xyleborus monographus*, native to Europe). This pest, which has recently been discovered in Napa, Sonoma, Lake, and Sacramento Counties, is attacking white oaks [particularly valley oak (*Quercus lobata*) and blue oak (*Quercus douglasii*)]. There is a high likelihood of the expansion of the beetle's range, since it can be moved in infested wood and the females can fly. The pest may be capable of establishing over much of California.

Strategies

A healthy forest landscape has the capacity for renewal and for recovery from a wide range of disturbances, while continuing to provide public benefits and ecosystem services. Threats to forest health include insects, disease, invasive plant and animal species, air pollution, and climate change. Assessments should identify high value forest landscape areas that are especially vulnerable to existing or potential, forest health risk factors, where forest management practices are most likely to prevent and mitigate impacts. Assessments should also identify areas where management could successfully restore impacted forests. Resource strategies should include feasible long term strategies for addressing forest health risks and

opportunities within important forest landscape areas (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goals of these strategies are to restore areas damaged by forest pests and to prevent, minimize and control to the extent feasible, future pest outbreaks, to maintain ecosystem health, preserve ecosystem services and avoid public safety hazards associated with large scale tree mortality events on wildland and urban forests. This strategy also addresses goals identified at the national and state level, as noted below.

National Goal Supported: Protect Forests from Harm

Montreal Protocol Supported:

MPC-3: Forest Health

State Assessment Theme: Restore forest pest-impacted areas; Protect ecosystem health and public safety from future forest pest outbreaks.

Strategies Overview

State Issues

- The large-scale mortality of trees in parts of the Sierra Nevada has resulted in state declared high hazard zones where people and structures are at risk from falling trees. In late 2015 Governor Brown made an emergency proclamation for dead and dying trees and created the Tree Mortality Taskforce. This body has since been renamed and given an expanded focus as the Forest Management Task Force. The State has had declared Zones of Infestation for native Bark Beetles since at least 1993.
- The growing number of exotic introductions of insects, diseases and invasive plants remains a risk to ecosystem health in California. The BOF has declared Zones of infestation (ZOI) for exotic pests, and including pitch canker disease, sudden oak death, and gold-spotted oak borer. The State continues to use agricultural inspection stations, early detection, eradication, and firewood or other quarantines to limit the spread of exotic insects and diseases.
- There has been great concern over a recent introduced invasive shot hole borer (ISHB) complex in Southern California coastal counties. The polyphagous shot hole borer (PSHB) and a closely related beetle the Kuroshio shot hole borer (KSHB) are invasive beetles that attack dozens of native and introduced tree species and infect them with fusarium dieback, a fungus that disrupts water and nutrient uptake by trees. Although no Zone of Infestation has been declared for PSHB and KSHB, state agencies are allocating resources to remove vector trees and are deploying traps into vulnerable areas to understand how and where these pests are currently spreading.
- Widespread overstocking of forests has created many areas vulnerable to forest pests and diseases. Projected climatic changes put many areas that are currently forested at risk of permanent type conversion to non-forest, through stand-replacing events and lack of forest tree regeneration. The State is using California Climate Investment (CCI) grant programs from cap and trade funding for forest management activities such as prescribed fire and thinning to improve forest carbon outcomes by preventing future outbreaks of forest pests in overstocked stands.

Cross-Cutting Issues

Strategies that prevent or control large scale forest pest outbreaks or limit the introduction and spread of exotic forest pests and invasive plant species involve forest management activities such as control of exotics; removal of infested, dead, dying, and diseased trees; using thinning operations to prevent future outbreaks of forest pests; and restoration of areas taken over by invasive plant species. These forest management activities support other themes and issues in the assessment.

- Wildfire Threats – Forest management activities can reduce hazardous fuel loads that feed wildfires.
- Climate Change – Forest management activities can yield additional climate benefits by protecting existing carbon stocks and producing more resilient forest stands.
- Emerging Markets – Forest management activities can improve stand health and increase growth of trees, allowing them to produce more wood fiber thus increasing wood product flow and biomass availability.
- Monitoring efforts to address exotic pests and invasive plants can benefit multiple strategies by collecting a broader range of forest health and vegetation related information as well as disturbance and management activities, while simultaneously reducing overall costs. Strategies that can benefit from a broader monitoring effort would include those that address wildlife habitat, climate change, emerging markets, water resources, sustainable forests, development impacts and wildfire.

Existing Supporting Plans and Programs

Supporting plans include:

Plans: California 2018 Strategic Fire Plan, California Forest Carbon Plan, California Air Resources Board Scoping Plan for the Global Warming Solutions Act of 2017, California Department of Fish and Game Wildlife Action Plan, National Fire Plan, Forest and Rangeland Assessment, the California Forest Action Plan, California Forest Pest Conditions Reports, CAL FIRE Urban & Community Forestry Program Strategic Plan 2019-2024.

Existing programs that support forest pest protection and restoration strategies include:

- California Forest Practices Rules – provides rules and procedures to avoid or lessen adverse effects on the environment from timber harvesting on local, state and privately owned lands.
- CAL FIRE Pest Management Program - forest pest specialists help protect the state's forest resources from native and introduced pests, conduct surveys and provide technical assistance to private forest landowners and promote forest health on all forest lands throughout the state.
- California Forest Improvement Program (CFIP) – improve productivity of non-industrial private timberlands and includes the improvement of other forest resources, including fish and wildlife habitat, soil, and water quality.
- California Forest Stewardship Program – Designed to promote long-term stewardship of private forest lands.
- California Forest Management Taskforce – a multi-agency taskforce established to restore forest health and reduce forest pest and wildfire threats to California Forestlands.
- University of California Cooperative Extension (UCCE) – Serves forest and range land owners through outreach efforts and technical assistance.

- The U.S. Forest Service plays several important roles in California: land manager, a provider of fire protection and prevention, private landowner assistance provider, and researcher as well as various technical support and evaluation monitoring programs.
- USFS Forest Health Protection has specialists in forest entomology and pathology, invasive plants, pesticide use, survey and monitoring, suppression and control, technology development and other forest health-related services that assist with protecting and improving the health of rural, wildland and urban forests.
- USFS Forest Health Protection has pest-specific funding to implement integrated pest management strategies on federal and state and private lands.
- NRCS – Emergency Watershed Protection (EWP), Conservation Stewardship program (CSP), Environmental Quality Incentives Program (EQUIP), Wildlife Habitat Incentives Program (WHIP).
- Community Assistance – Assistance to communities may include grants and technical assistance directly to local governments or non-profit organizations.

Key Stakeholders and Partners

Key stakeholders include land owners, Governor's Office (GO), Office of Emergency Services (OES), California Forestry Association, Governor's Office of Business and Economic Development, CAL FIRE, Department of Water Resources, State Water Resources Regional Control Boards, California Energy Commission, Air Resources Board, Department of Fish and Wildlife, University of California Division of Agriculture and Natural Resources, USFS, NRCS, US Fish and Wildlife Service.

Strategies and Supporting Actions

Strategy: 5.1. Restore forest lands impacted by current and historical forest pest outbreaks, air pollution, and invasive species.

Action A – Retain strong pest control, fuel reduction, forestry assistance and fire protection programs and further strengthen program delivery through seeking enhanced staffing levels in State and federal programs.

Action B – Provide landowner assistance through the delivery of programs that address technical and financial assistance, restoration, risk reduction, and stand improvements.

Action C – Support public and private nurseries to ensure that a reliable supply of seed for commercial and non-commercial tree species is available for appropriate genotypes, for reforestation and forest health improvement.

Action D – Implement policies that emphasize use of an appropriate mix of species, that are well-adapted to local conditions (i.e., from appropriate seed zone and elevation gradients) when reforesting areas after pest damage, harvest, or fire.

Action E – Enhance cooperation and coordination among agencies, landowners and groups with an interest in forests.

Action F – Expand research on pest control methods, including the potential for unwanted impacts on ecosystem health.

Action G – Develop a contingency plan for ecological impacts of climate change, including seed banks and land trades adjusted to ranges of vegetation types.

Action H – Implement effective training, education, and outreach programs to (1) inform landowners, government officials and the public, (2) develop a well-educated cadre of forest pest management professionals in California, and (3) ensure that the staff of State and federal forest pest and forest health programs have the best and most recently available scientific information about forest pests and their control, including through participation in professional meetings.

Action I – Continue to work with the California Air Resources Board and local Air Pollution Control Districts to address concerns over use of curtain burners, prescribed fire, and the particulate matter emissions from forest and range land management activities.

Action J – Maintain periodic assessments of impacts of ozone and other pollutants on forest and rangeland vegetation and aquatic resources.

Action K – Evaluate current tree seed zones in California and revise if needed to address current climate and anticipated future climates to assist in selection of appropriate seed for planting.

Action L – Create markets for low value biomass material resulting from large scale tree mortality events.

Action M – Use Emergency Proclamations and Executive Orders from the Governor to rally forest management stakeholders to help set priorities for state agencies, reduce barriers, improve coordination, and leverage state investments in addressing impacts from current and future forest pest outbreaks.

Strategy: 5.2. Reduce/prevent forest pest outbreaks and control their spread to maintain ecosystem health, preserve ecosystem services, and avoid public safety hazards associated with large scale tree mortality events.

Action A – Enhance forest resiliency through the strategic placement of stand improvement projects in high priority landscapes.

Action B – Streamline environmental review processes related to stand improvement projects.

Action C – Develop or improve monitoring and reporting systems for forest pests, including early detection. Quantitative data are needed for large-scale outbreaks to support planning for public safety and ecosystem health.

Action D – Provide landowner assistance to reduce forest susceptibility to future threats in priority landscapes and improve benefits from trees and forests.

Action E – Develop overall plan to guide forest and range land pest research and control, including public involvement.

Action F – Maintain California Department of Food and Agriculture quarantine capacity and the California Border Inspection Stations.

Action G – Enhance support for County Agricultural Commissioners, University of California researchers, and landowner participation.

Action H – Develop communication tools to inform the public of the positive benefits of active forest management.

Strategy: 5.3. Prevent the introduction and spread of new exotic pests and invasive plant species.

Action A – Develop an overall policy for California resources that integrates approaches to reduce overstocking, reduce fuel loading, increase tree health and vigor, increase fire protection and prevention, and control exotic pests and invasive plants.

Action B – Continue strong support for focused best management practices, such as restriction on movement of plant material, use of prescribed fire, and use of equipment to control or prevent the spread of exotic pests and invasive plants.

Action C – Support the implementation of the Noxious Weed Strategic Plan and treatment of Cooperative Weed Management Areas (WMA's).

Action D – Promote viable, diverse populations of native and valued non-native fish species by reducing risks of harm from invasive plant species and exotic pests.

Action E – Coordinate or integrate federal, state, university, and other diagnostic resources to support surveillance, detection, and identification efforts focused on preventing the introduction of new exotics into California.

Action F – Strengthen support for California Department of Food and Agriculture program on prevention, detection, eradication, education, and taxonomic identification.

Action G – Enhance support for county Agricultural Commissioners, University of California researchers, and landowner participation.

Action H – Use science-based approaches to evaluate, understand, and protect against the negative impacts of new and emerging threats such as exotic pests and invasive plants, and to develop and implement strategies to control them.

Action I – Develop and maintain a list of invasive plant species and exotic pests that have a reasonable likelihood of entering, or have entered, California for which an exclusion, detection, eradication, control, or management action by the state might be taken.

Action J – Create, consolidate and publicize a system for reporting sightings of invasive plant species and exotic pests and referring those reports to the appropriate agency.

Action K – Undertake educational and outreach activities to increase awareness of invasive plant species and exotic pest issues.

Action L – Develop an invasive plant species and exotic pest Action Plan, a statewide plan for dealing with invasive plant species and exotic pests, including a Rapid Response Plan.

Action M – Develop funding mechanisms for early detection, rapid response, eradication, and education projects.

Strategy: 5.4. Rapidly control or contain outbreaks of exotic forest pests and invasive plant species.

Action A – Maintain and improve early detection and rapid response capability.

Action B – Develop an overall plan to guide forest and range land pest research and control, including public involvement.

Action C – Strengthen the emergency response preparedness regarding exotic pests and invasive plant species including the network of responders, internal/external coordination, and the capability to quickly trace origins of outbreaks.

Action D – Implement effective training, education and outreach programs to inform landowners, government officials and the public. This includes developing a well-educated cadre of forest pest management professionals in California, including arborists, to address threats in urban forests.

Action E – Maintain quarantine capacity and other control services at the California Department of Food and Agriculture, including continued support and training for border protection stations that conduct inspections of agricultural products.

Action F – Focus on the development of control methods, both chemical and non-chemical.

Action G – Expand research on control methods.

Action H – Promote efficient and effective control programs and strategies characterized by efforts that address current outbreaks, prevent new invasions and quickly detect new occurrences so that the species may be removed or contained before spreading.

Action I – Aggressively and quickly address exotic pest outbreaks through (1) emergency harvesting of infected, infested or damaged trees; (2) sanitation removal of insect or disease attacked trees to maintain or improve the health of a stand; (3) salvage removal of trees killed by pests or other causes; (4) treatment of slash from timber operations in a manner that avoids build-up of insect pest populations; (5) implement and enforce regulatory control of movement of invasive infested materials and products and; (6) Use of traps to determine areas to which the pest has and has not spread.

Strategy: 5.5. Monitor forestland to quickly identify new, and evaluate current, outbreaks of exotic forest pests and invasive plant species to protect the most vulnerable and valued forest and range land assets.

Action A – Develop, improve and maintain monitoring and reporting systems for native and exotic forest pests and invasive plants, including early detection.

Action B – Update and maintain consistent, detailed vegetation and exotic species location maps across all ownerships in an efficient, collaborative, and cost-effective manner.

Action C – Update existing data for values and assets at risk utilizing GIS data layers and other mapping solutions.

Action D – Facilitate the sharing of all analyses and data collections across all ownerships for consistency in type and kind.

Chapter 6: Population Growth and Development Impacts

With 39.5 million current residents, California's population has been growing at just under 1% annually on average since the year 2000. This annual rate of growth is projected to continue or decrease slightly over the next few decades (⑥6.1).

Approximately 354,000 new residents are anticipated in California yearly (Demographic Research Unit 2017, Demographic Research Unit 2017).

INDICATORS

- ⑥6.1 Population trends
- ⑥6.2 Rangeland under Williamson Act
- ⑥6.3 Protected rangeland

Urban lands currently comprise 5.3 million acres, or about 5% of the total area of the state (U S Census Bureau 2012). Nearly 85% of urbanized land was converted from what was originally rangeland (although most had first been converted to intensive agriculture prior to urbanization). Population growth leads to increases in housing density and urbanized area, with new residential developments arising because of: (1) densification of existing housing in urbanized areas (in-fill) and urban housing redevelopment projects; (2) development of land currently under cultivation; and (3) development of forest or range characterized as open space (primarily near existing urban areas). While all three processes are at work in the state, only the last is addressed in this chapter.

For decades, there have been efforts to reduce the kind of development called “urban sprawl” in the state’s agricultural, forest, and range land. Agricultural lands under California Land Conservation Act (Williamson Act, or WA) contracts offer lower property tax assessments in return for an annually renewed 10-year agreement not to subdivide or develop. About 9.4 million acres of forest and rangeland (excluding croplands) were under WA contracts in 2015 (⑥6.2). That is down from about 10.5 million acres enrolled in 2005. It should be noted that in some counties, reporting of acreage under WA contracts has been sporadic since 2008.

More recently, forest and range landowners have been selling conservation easements (which restrict development) on their holdings (Merenlender, Huntsinger et al. 2004). Some owners have sold or donated land title directly to conservation organizations and resource conservation districts. Since the early 1990s, rangeland acreage in these categories has increased from under 400,000 to more than 1.4 million acres statewide (⑥6.3).

Overall, the 37 California counties designated as “metro” by the federal Census Bureau and the Office of Management and Budget (i.e. those with urban centers of 50,000 or more population) (Cromartie and Bucholtz 2011) account for almost 98% of the state’s residents. This makes California the most urbanized state in the Union (U S Census Bureau 2012). Nearly all population growth in the past ten years has occurred in these metro counties, and particularly around major urban centers such as the Bay Area, greater Sacramento area, and southern California. Consequently, forest and rangelands most at-risk of development in California are in these regions.

Metro counties account for about 72% (13.5 million acres) of private rangelands and 33% (2.9 million acres) of all private forestlands (including industrial timberlands) statewide. Historically,

private rangelands have been much more prone to development than private forestland, due to their desirable climate, lower elevations and access.

Since joining the Union in 1850, about 98,000 acres of California forest and rangeland per year on average have been converted to agriculture, housing development, and other uses. However, most acreage conversions, such as reclamation projects in the Central Valley for intensive agriculture, occurred during the first century of American statehood (1850–1950).

More recent conversion rates have been much lower than the long term historical average, with permanent forest and rangeland conversion to development currently estimated at around 25,000 acres per year. Statewide annual rates of forestland conversions *per se* are quite low - on the order of a few hundred acres per year. These are concentrated mainly in a few developed areas in forested lands (such as the I-80 and highway 50 corridors, greater Lake Tahoe, and mountainous areas of southern California around Lake Arrowhead and Big Bear).

To gauge the potential impact of population growth on the development of forest and rangelands we used California Department of Finance projections and other data to parameterize the U.S. Environmental Protection Agency's Spatially Explicit Regional Growth housing allocation model (SERGoM v3), and estimated the amount of acreage that would be parcelized into housing densities greater than or equal to 1 unit per 20 acres by the years 2040 and 2060. The result: nearly all such future development (excluding that on cropland) is projected to occur on rangeland (84%), with much less development on forestland (16%). Due to low amounts of available land, by 2060 the Bay Area and San Joaquin Valley regions emerge as those most likely to be impacted by new development, with on average nearly 63% of countywide forest and rangeland acres converted for development. Southern California counties would absorb about 22% of the total newly developed forest and rangelands, and the remaining development (~15% of the total) is shown as spread across the other regions of the state.

KEY FINDINGS

General and Regional Population Trends in California

①6.1 Indicator: Recent and Projected Population Trends

- ①🔑 California's population of 39.5 million has been growing at about 0.9% per year over the past decade. This amounts to about 354,000 additional residents per year. The same annual rate of growth, or slightly lower, is projected to continue over the next decades.
- ①🔑 About 98% of people live in California's 37 metropolitan counties, with the remaining two percent in the 21 rural counties. More than 99% of all new residents are projected to live in the metro counties in the coming decades.
- ①🔑 Metropolitan counties also contain about 72% of private rangelands, and 33% of private forestlands, and those within or close to urban areas are the most at-risk of new development.

- ① 🔑 With stable or declining populations, very little rural county forest and rangelands have recently been converted to development. Thus, these are considered at low risk. Projections suggest this is likely to continue.

Forest and Rangeland Protections from Development Threat

- 🔑 Statewide, acreage in the decade-long process of WA contract non-renewal (being withdrawn from the program) has apparently been increasing since 2002. Likewise, program participation appears to be decreasing in recent years, and at least one county (Imperial) has withdrawn from the program entirely. Since 2008, with the cessation of subvention payments from the state to county governments in lieu of WA contract land property taxes, some WA counties have not been diligent in submitting their WA records. Recent trends are more difficult to discern from sporadically incomplete data.
- 🔑 The number of land trust organizations active in protecting forest and rangelands in California has grown substantially in recent years. In 2017, the California Council of Land Trusts had nine statewide (or nationwide) member organizations, and 83 regional member organizations. The Council states that more than 220 organizations in total currently self-identify as land trusts in California.
- 🔑 Demand for conservation easements by owners of working rangelands currently exceeds the supply of willing and able purchasers. The USDA Natural Resources Conservation Service has acquired major acreage of forest and rangeland easements in California, but the trends and unmet demand signals that significantly more acreage is likely to be under such easements within the next decade.
- 🔑 Acreage within state-designated Timber Production Zones (TPZ), approximately 5.3 million acres of forestland, where immediate development options are extremely limited and a property tax reduction is provided to landowners, has been very stable through recent years. This is due in part to the level of difficulty required to rezone such areas at the county level.

①6.2 Indicator: Rangeland Under California Land Conservation Act (“Williamson Act”, or “WA”) Contracts

- ① 🔑 About 9.36 million acres of forest and rangeland were under WA contracts in California in 2015. This is an apparent net decrease of about 870,000 WA acres in these types since 2008* (average of -124,000 acres per year).
- ① 🔑 WA-enrolled rangeland engaged in the 10-year process of non-renewal totaled about 245,000 acres in 2015, about 17% fewer than the previous 4-year average. In recent years, about 25,000 acres of rangeland is losing its WA contract status per year on average.
- ① 🔑 Since the 2009 cessation of state government subvention payments to counties, both current and future WA contracts on rangeland statewide may be at-risk. Resumption of subvention payments to counties, and WA tax benefits, are important to keeping larger livestock operations in business.

①6.3 Indicator: Private Forest and Rangeland Under Conservation Easements, or Owned or Managed by a Conservation Organization

- ①🔑 In 2016, lands owned by conservation organizations included about 457,000 acres of rangeland and 154,000 acres of forestland. In the same year, lands with conservation easements comprised about 1.25 million acres of rangeland, and 452,000 acres of forestland.
- ①🔑 The amount of lands managed for conservation of rangeland has been increasing for the past 2 decades. Acreage of similarly managed forest lands has increased sharply in the past decade.
- ①🔑 In the past two decades, nearly 600,000 acres of desert rangelands and 8,300 acres of forest have been acquired by non-profits, and subsequently re-conveyed to government agencies permanently for management purposes.

Strategies

In many parts of the United States, forests and other open space are being fragmented and converted to non-forested uses including development. Forestry agencies can work with partners, stakeholders and communities to identify and protect priority forest landscapes through land acquisition, conservation easements, and land use policies. Forestry agencies can also provide technical assistance to communities to help them strategically plan for and conserve forests and other open space. Factors contributing to loss include residential, commercial and industrial development; expansion of utility infrastructure and transportation networks; and planning, zoning, and policies that favor conversion. Consequences include the outright loss of public benefits associated with forests or the marginalization of those values provided by contiguous forested landscapes. Fragmentation also includes “parcelization,” or the fracturing of large singular ownerships into numerous smaller ones. Assessments and strategies should attempt to identify, protect and connect ecologically important forest landscapes, and open space, thus maintaining a green infrastructure, particularly around and within areas of, population growth and development (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: Conserve, protect and connect ecosystems most threatened by development.

National Goal Supported: Conserving Working Forest Lands

Montreal Protocol Goals Supported:

MPC-3: Maintenance of Forest Ecosystem Health and Vitality

State Assessment Theme:

Threats to priority ecosystems from development due to population increase.

Strategies Overview

State Issues

- A growing population increases the demand for new housing and infrastructure. Much of the latter is occurring on working rangelands and agricultural lands near urbanized

areas, with some on forest lands as well. The southern Central Valley and Inland Empire are the fastest growing regions in the state.

- Rangelands and forests permanently converted to commercial, industrial and residential development lose most to all ecosystem services they provide. Conservation easements and fee title purchase by conservation NGOs retain ecosystem services and greatly lessen the threat of development on these lands.
- Williamson Act contracts reduce property taxes on agricultural land owners, and help reduce development on these lands. However, state funding for contracts has been suspended since 2008, and counties are facing mounting deficits as a result.
- Infill in existing developed areas can increase the housing supply while minimally affecting forests and rangelands. Numerous state legislative efforts (e.g., SB 375) are seeking to enable this more energy efficient pattern of new development.

Cross-Cutting Issues

Mitigating the effects of development on priority ecosystems relates to several other themes and issues presented in the assessment document. The most important are listed below:

- Wildfire Threat—ignition sources are a major factor in the frequency of large wildfires. Development, with the activities of people accessing new areas, increases the risk of ignitions and possibly the threat of wildfire. Human ignition sources are the predominant cause of wildland fires in California.
- Forest Pests and Other Threats to Ecosystem Health—New development can be primary points of entry into new areas for exotic plant pests. Predation on native species by domestic pets can adversely affect animal populations in formerly intact ecosystems.
- Water Quantity and Quality—Wetland ecosystem conservation in urban or ex-urban areas can help water quality through practices that mitigate flooding occurrence and damage by providing areas for stream overflow containment. These ecosystems also help recharge vital groundwater in more semi-rural areas that rely on wells for their water supply.
- Urban Forestry for Energy Conservation and Air Quality—Ecosystems most threatened by development are often in close proximity to existing urban or suburban areas. By conserving these ecosystems they can help provide the same air quality improvement and temperature-lowering benefits that occur with augmenting urban forestry.
- Plant, Wildlife and Fish Habitat Protection, Conservation and Enhancement—Conservation of high value ecosystems threatened by development benefits the local wildlife and fish in the watershed.
- Green Infrastructure for Connecting People to the Natural Environment—Ecosystems under threat of development most often occur in close proximity to areas already developed. Conservation of these areas would also provide opportunities to augment the green infrastructure in nearby and neighboring communities.

Existing Supporting Plans and Programs

A number of non-regulatory organizations, both public and private, are involved in influencing future development. Some operate more locally, while others are at regional, statewide or even national scales. Coalitions of groups, some including both public and private, have formed to help direct future development in specific regions.

California does not have an official strategic planning or vision document focused primarily on guiding the location of future development at a statewide scale. As a result, a coordinated

planning effort taking into account conserving threatened ecosystems across large scales is also lacking at the state level.

Codes, ordinances, programs, organizations and initiatives that have bearing on strategies for land use planning, ecosystem conservation and future development include (main types with some examples):

Public:

City zoning ordinances; Special districts (e.g. regional parks and recreation departments and open space districts); County General Plans, Local Area Formation Commissions (LAFCs) and Regional Transportation Planning Agencies (RTPAs); Metropolitan Planning Organization (MPO) Plans; State bond initiatives (e.g. Prop 50, Prop 84) have established funding for measures to help conserve important lands and remove the possibility of future development; CAL FIRE's Forest Legacy Program allocates monies annually for the purchase of land or conservation easements in areas of high-value ecosystems; The Strategic Growth Council coordinates state agencies with six main objectives, one of which is to "protect natural resource and agricultural lands."

Private:

National, regional, and local land trusts such as The Nature Conservancy, Pacific Forest Trust, Conservation Fund, etc.; Other non-governmental organizations (NGOs), including the Planning and Conservation League, Sierra Club, Audubon Society, Greenbelt Alliance and many more.

Coalitions of Private and Public:

Smart Growth Initiative (40 member organizations); North Sierra Partnership (5 member organizations); Bay Area Open Space Council (100+ organizations).

Governmental programs, laws, regulations and codes influencing future development:

USFS Scenery Management System; USFS and CAL FIRE Forest Legacy Programs; Forest Tax Reform Act; Williamson Act; Natural Communities Conservation Planning (NCCP) Act; Conservation and Mitigation Banking (DFG); Oak Woodlands Conservation Act; California Environmental Quality Act (CEQA); National Environmental Protection Act (NEPA); Clean Water Act; Clean Air Act; Select California State Bond Issues (e.g. Prop 50, Prop 84)

Key Stakeholders and Partners

Members of California Association of Councils of Government (CALCOG), municipal governments, special districts, county governments, metropolitan planning organizations (MPOs), other state and local governing bodies; Private land developers, contractors; Conservation-oriented Non-Government Organizations, including land trusts; State of California natural resource agencies and departments; U.S. government natural resource agencies.

Strategies and Supporting Actions

Strategy: 6.1. Reduce urban sprawl by promoting redevelopment and infilling of available land within the urban matrix, strengthening planning at the local level, capacity building, and improving access to tools and data sources.

Action A – Provide financial and other incentives for locating new development in areas already developed (redevelopment), or infilling those areas already developed. Support the passage of legislation similar to SB 50, the 2019 More HOMES Act (Wiener) to promote upzoning, particularly in areas of dense transit options.

Action B – Develop bond measures to provide funding for Action A.

Action C – Support the implementation of incentive-based SB 375 (Redesigning Communities to Reduce Greenhouse Gases (Steinberg) through curbing sprawl.

Action D – Continue support of Senate Bill 375 Strategic Growth Council (Steinberg) to coordinate actions towards improving the availability of affordable housing, improving transportation, encouraging sustainable land use planning, implementing urban greening plans and revitalizing urban and community centers in a sustainable manner.

Action E – Amend CEQA to streamline and facilitate timely environmental review requirements for infill development projects that are a part of approved regional plans.

Action F – In the process of meeting regional housing needs allocation (RHNA) code, update county general plans consistent with promoting and prioritizing redevelopment and infilling, also with emphasis on higher density housing.

Action G – Future development in priority landscapes should be located and designed to avoid, minimize, and mitigate impacts.

Strategy: 6.2. Aid in efforts to reduce development sprawl in rural communities.

Action A – Reinstate subvention payments from the state to county governments for property tax revenue lost due to Williamson Act contracts. Encourage counties and landowners to continue participation in the Williamson Act.

Action B – Support actions and incentives to curb the damaging impacts of sprawl into forested areas and other sensitive natural areas. Coordinate with other state agencies and nongovernmental organizations to develop regulatory guidance and incentives for local governments located in rural areas to plan development in a sustainable manner to curb impacts on forest and range landscapes.

Action C – Support the implementation of incentive-based SB 375 Redesigning Communities to Reduce Greenhouse Gases (Steinberg), the Strategic Growth Council, and other similar efforts at statewide and regional sustainability planning. Consider ecosystem priority landscapes from this study and others in the development project approval process.

Strategy: 6.3. Support comprehensive planning at state and regional scales that is coordinated with wildlife habitat conservation efforts.

Action A – Continue support of SB 732 Strategic Growth Council (Steinberg) to coordinate actions towards improving air and water quality, natural resource protection, and meet California Global Warming Solutions Act of 2006 goals.

Action B – Continue to support Natural Community Conservation Planning (NCCP) and Habitat Conservation Plans (HCPs) and their efforts to encourage cooperation between stakeholders to conserve natural communities at the ecosystem level while accommodating compatible land use.

Action C – Encourage involvement in the Oak Woodlands Conservation Program pursuant to the Oak Woodlands Conservation Act (2001) that offers landowners, conservation organizations and regional governments’ incentives to preserve oak woodlands.

Action D – Support other similar efforts at statewide and regional sustainability planning. Consider ecosystem priority landscapes from this study and others in the development project approval process.

Action E – Obtain Funds to map important areas for wildlife habitat connectivity in and around urban areas.

Action F – Support the recommendations and strategies of the California Department of Fish and Wildlife’s State Wildlife Action Plan (2015), and the companion plans tiered to it.

Strategy: 6.4. Support local and regional community efforts in preserving scenic landscapes.

Action A – Collaborate with local and regional groups and the NRCS engaged in preserving forest and rangeland open space through the purchase of conservation easements from willing landowners. Continue and increase funding for the state and federal Forest Legacy Programs that are key vehicles for putting easements (and in some cases fee titles) on the ground.

Action B – Help local governing bodies wanting to apply to CALTRANS for scenic highway designations, especially in making eligible scenic highways become officially designated as such. Aid efforts to develop the five legislatively required elements including the prerequisite Corridor Protection Programs.
http://www.dot.ca.gov/hq/LandArch/scenic_highways/scenic_hwy.htm

Action C - More generally support efforts to encourage and help counties in their general plan development to implement zoning and other restrictions to help maintain and enhance their current or planned scenic corridors.

Action D - Support communities applying for grant monies to the Federal Highway Administration of the Department of Transportation for National Scenic Byways Program to preserve/improve scenic byways. <http://www.scenic.org/byways>

Chapter 7: Climate Change

Climate greatly influences forest and range ecosystem dynamics, including the type, composition and productivity of vegetation, as well as the environmental services that they produce. Future climate change scenarios predict increases in temperature (①7.1), changes in the amount and distribution of precipitation (①7.2), changes in

spring runoff (①9.3), and increases in climatic water deficit (①9.4) - a measure of water stress. Altering these fundamental components of climate can result in changes in tree growth, range and distribution of species, and disturbance regimes. The latter include changes in the timing, frequency and extent of wildfires, pest infestations, and other agents of disturbance.

In California, climate change is leading to longer, hotter and drier summers, with more pronounced fire activity and increased tree mortality from pest outbreaks. Extreme weather events, hotter and longer droughts, and severe flooding are expected to occur more frequently with global warming. While California's forests currently have high carbon stocks (①7.3) and are functioning overall as a net sink (①7.4), in some forest types, current conditions combined with these increases in wide-scale disturbances have the potential to reduce carbon storage and affect the capacity of forests to continue operating as a net sink.

Management practices, including fire suppression policies, also influence the balance of carbon stored in forests and wood products, and involve trade-offs between carbon sequestration, carbon storage in live trees versus wood products, and risk of loss from wildfire and pests. The Sustainable Working Forests chapter (Chapter 1) described a range of observed timber management emphases, each with unique implications for carbon:

- High timber emphasis: Emphasize high tree growth and sequestration rates in live trees, set harvest rotations to avoid risk of loss, and increase carbon stored long-term in wood products.
- Medium timber emphasis: Longer rotations, uneven-aged management, higher ecosystem services, and accept potentially higher risk of forest carbon loss for gains in ecosystem benefits.
- Low/no timber emphasis: Allow forests to grow and store carbon naturally, emphasize ecosystem services, but accept potentially higher risk of loss and lower carbon stocks in wood products.

Scientific uncertainty exists in terms of how climate will change and how natural systems will respond. Under an uncertain future, it is possible that forest management fifty years from now could be quite different than it is today, in part to place a higher emphasis on managing for carbon. Research will be critical to explore new management paradigms, which could include changes in the use of genetic stock, planting a combination of species for different purposes (e.g., for wood products and for carbon storage), and innovative ways to reduce risk.

INDICATORS

- ①7.1 Temperature
- ①7.2 Precipitation
- ①7.3 Carbon Storage - Forests
- ①7.4 Carbon Sequestration

KEY FINDINGS

①7.1 Indicator: Average Annual Temperature

- ①🔑 Air temperatures have been increasing across California for decades.
- ①🔑 Statewide increases in air temperature are consistent with global trends (1–2°F).
- ①🔑 Minimum air temperatures are increasing faster than maximum air temperatures.

①7.2 Indicator: Annual Precipitation

- ①🔑 There is high interannual variability and no strong trend across the data record that covers more than 100 years of observational data.
- ①🔑 Wet years are commonly associated with El Niño events.
- ①🔑 Climate change is likely to create more extreme drought and flood events.
- ①🔑 Global Climate Models (GCM) have markedly different predictions for precipitation in future decades that vary with GCM and emissions scenarios.

①7.3 Indicator: Carbon Storage - Forests

- ①🔑 In 2015, total carbon storage in above and belowground living and dead plant materials in California's forests is nearly 2.2 billion metric tons.
- ①🔑 Over 65% of carbon storage is on federal, state, and other public lands.
- ①🔑 Total carbon storage is greatest across the Sierra/Cascades (1.01 billion metric tons), Klamath/Interior Coast Ranges (0.61 billion metric tons), and North Coast regions (0.28 billion metric tons).
- ①🔑 Carbon density (tons C/acre) varies by region and is greatest across the North Coast region.

①7.4 Indicator: Carbon Sequestration – Live Trees

- ①🔑 Based on changes in the aboveground live tree pool in 2015, California forests sequester 0.19 metric tons (MT) C/acre/year, equating to 6.2 million metric tons (MMT) C/year or 22.8 MMT CO₂e/year. For perspective, this would be equivalent to sequestering 5% of the total greenhouse gas emissions reported in the state for 2015.
- ①🔑 Federally-owned U.S. Forest Service (USFS) timberlands experience lower harvest rates than private timberlands, have higher growth rates than USFS reserve lands, and sequester 0.25 MT C/acre/year.
- ①🔑 Mortality outpaces growth on USFS reserve forestlands at the rate of -0.04 MT C/acre/year.
- ①🔑 On private corporate timberlands growth is high and exceeds removal from harvest and mortality, reflecting sustained yield. These lands sequester 0.17 MT C/acre/year and contribute the most to additional carbon storage in harvested wood products.
- ①🔑 On private non-corporate timberlands timber is not the primary objective. These lands show increasing inventories with the highest growth rates and net sequestration (0.68 MT C/acre/year).

Strategies

America's forests offset a significant portion of the nation's annual carbon emissions. Additional climate change mitigation benefits could be achieved through partnerships and management measures. These measures include supporting the development of markets for carbon offsets, utilizing woody biomass for energy, wood product substitution, and promoting tree growth in urban areas. Assessments should identify opportunities for promoting carbon emissions offsets through forestry.

The important benefits that forests provide, such as biodiversity, wildlife habitat, and water storage and flows are affected by climate change. Forest range, type and composition are projected to change significantly– with corresponding changes in wildlife habitat, biodiversity, water flows, and fire regimes. Assessments should consider how climate change will affect important public benefits from forests. Resource strategies should attempt to maintain and enhance resilient and connected forest ecosystems that will continue to provide public benefits in a changing climate (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: Promote actions to preserve and enhance carbon sequestration (i.e. mitigation) and actions to promote ecosystem health and resilience under changing climate conditions (i.e., adaptation).

National Goal Supported: Enhancing Public Benefits from Forests

Montreal Protocol Supported:

MPC-5: Forests and Climate

State Assessment Theme: Threats to forest carbon and long-term carbon sequestration; potential threats to key forest species

Strategies Overview

State Issues

- Climate change is leading to longer, hotter and drier summers, with more pronounced fire activity and increased tree mortality from pest outbreaks.
- Management practices, including fire suppression policies, influence the balance of carbon stored in forests and wood products, and involve trade-offs among carbon sequestration, carbon storage in live trees versus wood product, and risk of loss from wildfire and pests.
- High scientific uncertainty exists in terms of how climate will change and how natural systems will respond. Under an uncertain future, it is possible that forests and management practices fifty years from now could look quite different. This places a greater need for sustained investments in research and monitoring.

Cross-Cutting Issues

Priority landscapes were developed for threats to forest carbon from wildfire, forest pests, and development. However, the impacts brought on by climate change can produce a number of cross-cutting issues. These priority issues include:

- Wildfire – climate change expected to increase fire frequency and extent.
- Forest pests – increased frequency of outbreaks possible under warmer temperature scenarios.

- Shifts in species ranges – the distributions of tree, shrub, and herbaceous plant species are affected by climate; expected shifts will likely have secondary effects on vegetation composition, fire regimes, and wildlife habitat.
- Forest hydrology – warming conditions under future climate scenarios are expected to lead to declining snowpack and earlier snowmelt. This in turn will affect the timing and distribution of water and soil moisture in summer months.
- Interacting and synergistic effects.

Existing Supporting Plans and Programs

Supporting plans include: California Forest Carbon Plan; California Adaptation Strategy (CAS); Assembly Bill 32 Climate Change Scoping Plan; Forest Service Global Change Research Strategy, 2009–2019; Forest Service Strategic Framework for Responding to Climate Change.

Existing programs that support strategies include:

- California Forest Improvement Program (CFIP) - includes the improvement of all forest resources, including fish and wildlife habitat, soil, and water quality.
- Vegetation Management Program – supports fuels reduction actions to decrease the likelihood of high severity fire damage to forest carbon and to increase forest resilience to predicted increases in wildfires and pests.
- California Forest Stewardship Program – Designed to promote stewardship of private forest lands.
- Wildlife Conservation Board.
- NRCS - Environmental Quality Incentives Program (EQIP), Wildlife Habitat Incentives Program (WHIP), and Conservation Stewardship Program (CSP).
- USFS—Region V Best Management Practices Evaluation Program, State and Private Forestry programs.
- Voluntary markets for carbon offsets; Climate Action Reserve
- Possible future compliance markets under AB32, Western Carbon Initiative, or national programs.
- California Energy Commission’s AB 118 program.
- Air Resources Board’s Low Carbon Fuel Standard program.
- Renewable Energy Standard.
- USFS National Insect and Disease Risk Model program.
- Public Interest Energy Research Program – Climate Monitoring, Analysis, and Modeling

Key Stakeholders and Partners

USFS, California Natural Resources Agency, CALFIRE, Department of Fish and Game (DFG), California Energy Commission (CEC), California Air Resources Board (CARB), Natural Resources Conservation Service (NRCS), local government entities, industrial and non-industrial timber companies, Sierra Nevada Conservancy and other land conservancies, and NGOs.

Strategies and Supporting Actions

Strategy: 7.1. Protect and enhance the capacity of California’s forests to sequester carbon through reducing risk of loss from disturbance, protecting existing forest land, and expanding forest area through tree planting.

Action A – Implement key strategy elements from the Forest Carbon Plan and AB32 Climate Change Scoping Plan on Natural and Working Lands. Key actions include:

- A-1. Continued investments in Forest Health grants to improve resilience of forests from climate impacts.
- A-2. Avoid deforestation.
- A-3. Urban tree planting and maintenance for carbon sequestration and energy reductions from increased tree shading.
- A-4. Reduce risk of losses from wildfire by removing forest fuels and utilize materials for bioenergy. Increase pace and scale of fuel reduction – target 1 million acres treated annually.
- A-5. Protect existing carbon stocks through forest conservation and maintain and enhance carbon stocks through forest management.

Action B – Implement key elements from the Safeguarding California Plan: 2018 Update report that promote and enhance forest carbon sequestration.

Action C – Improve methods for conducting periodic inventories of forest carbon through AB 1504 reporting on carbon stocks and through current collaborative efforts to improve carbon accounting for forests and wood products on the Pacific coast of North America (California, Oregon, Washington, and British Columbia).

Action D – Prioritize fuel treatments in watersheds that support multiple benefits and address a range of cross-cutting issues (i.e. reduction of fire threat, watershed protection, forest health, habitat protection).

Action E – Implement strategies A - C from Board of Forestry Policy Statement that protect or enhance carbon sequestration (Criteria 5, Forests and Climate) and related actions.

- E-1. Promote conservation and management of forest lands and vigorous stands, which can significantly contribute to large-scale air pollution reduction. Maintain healthy forests which are vital to protecting resources from air borne waste impacts and which provide opportunities to contribute to pollution reduction through carbon sequestration.
- E-2. Promote forest health and conserve forest lands from land use changes by providing financial opportunities to land owners who are managing their lands in ways that positively influence sustainable carbon storage.
- E-3. Develop carbon protocols for fuels reduction to reduce wildfire emissions.
- E-4. Continue to evaluate life cycle carbon benefits of biomass utilization.
- E-5. Maintain existing ecosystem services/market infrastructure.

Action F: Support forest sector research and monitoring.

- F-1. Continued investment in Forest Health Research Grants.
- F-2. Expand monitoring for prescribed burning and related fuel reduction projects.

Strategy: 7.2. Support Adaptation Needs for Forests by Assessing Climate Vulnerabilities, Improving Institutional Capacity, and Promoting a Priority Research Agenda.

Action A – Implement long-term actions from the Safeguarding California Plan: 2018 Update report that address adaptation needs. Key elements include:

- A-1. Refine priority landscapes by promoting regionally-based vulnerability assessments.
- A-2. Implement forest and rangeland actions that create forest stands that are more resilient to expect future climate conditions.
- A-3. Building institutional capacity and decision support systems.
- A-4. Promoting local emergency response planning.

Action B – Support priority research needs identified in CAT Research Committee, CAS and Forest Service Global Change Research Strategy, 2009 – 2019.

- B-1. Support collaboration among land-based forest research institutions (USFS, DSF, PSW and UC demo forests (Blodgett)) to create representative geographic and elevation transects of forest habitats to monitor the effects of climate and of potential mitigation and adaptation actions.

Action C – Implement strategies E and F from Board of Forestry Policy Statement that support adaptation (Criteria 5, Forests and Climate).

- C-1. Maintain and adjust capacity and flexibility of emergency services related to natural process such as flooding, disease, and wildfire.
- C-2. Develop a contingency plan for ecological impacts of climate change, including seed banks and land trades adjusted to ranges of vegetation types.

Action D – Implement projects that demonstrate climate adaptation actions, such as reforestation of high severity wildfire burns, (e.g., Cuyamaca project with CA Department of Parks and Recreation, American Forest Foundation, Odwalla) and riparian flood plain forest.

Action E – Implement seed collection focused on southern populations of key tree species for likely future use in planting zones to the north. Make sure genetic material from threatened southern populations of conifers is conserved for possible assisted migration.

Strategy: 7.3. Support Actions that Maintain, Enhance, and Protect Ecosystem Functions to Promote Biodiversity and Increase Resilience to Climate Change.

Action A – Implement strategies identified in the Safeguarding California Plan: 2018 Update related to forestry.

Action B – Maintain connectivity across forest landscapes by reducing fragmentation and identifying important habitat corridors and key linkages between conservation areas.

Action C – Restore degraded forest and rangelands to enhance biodiversity and related ecosystem services.

Action D – Support restoration actions to reduce impacts from invasive species and pest outbreaks on forest and rangelands.

Action F – Evaluation, monitoring and protection of habitat that serves as key refugia on forests and rangelands.

Chapter 8: California's Non-Metro Regional Economy

For nearly two decades, non-metro California has outperformed U.S. non-metro in terms of employment, population percent change, and changes in personal income (①8.1). The 46-year span of history (1970–2016) trends upward for California's non-metropolitan economy (①8.2). Compared to the rest of the nation's non-metro regions, California has demonstrated remarkable signs of health and prosperity (①8.1). This does not trivialize the plight of many rural communities and families; the economy does not seem to work well for some white male Americans in midlife, or for some tribal communities. Short and long-term unemployment has been an adverse force in the lives of many (①8.4), and environmental degradation has harmed us all. But the region is not well characterized by crisis (①8.1). Still, there is too much inequality and not enough upward mobility. In addition, the lingering effects of the Great Recession are still felt. However, the structural makeup of the region indicates several years of consecutive job growth, and nationally the financial sector has stabilized (①8.5). In short, the story of the region is one of progress, moving toward full recovery from the Great Recession (①8.1).

INDICATORS

- ①8.1 Relative Performance
- ①8.2 Economic Trends
- ①8.3 Economic Prosperity
- ①8.4 Economic Stress
- ①8.5 Economic Structure

Economists agree that we cannot freeze the structure of the employment opportunities to some past level. The shift from natural-resource-based livelihoods to service-based industries has been underway for many years. Well into the twentieth century, rural industries such as ranching, farming, mineral extraction, and logging contributed significantly to California's Gross Domestic Product (GDP). Today those mature industries provide an increasingly smaller fraction of employment and income opportunities in those areas (①8.5). This decline has been alarming to many rural residents, ushering a feeling that something vital is being lost because of this transition. That transition is characterized by a "New West" where most employment is services-related (①8.5). The New West is characterized by an "amenity gold rush" that attracts people in search of high quality living environments, which "has replaced the 'Old West' based on commodity extraction". The purpose of this chapter is to foster an understanding of rural area economies, which in turn may provide a useful place to start discussing important policy options that often have been accompanied by difficult political discourse.

KEY FINDINGS

①8.1 Indicator: Relative Economic Performance

- ① Based on economic data from 2000–2016, non-metro California surpasses U.S. non-metro in terms of employment, population percent change, and increases in personal income.

①8.2 Indicator: Economic Trends

- ①🔑 From 1970–2016, population grew from 425,424 to 832,574 people, a 96% increase.
- ①🔑 From 1970–2016, employment grew from 171,829 to 406,789, a 137% increase.
- ①🔑 From 1970–2016, personal income grew from \$10,833.9 million to \$37,193.9 million (2016 dollars), a 243% increase.
- ①🔑 From 2000–2016, migration (intranational and international) contributed to 57% of population growth.

①8.3 Indicator: Economic Prosperity

- ①🔑 From 1970–2016, average annual earnings per job grew from \$46,331 to \$47,790 (2016 dollars), a 3% increase.
- ①🔑 From 1970–2016, per capita income grew from \$25,466 to \$44,673 (2016 dollars), a 75% increase.

①8.4 Indicator: Economic Stress

- ①🔑 Since 1990, the *annual* unemployment rate ranged from a low of 6.2% in 2000 to a high of 16.3% in 1982. As of July 2017, the *monthly* unemployment rate for the region was 5.8%.
- ①🔑 In 2016, people with disabilities accounted for 18 % of non-institutionalized civilian population compared to 15.1% for the nation as a whole.
- ①🔑 In 2016, people without health insurance accounted for 11.6% of non-institutionalized civilian population.

①8.5 Indicator: Economic Structure

- ①🔑 In 1970, proprietors (the self-employed) represented 23% of total employment (full & part-time jobs). By 2016, proprietors represented 31% of total employment.
- ①🔑 From 1970–2016, proprietors grew from 39,099 to 124,201, a 218% increase.
- ①🔑 In 1970, proprietors represented 23% of total labor earnings. By 2016, proprietors represented 31% of total labor earnings.
- ①🔑 From 1970–2016, dividends, interest and rent grew from \$3,328.8 million to \$17,954.5 million (2016 dollars), an increase of 439%.
- ①🔑 From 1970–2016, age-related transfer payments grew from \$641 million to \$5,187 million (2016 dollars), an increase of 708%.
- ①🔑 From 1970–2016, income maintenance transfer payments grew from \$507 million to \$3,446 million (2016 dollars), an increase of 580%.
- ①🔑 In 2016, total non-government service-related employment accounted for approximately 77% of the total share of jobs.
- ①🔑 From 2001–2015, the three industry sectors that added the most new jobs were health care and social assistance (12,480 new jobs); real estate, and rental and leasing (2,797 new jobs); and other services, except public administration (2,430 new jobs).

- ① 📌 From 2001–2015, earnings in services-related industries grew from \$6,127.3 million to \$8,854.6 million (2015 dollars), a 45% increase.
- ① 📌 From 1998–2015, timber employment shrank by 5,927 jobs, a 50% decrease.
- ① 📌 From 1998–2015, industries associated with travel and tourism in the region grew by 8%; non-travel and tourism industries shrank by 4%.

Strategies

To support and enhance local economies and communities, assessments should identify forest landscape areas where there is a real, near term potential to access and supply traditional, non-timber, or emerging markets such as those for biomass or ecosystem services. These might be areas where necessary infrastructure currently exists, is planned or developing, where group certification of landowners has created market supply aggregation potential, or where retention and management of forest cover presents a money saving alternative to an engineered fix – such as a water filtration facility. Strengthening and developing new market opportunities for forest products and benefits provide incentives for forest stewardship and conservation (based on the US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies).

GOALS:

National Goals Supported: Conserve Working Landscapes; Enhance Public Benefits from Trees and Forests

Montreal Protocols Supported:

MPC-6: Maintenance and enhancement of long-term multiple socio-economic benefits to meet the needs of societies/Socio-economic Well-Being

MPC-7: Legal, institutional, and economic framework for forest conservation and sustainable management/Governance

State Assessment Theme: Non-Metro Regional Economies

STRATEGIES OVERVIEW

State Issues

- Non-metro counties are quite distinct from Metro in terms of their economic and demographic characteristics, as is true in a number of other western states.
 - In the “New West” areas, local economies are generally strong, demographics are stable, and connectivity to metro counties is robust (airports, highways).
 - Often these areas of major ski resorts, other major tourist activities, wealthy retirement communities, and summer homes of metro county residents
 - In areas where the “Old West” remains locally dominant (absent factors above):
 - Demographic generally older and whiter, and economically less well-off; those that work for income often must have more than one job to be solvent.

- Opioid abuse occurred earlier and has been higher per capita than in metro areas. Indicates higher levels of mental health issues.
- Higher percentage of income overall from transfer payments (non-job income) per capita; i.e., more reliance on food stamps, pensions, and government programs such as Social Security and Medicare.
- Remote geographic locations:
 - Often poor access and long distances to health care facilities
 - Poor access to high speed Internet connectivity, larger commercial airports
 - Reliant on smaller set of industries for employment
 - Difficult to attract new industries and younger workers and families
- The legalization of cannabis may significantly improve the economies of remote counties such as Trinity, while others (e.g. Siskiyou) have made it clear they are not interested in that industry growing within their borders.

Cross-cutting Issues

- Starting around the year 2000, population growth rates have been much slower in non-metro counties relative to metro counties, and in some cases populations in the former have been shrinking. This poses long-term economic challenges in these areas.
- The decline of permanent jobs in forestry has hit non-metro economies harder than others in those counties with significant forests.
- Many areas of wildland-urban interface and intermix (WUI) occur in more remote non-metro counties. This exposure to possible loss, and limited means of residents, compounds the challenge of managing WUIs and dealing with the aftermath of severe wildfires.
- Most headwaters of municipal water supply systems are in remote non-metro counties. Metro counties thus have a strong interest in what occurs that might affect the water quality or yield of these prized watersheds, including the effects of climate change.
- The growing sustainable energy industry sector (wind, solar) offers increased employment opportunities with stable, well-paying jobs in (often) remote rural areas.
- Volatility in the price of beef cattle, and the continual increasing disparity between the costs to raise cattle and prices received for them, has made it difficult for rural ranchers to survive economically. Nearly all ranchers have a second, non-agriculture based source of income per household.
- Emerging, new wood products, such as mass timber, may create new demands for timber and new wood products manufacturing jobs in rural areas.
- Similarly, expansion of bioenergy plant capacity in rural areas can increase employment for biofuel production and procurement and for plant operations.
- Increased fuels reduction and other forest management activities for improved forest ecosystem health can create additional investment opportunities and rural jobs.

Existing Supportive Plans and Programs

- The California Forest Improvement Program grants (CFIP) provide funding for management on private forest lands, which often occur in non-metro counties.
- The USDA NRCS provides significant programmatic funding to agriculture in rural counties through the Environmental Quality Improvement Program (EQIP) and others.

- USDA Rural Development programs offer a wide variety of means of supporting rural economies through grants and loans. <https://www.rd.usda.gov/ca>
- Land trusts (many active across the state) and the Nature Conservancy are helping non-metro agricultural operations to obtain conservation easements and other funding that often go to land parcels in non-metro counties, providing property tax relief and some income for livestock ranchers.
- The Williamson Act continues to provide property tax relief to agricultural properties including many acres of rangeland. However, unless the state government resumes subvention payments to counties in lieu of their lost tax base, it is likely that this program and benefit to rural farmers and ranchers will soon be gone.

Key Stakeholders and Partners

- The California Association of Resource Conservation Districts (CARCD) is comprised of 95 districts statewide. They are self-funded NGOs and must raise money and obtain grant funding to provide their district with program funding.
- The Rural Counties Representatives of California (RCRC) is a non-governmental organization comprised of 37 member counties across the state. It represents an important voice for the interests of rural counties (all non-metro counties are rural) before the state government agencies. <https://www.rcrcnet.org/>
- The California Rangeland Conservation Coalition also provides rural property owners with a venue to learn about opportunities for funding and rangeland improvement.
- The Board of Forestry and Fire Protection's Range Management Advisory Committee (RMAC) was statutorily created to advise the Board, the Resources Agency, the California Environmental Protection Agency, and the California Department of Food and Agriculture on rangeland resources.
- The recently established [Joint Institute for Wood Products Innovation](#) under the Board of Forestry and Fire Protection. Per its website, the Institute "is dedicated to providing California forest product information, research, and analysis to increase economic drivers for healthy forests. Institute work focuses on long-term ecological and economic sustainability; education and outreach; increased forest resilience, long-term carbon storage, and local economies; and industry retention and development in California."

STRATEGIES AND SUPPORTING ACTIONS

Strategy 8.1 Economic-, employment-, and housing-related Measures

Strategy 8.1.1 Support income stability

Action A - Stabilize revenue streams with industrial diversification, enhanced economic structure, and advantageous characteristics. State and local governments may influence stability and structure through union membership, increases in the minimum wage and limited reliance on single sector employment (Gude, Rasker et al. 2012, Acemoglu and Restrepo 2017, Krueger 2017, Shambaugh, Nunn et al. 2017).

Strategy 8.1.2 Support Provisioning Services

Action A - Incentivize and encourage certified sustainable wood products by global private governance (e.g., FSC). Certification programs provide a direct assessment of forest

management that are generally thought to be ecologically, socially, and environmentally responsible.

Strategy 8.1.3 Maintain wildland and forestry job safety

Action A – Improve understanding of firefighter safety zones.

A-1. Implement pedestrian sensors on firefighters to better understand travel rates over various terrain and slopes.

A-2. Better estimate wildland firefighter escape route travel time to safety and the time it will take for search and rescue to reach an accident victim.

A-3. Use geospatial modeling to identify and evaluate wildfire safety zones using lidar data, which could be used to highlight areas where safety zones could be created using controlled burning and vegetation management.

Action B - Improve health and safety in the timber and forestry industries.

B-1. Logging is the most dangerous job in America, with the highest fatality rates among US civilian workers. Efforts to improve this should include safety management and leadership video trainings via mobile devices.

B-2. Adhere to the most recent Occupational Safety and Health Administration (OSHA) guidelines; which include (1) safety leadership, (2) worker participation, (3) hazard identification and assessment, (4) hazard prevention and control, (5) education and training, (6) program evaluation and improvement.

Strategy 8.1.4 Support Recreation and avoid the commercial development of wildlands.

Action A - Preserve recreational assets by helping to preserve, protect and enhance the state's biological diversity and preserve a high quality of life.

Strategy 8.1.5 Support sustainable residential housing, with stability, affordability and wildfire safety

Action A - Incentivize structure hardening and retrofitting of roofs, walls and decks of existing homes in low density landscapes. Apply the same strict ordinances that city firefighters do when upgrading municipal water infrastructure for fire hydrants and sprinklers.

Action B - Add enough full-time inspectors to conduct building inspections in all communities (not just wealthy) every 3 years.

Action C - Develop a WUI Building Risk Index and theory of wildfire spread in WUI to assess wildfire risk at the building scale.

Action D - Disaster response should include potentially vulnerable populations. When government funding is used in planning, inclusion of these groups should be required.

Action E - Facilitate an increased supply of affordable housing. Tools could include:

E-1. Reduce taxes when purchasing a new home or apartment

E-2. Reduce the down payment by providing land and loans

E-3. Reform disaster relief housing programs to serve those most in need

E-4. Where possible, encourage urban densification and limit urban expansion.

E-5. Like the *Drought Housing Relocation Assistance Program*, similar efforts should be available to those unable to find affordable housing post wildfire.

E-6. Consider cooperative housing by non-profits, where the owner owns the home, but not the land

Action F - Curtail migration of people towards wildfire-prone areas.

F-1 - Institute building codes that require ignition resistant materials for new home and retrofits

F-2 – enable and enforce defensible space inspections, ensuring that property owners manage hazardous vegetation and maintain their properties.

F-3 - Map areas of high wildfire exposure and potentially vulnerable households and populations. Conduct a vulnerability assessment of infrastructure and capital to derive risk. Work with the NASA Socioeconomic Data and Applications Center (SEDAC) operated by the Center for International Earth Science Information Network, and others, to validate and refine approach and results.

Strategy 8.2 Resident-related strategies

Strategy 8.2.1 Develop online tools to identify, estimate and warn vulnerable populations exposure to wildfire smoke.

Action A - Based on those results, estimate adaptive capacity of healthcare institutions to provide care (e.g. training, guidance or access to healthcare).

Action B - Encourage work from home programs to limit exposure to wildfire smoke.

Strategy 8.2.2 Expand medication assisted treatment via telemedicine to combat the opioid crisis, and provide training for state and local law enforcement, to bolster our mental health workforce, educate the public on what they can do to prevent or react to the crisis, and address the availability of a range of treatment options.

Action A – Promote virtual reality applications of pain reduction

Action B – Promote three-phase recovery programs

Action C - Combine psychological therapy with opioid replacement prescriptions

Strategy 8.2.3 Stabilize rural hospitals financially, to maintain 24/7 emergency department services. Model the impact of policies and interventions and evaluate what would be most helpful in improving rural hospitals financial condition.

Strategy 8.2.4 Improve educational opportunities and quality to rural residents. Increase post-secondary access to career and vocational schools, liberal arts colleges, research

universities and community colleges. Seek to understand the motivational choices of residents, which may better inform policy and programs.

Strategy 8.2.5 Build the strength of social capital and community relationships.

Strategy 8.3 Politics- and Governance related strategies

Strategy 8.3.1 Create better access to broadband connectivity for residents of rural areas.

Action A - Streamline and expedite permitting process to address delayed applications.

Action B – Create accurate maps of assets (transmitters etc.) to better facilitate the co-location of critical infrastructure and sharing.

Action C - Embrace a “Dig Once” rule on all federal, state and public rights of way adjoining roads, requiring the installation of conduits for broadband equipment whenever roads are dug up for any reason.

Chapter 9: Water Resources

Forested watersheds and headwater areas are the origin of much of California's water supply and are a critical part of the natural infrastructure. Healthy forests play an important role in the hydrologic cycle, promoting infiltration and maintaining the delivery of high quality water to streams and downstream uses. While headwater forests play a pivotal role in maintaining the delivery of

high quality water, these resources have been undervalued. Water quality is generally good on forested streams (①9.1), but there are many threats. In many watersheds, the type of management and fire exclusion has resulted in forests that are either at risk from fire and pests, or have already been impacted. Forested watersheds are also at risk from drought (①9.4), warming climate (①7.1), and declining snowpack (①9.2). Post-fire erosion from large wildfires and other disturbances can negatively impact water quality, downstream water storage, and other critical water infrastructure. The impact of grazing on water quality has mixed findings, but water quality indicators (①9.1) on rangelands show a higher level of impairment.

INDICATORS

- ①9.1 Water Quality
- ①9.2 Snowpack
- ①9.3 Spring Runoff
- ①9.4 Cumulative Water Deficit

Indicators and key findings suggest that California's climate and hydrology are changing, resulting in warmer annual temperatures, with increases in both maximum and minimum temperatures (①7.1). In turn, warmer temperatures are causing declining snowpack (①9.2) and altering the timing of spring runoff (①9.3). As snowmelt begins to occur early in the spring we are likely to see longer dry periods and more frequent severe fire weather (①4.5). In addition, as evapotranspiration from forest vegetation increases under warmer temperatures, vegetation becomes further stressed by prolonged drought conditions (Luce, Pederson et al. 2016). When these climatic factors are combined with high risk stands, forests face increased threat from both pests and severe wildfires. Changes in the frequency and severity of large scale disturbance can influence the timing and delivery of water downstream, which may necessitate changes in water storage, increased storage capacity, and altered flood management practices. Higher levels of plant evapotranspiration due to higher temperatures may reduce the runoff available for water supply.

KEY FINDINGS

①9.1 Indicator: California Stream Condition Index

- ①🔑 Forest streams are mostly in a good biological condition. There are a higher percentage of forest streams in good condition (62%) compared to rangelands (34%).
- ①🔑 California Stream Condition Index scores show some interannual variability across the monitoring period (2000–2012), but forest streams consistently show that 50% or more streams are in good condition in the more recent sampling period.
- ①🔑 Rangeland streams show a greater percentage of streams in poor (21%) and very poor (21%) condition.

①9.2 Indicator: April 1st Snow Pack

- ①🔑 The data shows high interannual variability, but no strong historical trend.
- ①🔑 Snowpack has declined substantially under the current drought conditions (2012–2015), but rebounded in 2016.

①9.3 Indicator: Spring Runoff

- ①🔑 Spring runoff shows a steady decline from 1997–2015.
- ①🔑 Trends show a decreased water availability in summer months for downstream uses.
- ①🔑 Decreased water availability likely increases moisture stress for vegetation, and drier vegetation can lead to more extreme wildfire behavior, and greater insect mortality.

①9.4 Indicator: Cumulative Water Deficit (CWD)

- ①🔑 The increasing trend in CWD since 1980 suggests drier conditions and increased moisture stress.
- ①🔑 Cumulative Watershed Deficit for the recent drought (2012–2015) has created moisture deficits that are as or more severe than the drought of the 1970s.
- ①🔑 Many forested watersheds that provide water supply for the state have experienced high CWD.
- ①🔑 There appears to be a positive relationship between CWD and elevated levels of tree mortality.

Strategies

Forests and forestry practices can help protect, restore, and sustain water quality, water flows, and watershed health. Healthy urban and rural forested watersheds absorb rainfall and snow melt, slow storm runoff, recharge aquifers, sustain stream flows, and filter pollutants. Assessments should identify watersheds where continued forest conservation and management is important to the future supply of clean municipal drinking water, or where restoration or protection activities will improve or restore a critical water source. Resource strategies should include actions for managing and conserving these priority watersheds for water quality and supply, and other ecosystem services (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goals of these strategies are to maintain and enhance water supply and water quality in forested watersheds that support a broad range of downstream uses.

National Goal Supported: Enhancing Public Benefits from Forests

Montreal Protocol Supported:

MPC-4: Conservation and Maintenance of Soil and Water

State Assessment Theme: Water Resources: Assessing Water Quantity and Water Quality

Strategies Overview

State Issues

- Headwater forests are at risk from increasing severity of wildfire, elevated tree mortality. Investment in headwater forests is critical to maintain high quality water supply for downstream uses., including municipal, agricultural, and industrial uses.
- Climate change is altering snowpack and forest hydrology.

Cross-Cutting Issues

There are a number of cross-cutting issues that include:

- Protecting Ecosystems from Wildfire Threat – High severity wildfire can directly affect water quality. Priority areas for protecting ecosystems from wildfire threat have substantial overlap with priority areas for protecting water resources.
- Meadow and Riparian Forest Restoration – Restoring riparian forests and meadows can enhance water supply, water quality, flood protection, wildlife habitat and carbon sequestration.
- Development – Increased development in forested landscapes can impair the quality of water from source watersheds.
- Climate Change – Climate modification is expected to lead to substantial declines in Sierra snowpack; this in turn will affect the timing and delivery of water from upper elevation watersheds.

Existing Supporting Plans and Programs

Supporting plans include:

California Water Plan Update, California Water Action Plan (Natural Resources Agency, CalEPA, and Department of Food and Agriculture), Board of Forestry and Fire Protection's Effectiveness Monitoring Committee's Strategic Plan, Regional Water Quality Control Board Basin Plans, California Rangeland Water Quality Management Plan, USFS Region V Water Quality Management Program, including its Best Management Practices Evaluation Program.

Existing programs that support the water quantity and quality strategies include:

- Board of Forestry and Fire Protection – Forest Practice Implementation and Effectiveness Monitoring Program (FORPRIEM), evaluates the implementation and effectiveness of selected Forest Practice Rules on protecting water quality.
- Surface Water Ambient Monitoring Program (SWAMP) – Responsible for assessing water quality in California's entire surface waters (http://www.swrcb.ca.gov/water_issues/programs/swamp/).
- DWR river runoff, precipitation and snowpack monitoring.
- Regional Water Quality Control Board Waiver monitoring programs.
- TMDL implementation plans for sediment and water temperature listed waterbodies.
- CDFW's Fisheries Restoration Grant Program.
- NOAA Fisheries Restoration Center.
- California Forest Improvement Program (CFIP)
- California Forest Stewardship Program – promote stewardship of private forest lands.
- California Wildlife Conservation Board

- NRCS Programs – Environmental Quality Incentives Program (EQIP), Wetlands Reserve Program (WRP), and Wildlife Habitat Incentives Program (WHIP). (<http://www.ca.nrcs.usda.gov/programs/>)
- USFS–Region V Best Management Practices Evaluation Program.
- CDFW – Lake and Streambed Alteration Program.
- Board of Forestry and Fire Protection’s Forest Practice Rules
- Board of Forestry Effectiveness Monitoring Committee
- Instream Monitoring Programs

Key Stakeholders and Partners

USFS Region V; USFS-PSW; US EPA; NOAA Fisheries; USFWS; NRCS; SWRCB; RWQCBs, DFG; CAL FIRE; Conservation-CGS; UC Berkeley, HSU, Cal Poly SLO, industrial and non-industrial timber companies; Sierra Nevada Conservancy and other land conservancies; DWR Integrated Regional Water Management; NGOs including National Fish & Wildlife Foundation; Sierra Nevada Alliance, Pacific Forest Trust, and other community based wildfire and watershed protection groups.

Strategies and Supporting Actions

Strategy: 9.1. Promote Watershed Protection and Watershed Restoration.

Action A – Promote restoration, enhancement and management of mountain meadows to enhance timing and delivery of runoff. This includes financial and technical assistance to private landowners willing to voluntarily restore mountain meadows and riparian habitats. Utilize easements and provide other financial incentives as needed where livestock are temporarily or permanently excluded from grazing in mountain meadows.

Action B – Promote restoration of riparian forests to enhance flood protection, water quality, recovery of aquatic habitat, terrestrial wildlife habitat and carbon sequestration.

Action C – Increase public awareness of existing landowner incentives, and expand incentives where possible; for restoration projects in high priority watersheds that maintain and enhance high quality water supply to downstream users.

Action D – Enhance watershed protection through the strategic placement of fuel reduction projects in high priority water supply watersheds.

Action E – Implement resource management strategies for forest management consisted with those stated in the California Water Plan Update – Forest Management Strategies.

Action F – Increase funding for monitoring runoff in upper elevation watersheds that are a priority for water supply.

Action G – Conduct necessary research to improve understanding of wildfire and pre-fire management effects on forest hydrology.

Strategy: 9.2. Improve Water Quality through Implementation of Best Management Practices and Monitoring in High Priority Watersheds.

Action A – Implement strategies from the Board of Forestry and Fire Protection's Policy Statement that protect beneficial uses of water (Criterion 4, Soil and Water Quality; objective 2). Board of Forestry and Fire Protection Strategic elements are paraphrased as:

- A-1. Continue support for watershed assessments using common watershed models and risk assessment methods to advance understanding of cumulative watershed effects.
- A-2. Continue monitoring, especially to link in-stream conditions to hillslope processes.
- A-3. Increase options for long-term plans by forest and range landowners and connect plans to ease regulatory process requirements and improve resources protection at the plan level.
- A-4. Foster collaboration between regulatory agencies, the general public, and private landowners including integrating Timber Harvest Plan review and rules and Total Maximum Daily Load requirements.
- A-5. Maintain funding and increase landowner incentives for restoration projects and maintain support for urban stream restoration.
- A-6. Use the Demonstration State Forests as a venue for testing and demonstrating watershed assessment approaches and restoration techniques.
- A-7. Conduct focused research on the dynamics of fish populations and their linkages to instream conditions and land uses.

Action B – Implement recommendations in the Board of Forestry and Fire Protection Effectiveness Monitoring Committee's Strategic Plan to evaluate the implementation and effectiveness of forest practices related to water quality.

Action C – Continue state and federal support for long-term watershed studies that lead to increased understanding of cumulative watershed effects; changes in annual water yield, peak flows, and summer low flows; changes to water quality parameters; and impacts of current forest management practices to key aquatic habitat metrics.

Action D – Support implementation and evaluation of the effectiveness of Best Management Practices (BMPs) for protection of water quality on both private and federal lands.

Chapter 10: Wildlife Habitat

Indicators and key findings suggest that California's plant and animal species are experiencing increasing pressure. Of the approximately 7,500 native plants and animals found in California, 252 are listed as threatened or endangered under the California and 320 under the Federal Endangered Species Act (CESA and ESA respectively), for 408 total listed (⑩10.1). A more comprehensive account of at-risk species in the state is provided by the California Department of Fish and Wildlife (CDFW) under the Species of Greatest Conservation Need (SGCN) (California Department of Fish and Wildlife 2015), which consists of 1,153 species (264 invertebrates, 414 fish and wildlife and 475 plants). Most of the SGCN are identified as threatened, endangered, rare, endemic or vulnerable under ESA, CESA or other dependable species status ranking systems by NatureServe, California Native Plant Society and CDFW.

INDICATORS

- ⑩10.1 Species at risk
- ⑩10.2 Habitat Structure
- ⑩10.3 Habitat Degradation
- ⑩10.4 Habitat Vulnerability to Climate Change
- ⑩10.5 Protected Habitat

The extent, structure and quality of forest and rangeland habitats have significantly changed due to human activities such as development, road construction, exotic species introduction and invasion, and land management (see Chapter 6 - Population Growth and Development Impacts). Currently only around 13% of California conifer forests are over 200 years old (⑩10.2), and these are mostly high elevation types such as western white pine and lodgepole pine, as defined in the California Wildlife Habitat Relationships (CWHR). Vegetation types that have gone through extensive human disturbances (low intactness) are primarily coastal, foothill, and riparian types. They include Valley Foothill Riparian (71% low intactness), Desert Riparian (49%), Valley Oak Woodland (43%), Coastal Scrub (33%), Montane Riparian (26%), Coastal Oak Woodland (26%), and Redwood (21%) (⑩10.3). These habitat alterations have inevitably influenced the livelihood of many forest and rangeland obligate species.

Climate change imposes an additional threat to wildlife habitats. Modelling efforts under different climate scenarios show that the quantity, quality, and spatial distribution of habitat types will likely change. For example, red fir is projected to be severely impacted in its extent under the hotter and dryer climate model. Within the Sierra regions, oak woodlands show the highest potential loss in the southern counties both under the hotter and dryer and the warmer and wetter models (⑩10.4). In drier regions, Joshua Tree is projected to be severely impacted in its extent under the two climate models, and impacts are already being observed.

Total extent of forest and rangeland have consistently been around 80 million acres since CAL FIRE's Fire Resource and Assessment Program started reporting in 1979. Just under 58% of these habitats are protected from conversion through acquiring fee title by public or conservation organizations, or more recently by a growing trend of purchasing conservation easements on private lands (⑩6.4). The least protected habitat types under CWHR are Blue Oak and Valley Oak Woodland (17% each), Annual Grassland (20%), Blue Oak Foothill Pine (24%) and Valley Foothill Riparian (36%). Within the conifer forests, they are Redwood (30%), Ponderosa Pine (49%) and Montane Hardwood Conifer (49%) (⑩10.5). There are also habitat types that have experienced major habitat losses over time, for example, Fresh Emergent Wetland (68% protected) and Coastal Scrub (42% protected). Countless native species,

including SGCN, rely on these diverse ecosystems; thus, protecting what remains as an attempt to ensure their survival is a highest conservation priority of the state.

CAL FIRE supports the opportunities and conservation priorities outlined in the California Wildlife Action Plan 2015 Update. In addition, other opportunities include partnership agreements, state regulations and laws, private landowner incentives, and conservation efforts. A more detailed list of conservation opportunities is found at the end of the chapter.

KEY FINDINGS

Key findings and indicators are grouped below for the five chapter topics.

Species at Risk

🔑 Of the 32 salmonid populations recognized in California (21 anadromous and 11 non-anadromous), one is now extinct and 14 others are listed as threatened or endangered under ESA and/or CESA. The most threatened salmonid populations in California are the southern steelhead, the winter-run Sacramento River Chinook salmon and the Central California Coho salmon. Forty-five percent of California salmonid populations are likely to be extinct in the next 50 years. In 100 years, 23 of the remaining 31 populations (74%) are likely to be extinct if present conditions continue (Moyle and Lusardi 2017).

① 10.1 Indicator: Number of Threatened and Endangered Species Listed Under the California (CESA) and/or Federal Endangered Species Act (ESA)

① 🔑 Since 2010, 15 species have been added to the list of threatened or endangered species, mostly under ESA. The amphibian taxa saw the largest increase in listing with 7 new species added. Currently, 252 species are listed as threatened or endangered under the CESA and 320 under the ESA, with 408 total species listed (2017).

① 🔑 California Department of Fish and Wildlife has compiled a list of Species of Greatest Conservation Need (SGCN) that includes threatened, endangered, rare, endemic, vulnerable and species of special concern, under ESA, CESA or other dependable species status ranking systems by NatureServe, California Native Plant Society and CDFW. Of the 1,153 species on the list, there are 264 invertebrates, 414 fish and vertebrates, and 475 plants.

Habitat Structure

① 10.2 Indicator: Forest Stand Age Class by Ownership

① 🔑 Late seral (201+ years stand age): Nearly 32% of public reserve conifer land is late seral stage defined as being over 200 years old, and 12% of public non-reserve conifer land is late seral. Nonindustrial private conifer land is made up of less than 3% stands over 200 years old, and well under 1% on forest industry land. Western white pine (62%) and lodgepole pine (40%) currently have the largest proportion of older stands.

① 🔑 Mid-late seral (161-200 years stand age): Public reserve lands have the largest percent of mid-late seral tree stands between 161 and 200 years old (13% for conifer and 6% for hardwood lands). For forest industry, the percentages are 1% of conifer and 1% of hardwood.

- ① 🗝️ Early seral (<80 years stand age): 67% of California's conifer stands and 73% of hardwood stands on forest industry forestland are less than 80 years old. Nonindustrial private lands are comprised of 52% conifer stands under 80 years old and 61% hardwood stands under 80 years old. On public land, 25% of the conifer stands and 49% of hardwood stands are less than 80 years old.

Habitat Degradation

- ① 10.3 Indicator: Terrestrial Intactness of California Wildlife Habitat Relationships (CWHR) Types Based on Human Impacts
 - ① 🗝️ Vegetation types with the lowest percentage of intactness are primarily coastal, foothill and riparian vegetation types such as Valley Foothill Riparian (71% low intactness), Desert Riparian (49%), Valley Oak Woodland (43%), Coastal Scrub (33%), Montane Riparian (26%), Coastal Oak Woodland (26%), and Redwood (21%).
 - ① 🗝️ By this measure, the Central Valley (65% low intactness), Central Coast (34%) and South Coast (43%) ecological regions (i.e., USFS Bailey's classification system) have the highest level of habitat degradation while the Great Basin (79% high intactness), South Sierra (75%) and South Interior (68%) have the highest level of intactness.
 - ① 🗝️ Desert and high elevation vegetation types (Desert Scrub, Low Sage, Joshua Tree, Pinyon-Juniper, Palm Oasis, Alpine Dwarf Shrub, Aspen, Lodgepole Pine, Red Fir and Subalpine Conifer) have the most intactness, all with over 75% of habitat extent in the high intactness class.

Habitat Vulnerability to Climate Change

- 🗝️ We can assume that habitat, species, community composition, and spatial distribution will continue to change into the future under a changing climate. With the understanding that there is uncertainty in climate change models, they still provide valuable information for systematic conservation planning that can be adjusted through adaptive management, incorporating future findings from monitoring and research.
- ① 10.4 Indicator: Projected Impacts of Climate Change on the Extent of California Wildlife Habitat Relationships (CWHR) Types
 - ① 🗝️ Habitat types such as Joshua Tree are projected to have severe impacts under the two hotter/drier and warmer/wetter climate scenarios.
 - ① 🗝️ For habitat types such as Red Fir, the future climate scenario has major implications in terms of loss.
 - ① 🗝️ Habitat types such as Coastal Scrub are projected to have minimal impact in extent under either climate scenario. However, there may be qualitative impacts, for example, changes in species composition within Coastal Scrub habitats.
 - ① 🗝️ In addition to losses, changing climate provides opportunities for habitat types to "migrate" into areas that were climatically unsuitable in the past. Whether these opportunities are realized will depend on the adaptability of the habitat types to these

new sites, for example, intactness, soil conditions, competition with other vegetation, and disturbance regimes found in the potential sites.

Protected Habitat

① 10.5 Indicator: California Wildlife Habitat Relationships (CWHR) Types Protected from Conversion

- ① 📌 High elevations types such as Red Fir, Subalpine Conifer and Aspen are at least 85% protected in almost all regions where they are found.
- ① 📌 Nearly 80% of hardwood woodland habitat types are on private lands and thus have low protected status, e.g., Blue Oak Woodland (17%), Valley Oak Woodland (17%), Foothill Pine (24%) and Coastal Oak Woodland (35%). Hardwood forests are less protected than other vegetation types in all regions across the state, with the exception of Aspen and Joshua Tree. The Central Coast and Central Valley regions have the lowest percentage of protected hardwood woodlands.
- ① 📌 Of the 65% of conifer forestlands that are protected, most are managed by the US Forest Service. Some conifer forest habitats are not as well protected, depending on the specific regions. Coastal Redwood habitat is almost exclusively found in the North Coast but only 28% is protected, and less than 50% of Ponderosa Pine habitat is protected in the North Coast, North Interior and North Sierran regions.

Strategies

Protection, conservation, and restoration of forested wildlife habitat are critical to maintaining and enhancing the rich biodiversity of our nation. Major threats to fish and wildlife habitat include the patchwork of public-private ownership, threats associated with urbanization and uncharacteristic wildfire. Assessments and resource strategies should identify forest landscapes that represent or contribute to viable wildlife habitats (contiguous or connected), contain high species richness, endemism, and/or that represent core habitat for focal conservation species (i.e. species of concern, threatened and endangered species or keystone species that are representative of a healthy ecosystem). Assessment and resource strategies should incorporate State Wildlife Action Plans. Resource strategies should include actions for conserving and enhancing habitat attributes in priority landscape areas (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goals of these strategies are to address the broad need to protect and conserve wildlife and fish habitat to enhance high species richness, endemism, and core habitat. The strategies are also intended to address the more focused issue of restoring wildfire-impacted lands and reducing risk of wildfire impacts on protected lands that the priority landscape has identified. These strategies also address goals identified at the national and state level, as noted below.

National Goal Supported: Enhancing Public Benefits from Trees and Forests

Montreal Protocol Supported:

MPC-1: Conservation of Biological Diversity

State Assessment Theme: Conserve Fish and Wildlife Habitat

Strategies Overview

State Issues

- The Federal Endangered Species Act 2019 rule changes will decrease the ability to utilize regulations to protect species, and the list will become a poor indicator of endangered species.
- Marijuana cultivation has increased with legalization, but enforcement of regulations has not been able to keep up with farming, so illegal forest conversion, chemical application, and stream alteration is occurring at an extreme rate.
- Population growth continues to contribute to more habitat loss from development, urban growth, and pollution.
- Species vulnerability continues to increase as climate becomes more unpredictable. A number of California forest flora and fauna are already showing a decrease in populations due to climate, and species climate models are predicting that more species will continue to lose habitat and become more vulnerable to loss in the future.

Cross-Cutting Issues

- Wildfire and Forest Pests – Preventing catastrophic wildfire and forest pest threats to maintain healthy ecosystems and restoring the ecosystems if they are negatively affected by these threats is a direct benefit to wildlife.
- Development – Development and fragmentation are rapidly depleting valuable habitat throughout the state. Land management plans must ensure long-term protection of biological resource values in addition to community development and economic growth.
- Climate Change – The effects of climate change will influence plant and animal species distribution, critical resources, and available habitat. Strategies to capture and conserve forest carbon stocks may also simultaneously protect wildlife habitat into the future.
- Water Quality/Quantity – Water quality and quantity can directly affect native fish populations and associated terrestrial plant and wildlife habitat. Watershed protection and improved water quality will benefit fish and wildlife and their habitat.

Existing Supporting Plans and Programs

Supporting plans include:

California Wildlife Action Plan, Landscape Conservation Planning (Natural Community Conservation Plans and Habitat Conservation Plans). California Partners in Flight Conservation Plans, State Fire Plan, National Fish Habitat Action Plan, Bay Delta Conservation Plan, USF&WF Recovery plans, Sierra Nevada Conservancy Strategic Plan, California Desert Renewable Energy Conservation Plan,

State programs:

- California Department of Fish and Game programs include:
 - California Waterfowl Habitat Program (CWHP)
 - California Winter Rice Habitat Incentive Program
 - Cannabis Restoration Grant Program
 - Ecosystem Restoration Program (ERP)
 - Endangered Species Conservation and Recovery Grant Program
 - Endangered Species Conservation and Recovery Land Acquisition
 - Fisheries Restoration Grant Program

- Natural Community Conservation Planning (NCCP) and Habitat Conservation Planning (HCP) Local Assistance
 - Significant Natural Areas Program
 - Regional Conservation Investment Strategies
 - Wetlands Restoration for Greenhouse Gas Reduction Program
 - Conservation and Mitigation Banking
 - Timberland Conservation Program
 - Endangered Plant Program
- California Department of Conservation Programs
 - Division of Land and Resource Protection
 - Sustainable Agricultural Lands Conservation (SALC) Program
- University of California Cooperative Extension (UCCE)
- CAL FIRE programs include:
 - Vegetation Management Program.
 - California Forest Practice Rules
 - Forest Improvement Program (CFIP).
 - Forest and Range Assessment Program.
 - Forest Practice and Anadromous Salmonid Protection Rules
 - Forest Legacy Program
 - California Forest Improvement Program
 - California Forest Stewardship Program
- Inland Wetlands Conservation Program – Administered by the Wildlife Conservation Board to implement the Central Valley Habitat Joint Venture in wetland acquisition and restoration.
- Habitat Conservation Fund – Funding from Proposition 117 to acquire or develop wildlife corridors and trails, and to provide for nature interpretation and other programs which bring urban residents into park and wildlife areas.
- CA Department of Pesticide Regulation – Endangered Species Project.
- Sierra Nevada Conservancy Watershed Improvement Program (WIP) Grant Program (Proposition 1 and 68)

Tribal Programs

- Round Valley Natural Resources Department
- Yurok Tribal Fisheries and Wildlife Programs
- Hoopa Valley Tribe Fisheries and Forestry Departments
- Karuk Department of Natural Resources

Federal programs:

- NRCS – Emergency Watershed Protection (EWP), Healthy Forest Reserve Program (HFRP), Conservation Stewardship program (CSP), Environmental Quality Incentives Program (EQIP), Wildlife Habitat Incentives Program (WHIP), Conservation Reserve Enhancement Program (CREP), Wetlands Reserve Program (WRP). Wetland and stream restoration projects, fuel reduction projects when developed and implemented for fish or wildlife benefit. Regional Conservation Partnership Program (RCPP), Agricultural Conservation Easement Program (ACEP)
- BLM – California Desert Conservation Area Plan (CDCA), Resource Management Plans (RMPs).
- US Fish and Wildlife Service – Endangered Species Program, Partners for Fish and Wildlife Service Program, Safe Harbor.
- USFS - Sierra Nevada Forest Plan.

- USFWS – Working Lands for Wildlife
 - The Wildlife Conservation and Restoration Program – Congressionally authorized federal funding program for wildlife conservation and related recreation and education. While the program has been authorized, it is not currently receiving any funding.

Non-profit organization programs:

- California Native Plant Society's Conservation Program - preserves native plant species and their habitats on public and private lands in California by advocating for the maximum protection of native plants and promoting science-based and ecologically-sound land management practices.
- CalFish - Calfish is the leading source for California anadromous fish and stream habitat data, as well as the standards and tools needed to collect, understand, manage, analyze, and share those data.
- Land conservation and land trust organizations.

Key Stakeholders and Partners

State partners: DFG, CAL FIRE, State Water Resources Control Board, California Department of Parks and Recreation, California Energy Commission, Department of Conservation, California Resources Agency, Wildlife Conservation Board, Cal Trans, Universities, state conservancies.

Native American tribes.

Federal partners: USFS, BLM, NPS, NRCS, USFWS, USGS, Army Corp of Engineers. NGOs.

Strategies and Supporting Actions

Strategy: 10.1. Reduce the loss and modification of habitat that supports wildlife, and maintains California's unique biodiversity.

Action A – Increase land conservation and long-term land protection incentives, particularly focusing on areas of high biodiversity and that contain species of greatest conservation need.

Action B – Target funding to recover sensitive species through improved data collection strategies, conservation planning on private lands, and effectiveness monitoring to validate selected avoidance and mitigation measures.

Action C – Develop a continuous funding mechanism for restoration and maintenance of conservation areas.

Action D – Continue to support funding for increased warden presence and effectiveness throughout the state.

Action E – Map, monitor and effectively eradicate invasive plant and animal species.

Action F – Reduce excessive grazing in montane meadows, aquatic riparian habitat, blue oak woodlands, and bighorn sheep habitat.

Action G – Ensure that hydropower projects provide adequate flow regimes for aquatic species and ecosystems.

Action H - Support and coordinate with other agencies to continue and increase development, implementation, and enforcement of regulatory programs, including monitoring, to ensure that cannabis cultivation and associated activities do not adversely impact fish and wildlife.

Strategy: 10.2. Develop policies and incentives to facilitate better integration of wildlife conservation considerations into local and regional planning and land-use decision making.

Action A – Develop policies and incentives to facilitate better wildlife conservation needs into local and regional land-use planning and decision making. For example, securing sensitive and key linkage habitat and ensuring infrastructure and transportation development avoid sensitive species habitat.

Action B – Implement resource management strategies for wildlife management as stated in the California Wildlife Action Plan (CWAP). Maintain a current CWAP by regularly updating it. (<http://www.dfg.ca.gov/wildlife/WAP/docs>). Continue working toward the completion of the Areas of Conservation Emphasis project.

Action C – Support ongoing vegetation and species mapping, monitoring, technological (GIS and remote sensing) and field data improvements. Continue support to collect baseline inventory and life history information on priority species and their habitats, and maintain current assessments of wildlife species of greatest conservation need.

Action D – Improve public awareness of California’s unique natural resource values and the strategies needed for their protection. Enhance communication, education, and outreach.

Action E – Develop and enhance partnerships that protect and conserve wildlife habitat. Support collaboration between regulatory agencies, communities and organizations in addressing fish, wildlife and rare plant concerns.

Action F – Implement future regional development plans that consider wildlife habitat, fire prevention and long-term maintenance of associated conservation lands.

Action G – Implement planning efforts aimed to establish a system of sustainable habitat connectivity and wildlife corridors, reduce mortality from roadways and increase fish passage. Refine existing large-scale tools, such as DFG and CalTrans California Essential Habitat Connectivity Project, so that they can be used for regional and local planning efforts.

Action H – Utilize an adaptive management approach to optimize decision making in implementing conservation programs by adjusting existing management strategies as information is improved through monitoring and research.

Action I - Develop comprehensive watershed management programs that aim to bring private and public stakeholders together to work cooperatively towards an environmentally healthy watershed. This may include conserving, protecting and

restoring aquatic systems, riparian and sensitive habitat, and identifying and controlling stressors and pollutant sources.

Action J – Evaluate current regulatory framework regarding wildlife and habitat. Recommend legislative changes as necessary.

Action K – Promote agricultural and rangeland management practices that are compatible with wildlife and habitat conservation.

Action L – Consider predicted climate change effects during conservation planning and restoration.

Strategy: 10.3. Sustain healthy forest ecosystems to maintain California's unique biodiversity.

Action A – Include the role of fire in forest and range habitats through managed fire and fire surrogate projects that reduce the potential of high severity wildfire.

Action B – Include minimum impact fire suppression tactics to reduce environmental degradation to natural resources during fire suppression activities.

Action C – Re-establish the landscape to support appropriate fire regimes: reduce fuel accumulation through mechanical and prescribed fire treatments, while minimizing loss of property and life.

Action D – Maintain mixed age, multi-story and ecologically complex forests through landscape management, planning and protection policies, developing old-growth characteristics in forest and plantations and minimizing catastrophic fires to enhance biodiversity.

Action E – Increase environmental monitoring and testing of herbicide application in forest management practices and enforce herbicide and pesticide regulations and oversight under the US EPA and California Department of Pesticide Regulations (DPR).

Action F – Improve building codes for new and expanding communities in fire-adapted landscapes to be more fire compatible in order to allow the state to reduce the need for fire suppression.

Action G – Ensure that the best available science is used to develop timber harvesting plans and that they consider cumulative impacts to protect aquatic ecosystems and conserve wildlife habitat for each watershed.

Action H – Continue to support studies intended to better understand the effects of vegetation treatments in regards to wildfire risk, forest health, wildlife, and water quality, such as the Sierra Nevada Adaptive Management Project.

Action I - Encourage self-sustaining systems that can adjust to unpredictable changes, including climatic related changes, through adaptive management planning.

Action J - Work to avoid salmonid population extinctions by prioritizing protection and restoration of mountain meadows, floodplains, coastal lagoons, estuaries, groundwater,

springs, spring-fed rivers and the most fully functioning, intact watersheds and river ecosystems. Continue support of working landscapes and management models that integrate innovative, science-based solutions that influence statewide policy relating to salmonid conservation. Bring together stakeholders, agency partners, and diverse communities of fishery, farming, ranching, and commerce to improve fish passage in historical and new spawning and rearing habitats by removing dams and other barriers, improving genetic management, and reducing gene flow between hatchery and wild salmonids.

Action K - Utilize species distribution and climate models to help inform land management and policy decisions and to maintain insight into mechanisms and awareness of the potential magnitude of climate change effects.

Chapter 11: Reducing Community Wildfire Risk

In California, severe fire years can potentially lead to the loss of lives and thousands of structures, and the historical trend shows the problem is getting worse (①11.1). This is consistent with trends from the wildfire chapter (Chapter 4) - increasing wildfire activity (①4.3) and severity (①4.2).

Development patterns have created a fire environment where about 3 million housing units are within Fire Hazard Severity Zones (FHSZ) and are potentially at risk (①11.2). This includes 2.2 million housing units within the Wildland Urban Interface (WUI), 83% of which are in dense Interface, and 17% of which are in more sparsely populated Intermix (①11.3). In addition, 67% of Interface and 73% of Intermix housing units are within High or Very High FHSZ.

INDICATORS

- ①11.1 Structure Loss
- ①11.2 Housing Units by Hazard Class
- ①11.3 Housing Units in WUI
- ①11.4 Community Planning

The National Cohesive Wildland Fire Management Strategy includes a goal of creating Fire Adapted Communities, which recognizes the importance of various programs and actions such as community planning, land use planning, education programs, and homeowner responsibility. Communities are encouraged to take collective action to analyze their unique fire environment, identify appropriate solutions, and commit resources to mitigate risk and raise community awareness. Two ways this can be accomplished are by creating a Community Wildfire Protection Plan (CWPP), or by becoming a Firewise community. Currently, of 1,338 communities identified as Communities at Risk (CAR), 66% (881) are covered by a CWPP (individual, regional or countywide) and/or are recognized by the Firewise program (①11.4). Numerous other communities are at various stages of CWPP development.

The CAL FIRE Land Use Planning Program and the Board of Forestry and Fire Protection work with local government to address wildfire risk as part of the safety element in city and county general plans, as required in Government Code 65302. Land use planning includes considering wildfire risk in the location, arrangement, and composition of new development. There are opportunities to reduce overall fire risk through new development that meets current code and standards for fire resistive construction, infrastructure upgrades such as increased roadway and water flow standards, and fuel modification requirements.

Additional components of community safety are education programs such as Ready, Set, Go!, and individual homeowners taking responsibility to reduce their individual risk. A recent sample of almost 19,000 CAL FIRE defensible space home inspections indicates that 76% passed on the first visit; within Firewise communities the pass rate increased to 84%.

KEY FINDINGS

①11.1 Indicator: Number of Structures Destroyed by Wildfire Annually

- ① Since 1989, there were seven years in which there was a loss of more than 900 structures (residences, commercial properties, outbuildings) in CAL FIRE/Contract County Direct

Protection Areas (DPA). Five of these years were in the second (more recent) half of this period. In bad fire years, this number can exceed 5,000, as in 2003.

- ① In all jurisdictions, the top 20 most damaging fires on record destroyed 18,143 structures. More than half of these losses occurred since 2000.
- ① The National Fire Information Reporting System has complex requirements for reporting structure loss due to wildfire. Structure losses on lands protected by local agencies are not always reported.

①11.2 Indicator: Housing Units by Fire Hazard Severity Zone (FHSZ) Class

- ① In 2010, in all counties, about 3 million housing units (HU) were in FHSZ and potentially at risk from wildfire. This includes about 1.2 million HU (41%) in the Very High class.
- ① Over 460,000 HUs were added within FHSZ between 2000 and 2010. This includes 144,000 HU added to the Very High class.
- ① A large proportion of the HU within FHSZ are in the southern portion of the state. The top five counties for FHSZ HU, all in southern California, contain about half of all statewide HU in FHSZ, and 62% of the HU in the Very High class. However, this is clearly a statewide problem – 37 of 58 counties have at least 10,000 HU in FHSZ.

①11.3 Indicator: Housing Units and Wildfire Threat within the Wildland Urban Interface (WUI)

- ① In 2010, in all counties, about 2.2 million housing units (HU) were in WUI, with 17% in Intermix and 83% in Interface.
- ① County development patterns create unique fire risk environments. Urban counties like Los Angeles and Orange tend to have areas of dense development next to unpopulated open space, and HU are primarily in the Interface (97% and 99%). Conversely, numerous counties provide a rural lifestyle that includes low density Intermix dispersed within wildland fuels, where about half of HU are in Intermix (e.g., Butte, Eldorado, Santa Cruz, and Sonoma counties).
- ① The difficulty in protecting HU from wildfire in California is demonstrated by the fact that 67% of Interface HU and 73% of Intermix HU are in High or Very High fire hazard classes.
- ① Statewide, the 2010 WUI footprint is 17.7 million acres, including 1 million acres of Interface, 1.3 million of Intermix, and a 15.3 million acre influence zone.

①11.4 Indicator: Number and Percent of Communities at Risk (CAR) that are Firewise Communities or Covered by a Community Wildfire Protection Plan (CWPP)

- ① There are 1,338 individual communities represented by the Communities at Risk (CAR) list. Of these communities, 66% (881) are covered by a CWPP (individual, regional or countywide) and/or are recognized by the Firewise program. Numerous other communities are at various stages of CWPP development.

Of the CARs communities, 16% (213) are covered by individual CWPPs or the Firewise program. Individual CWPPs typically provide the finest detail for project-level planning; however, many county-level plans are very detailed, while others serve more generally as an umbrella for individual CWPPs.

Strategies

Assessments should identify communities where State and Private programs can substantially mitigate the risk of catastrophic wildfire occurrence and associated risks to human safety and property. Assessments should incorporate existing CWPPs and identify communities in especially vulnerable areas that need a CWPP. Resource strategies should include a plan for effectively addressing those communities that are most at risk (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goal of this strategy is to increase the number of communities directly involved in coordinated wildfire planning and the number of community wildfire protection plans where needed to reduce wildfire risks. This strategy also addresses goals identified at the national and state level, as noted below.

National Goal Supported: Enhancing Public Benefits from Trees and Forests

Montreal Protocol Supported:

MPC-6: Socio-Economic Well-Being

State Assessment Theme: Community protection to reduce costs and losses due to wildfire.

Strategy Overview

State Issues

- Recent trends have shown an increase in the number of ignitions, area burned, and impacts to ecosystems. Ignitions, which are correlated to increased workload, have been on the rise since 2007.
- California has seen a number of catastrophic wildfires in recent years under severe fire weather conditions.
- California has a substantial number of communities and structures that are embedded in WUI and vulnerable to wildfire.

Cross-Cutting Issues

Priority landscapes identified wildfire threats to ecosystem health and community safety, however there are a number of cross-cutting issues that include:

- Emerging Markets – Promote hazardous fuel reduction by improved utilization through forest products, small logs, urban green waste and biomass facilities. Promote the use of state and federal incentives for utilization of biomass harvested during wildfire hazard reduction activities, such as the California Energy Commission's Renewable Energy Program (California Renewable Portfolio Standard), and the USDA's Biomass Crop Assistance Program (BCAP). Improve community understanding about inherent economic incentives and examples for collection and delivery to biomass processing plants where the cost of transportation is defrayed by selling the biomass.
- Wildfire and Forest Pest Threats to Community Safety – Reduce the occurrence of catastrophic mortality from future forest pest outbreaks or wildfire to protect public safety from fire and falling trees through efficient and effective fire protection planning and suppression and financial management.

Existing Supporting Plans and Programs

Supporting plans include:

2018 California Strategic Fire Plan (Unit Plans may serve as CWPP certification for some communities, depending on participation), California State Disaster Mitigation Plan

and local disaster mitigation plans, National Fire Plan, county fire plans and regional fire plans.

Existing programs that support strategies include:

- The California State Forester (CAL FIRE Director) maintains a list of “communities at risk” from wildfire: <https://osfm.fire.ca.gov/divisions/wildfire-prevention-planning-engineering/fire-plan/communities-at-risk/>
- The Fire Safe Council <http://www.firesafecouncil.org/index.cfm> provides information on forming local Fire Safe Councils, homeowner information, educational tools for communities, and serves as a grants clearinghouse.
- County Fire Safe Councils address multiple community wildfire planning needs.
- Local Fire Safe Councils serves individual or clustered communities.
- CAL FIRE Unit fire planning process assesses wildfire risks and assets for CAL FIRE Direct Protection Areas, designs and prioritizes fuel hazard reduction projects while encouraging community involvement.
- National Fire Plan may fund local fuel hazard reduction projects that would reduce risks to adjacent lands under federal ownership.
- Joint Fire Science Program develops information to help communities understand how to accomplish community fire planning.
- The Sierra Nevada Conservancy funds hazard mitigation projects.
- California Tahoe Conservancy.

Key Stakeholders and Partners

Local governments, tribes, County Fire Safe Councils, Local Fire Safe Councils, USFS, BLM, DFG, Sierra Nevada and Tahoe Conservancies and other land conservancies, NGOs including Sierra Nevada Alliance and other community based wildfire and watershed protection groups.

Strategies and Supporting Actions

Strategy: 11.1. Promote formation of Local Fire Safe Councils for priority communities.

Action A – Assess outreach efforts and recommend enhancements.

Action B – Increase funding for community planning administration.

Strategy: 11.2. Promote participation in the National Firewise/USA program.

Action A – Assess current outreach efforts and recommend enhancements.

Strategy: 11.3. Establish a statewide comparative database of community wildfire planning.

Action A – Determine key indicators needed for monitoring.

Action B – Develop a web-based and spatially-enabled information system for knowledge sharing.

Strategy: 11.4. Reduce wildfire risk to communities.

Action A – Increased inspection and compliance of vegetation clearance around homes.

Action B – Invest in home hardening to reduce wildfire risk

Action C – Continued investment in fuel reduction projects within the Wildland Urban Interface

Action D – Ensure that CAL FIRE Unit Fire Plans are updated regularly and include local participation in their development.

Chapter 12: Renewable Energy

Based on current trends (①12.1), meeting the Governor's Renewables Portfolio Standard (RPS) mandate (SB350) that utilities procure 50% of their electricity from renewables by 2030 will require a

continuing major expansion of solar and wind energy production. Where additional facilities are located, and how they are configured and maintained, will determine the level and type of impacts on forest and range operations and ecosystems. Public policies can affect the balance of large solar and wind facilities versus small-scale use for homes and businesses, and whether large projects are sited on leased public lands or private lands with lower ecological values (e.g., marginal agricultural lands).

INDICATORS

①12.1 Renewable Energy

①12.2 Biomass Energy

Current trends (①12.2) also suggest that biomass energy will not be a significant contributor to renewable energy expansion. Declining natural gas prices and cost reductions for solar and wind technologies (Mohan 2015, Kaplan 2016) have made biomass less competitive, leading to a decline in the number of biomass facilities and total megawatts produced (①12.2). At least 15 biomass facilities have closed or have contracts that expire in 2016. On the other hand, about 9 small biomass plants are in various stages of development because of SB1122 (2012), known as the Bioenergy Market Adjusting Tariff, or BioMAT. SB 1122 provides support for plants that supply 3 MW or less to the grid and use for feedstock forest residues from sustainable forest management, fire threat reduction, or defensible space clearance activities. If the maximum 50 MW capacity from small plants is ever realized, it is roughly equivalent to a single large plant, but could be important in those localities for improving forest health, reducing fire risk, and supporting rural economies.

Biomass energy has been an important factor for maintaining healthy forests, reducing fire risk, supporting rural economies, and reducing greenhouse gas and black carbon emissions. The future of forest biomass in the energy sector depends in part on federal and state policy and legislation that considers the non-monetary benefits of biomass power, continued technological improvements to improve competitiveness versus other energy sources, and emerging potential uses such as advanced biofuels (e.g., renewable diesel) that could be part of the state's Low Carbon Fuel Standard program.

KEY FINDINGS

①12.1 Indicator: Contribution of renewable energy sources to California electricity generation

- ①👉 California appears to be on schedule to meet 2020 RPS targets, mainly due to increases in the contributions of wind and solar. Over the period 2010-15, the contribution to state total system power from wind power increased 78%, and solar increased over 1700%.
- ①👉 Based on current trends, meeting RPS targets in the longer term will require a continued major expansion of wind and solar energy. Where additional facilities are located, and how they are configured and maintained, will determine the level and type of impacts on forest and range operations and ecosystems.

①12.2 Indicator: Contribution of forest biomass to California electricity generation.

- ①🔑 There is a downward trend for electricity generation (17% decline) as well as number of facilities that primarily utilize forest biomass (24 to 17) over the 2001 to 2015 period.

Strategies

Assessments should identify forest landscape areas where there is a real, near term potential to access and supply traditional, non-timber, or emerging markets such as those for biomass or ecosystem services. These might be areas where necessary infrastructure currently exists, is planned or developing, where group certification of landowners has created market supply aggregation potential, or where retention and management of forest cover presents a money-saving alternative to an engineered fix – such as a water filtration facility. Strengthening and developing new market opportunities for forest products and benefits provide incentives for forest stewardship and conservation (excerpted from the [US Forest Service State and Private Forestry Farm Bill Requirement and Redesign Strategies](#)).

GOALS: The goals of these strategies are to facilitate the sustainable development of renewable energy resources from forest and rangelands to lower greenhouse gas (GHG) emissions, contribute to local economic development, reduce fuel hazard, improve ecosystem health, and minimize negative environmental impacts.

National Goals Supported: Enhance Public Benefits from Trees and Forests, Conserve Working Forest Lands, protect Forests from Harm

Montreal Protocol Supported:

MPC-6: Maintenance and Enhancement of Long-Term Multiple Socio-economic Benefits to Meet the Needs of Societies

MPC-2: Maintenance of Productive Capacity of Forest Ecosystems

MPC-3: Maintenance of Forest Ecosystem Health and Vitality

MPC-5: Maintenance of Forest Contribution to Global Carbon Cycles

State Assessment Theme: Renewable Energy

Strategies Overview

State Issues

- Forest biomass as a form of renewable energy competes with other types of renewable energy that are less expensive.
- The relatively small number of plants to process forest biomass increases costs and reduces utilization of forest biomass.
- There are some public concerns over sustainability of using forest biomass for energy. In addition, the location of biomass plants (existing and potential new facilities) can raise air quality concerns for nearby communities.
- Meeting Renewables Portfolio Standard (RPS) targets while minimizing the impact of renewable energy development on forest and range ecosystems
- Developing an energy distribution system that is inclusive of various renewable energy sources, able to meet the continuous need of consumers, and minimizes the potential for contributing to fire risk.

- Balancing the need for economic growth and jobs from both renewable and nonrenewable energy sources versus their relative environmental impacts
- Sustaining and enhancing various funding sources used to support research, training, and education programs that advance more effective development and deployment of renewable energy sources

Cross-Cutting Issues

- Balancing demand for renewable energy facilities on forests and rangelands to reduce GHGs and contribute to local economies, versus environmental impacts such as on wildlife habitat
- Proper location and design of new projects through a collaborative process
- Economic viability of the biomass energy sector and ability to facilitate projects that reduce wildfire risk and improve forest health.
- Continued research to improve the competitiveness of renewable energy sources, and minimize environmental impacts.

Existing Supporting Plans and Programs

Supporting plans include:

- Integrated Energy Policy Report (IEPR), published by the CEC bi-annually, provides strategies to address California's pressing energy needs and issues.

Existing programs that support the emerging markets strategies include:

- The Cap-and-Trade Program, administered by ARB, funds the California Climate Investments (CCI) initiative that puts billions of Greenhouse Gas Reduction Fund (GGRF) dollars to work reducing greenhouse gas emissions, while improving the economy, public health and the environment.
- CAL FIRE's Forest Health Grant Program uses CCI funds to support forest health improvement projects.
- CAL FIRE's CCI-funded Fire Prevention Grant Program, supports projects to reduce wildfire risk, increase carbon sequestration in healthy forests, and reduce GHG emissions by wildfires.
- The Bioenergy Feed-in Tariff Program or the Bioenergy Market Adjusting Tariff (BioMAT) provides support for plants that supply 3 MW or less to the grid and use for feedstock forest residues from sustainable forest management, fire threat reduction, or defensible space clearance activities.

Key Stakeholders and Partners

California Energy Commission, California Air Resources Board, California Department of Fish and Wildlife, U.S. Forest Service Region 5 and Pacific Southwest Research Station, Bureau of Land Management; Natural Resources Conservation Service, U.S. Fish and Wildlife service, CAL FIRE, industrial and non-industrial timber companies, ranchers, and other landowners; California Biomass Energy Alliance.

Strategies and Supporting Actions

Strategy: 12.1. Continue to Develop, Implement, and/or Expand Initiatives that Reduce Demand for Large Renewable Energy Projects on Forests and Rangelands

Action A – initiatives to improve energy efficiency, e.g. the CEC 2019 Building Energy Efficiency Standards.

Action B – initiatives to support expanded use of small-scale renewable energy (e.g., rooftop solar) for private residences, businesses, farms, and ranches.

Strategy: 12.2. Promote Continued Research to Improve Guidelines, Standards, and Best Practices to Minimize the Environmental Impact of Large Renewable Energy Projects on Forests and Rangelands

Action A - continue and expand efforts to reduce impacts on wildlife, such as the U.S. Fish and Wildlife Service Land-Based Wind Energy Guidelines

Strategy: 12.3. Support expansion of transmission infrastructure for emerging renewable energy generation from sources such as biomass, wind, hydro and solar in a way that minimizes environmental impact on forests and rangelands.

Action A – Support improvements to California's electric transmission infrastructure, while avoiding new infrastructure development in areas that are environmentally sensitive.

Action B – Encourage a transparent, stakeholder based planning process that includes environmental organizations, regulatory and permitting agencies, public health experts, major transmission providers renewable energy generators, and the public.

Strategy: 12.4. Support Development of Biomass Energy Projects that can Facilitate Projects to Reduce Wildfire Risk and Improve Forest Health

Action A – Continue efforts to establish policies and enact necessary laws to monetize the external benefits of biomass energy (e.g., wildfire risk reduction, improved forest health) through tax credits, price supports, loan guarantees, and other financial incentives.

Action B – Encourage replacement and/or upgrade of existing power facilities with more advanced systems to improve competitiveness, increase efficiency, and reduce environmental impacts.

Action C - Support biomass research and development, including life cycle analysis, best management practices, monitoring and sustainability.

Action D - Support education and training to inform citizens, consumers, and decision makers, and to develop well-trained biomass industry professionals in California.

Appendix A: Proclamations, Plans, Programs, Organizations, and Support

State of California

[Office of the Governor](#)

Proclamations

[Drought State of Emergency \(2014\)](#)

With California facing water shortfalls in the driest year in recorded state history, Governor Edmund G. Brown Jr. proclaimed a State of Emergency and directed state officials to take all necessary actions to prepare for these drought conditions.

[State of Emergency \(Tree Mortality, 2015\)](#)

Due to drought-caused, widespread forest mortality in the Sierra Nevada, Governor Brown issued a proclamation to increase resources and expedite efforts to ascertain which dead trees pose hazards to public safety and should be promptly removed.

[Executive Order N-05-19](#)

In January 2019, citing extreme peril posed by wildfire risk, Governor Newsom issued this executive order directing CAL FIRE to immediately implement projects to protect lives and property. To ensure necessary work can get underway immediately, the order suspended certain requirements and regulations as needed to carry out its directives. CAL FIRE responded by publishing the “45-day Report”.

[Executive Order B-52-18](#)

Executed by Governor Jerry Brown following the devastating 2018 fire season, this order directed the Natural Resources Agency to take all necessary steps to increase the level of fuels treatments on nonfederal lands to at least 500,000 acres per year, consistent with the California Forest Carbon Plan. It directed agencies and departments to reduce the barriers to executing fuels reduction projects, including addressing air quality permitting challenges for prescribed fire projects. It directed CAL FIRE to develop a training and certification program to increase the number of qualified individuals available to conduct prescribed fires. It also directed the Board of Forestry and Fire Protection to establish a Joint Institute for Wood Products Innovation.

Subnational Memorandums of Understanding

[Memorandum of Understanding \(MOU\) on Pacific Coast Temperate Forests](#)

This MOU was executed in December 2018 by California, Washington, and British Columbia. Oregon became a signatory in November 2019. It recognizes that the Pacific Coast’s temperate forests have the capacity to sequester more carbon per acre than any other forest type around the world. The MOU calls out the value of the parties working together to better understand forest carbon dynamics and forest responses to climate change. The partners pledge to work together to explore innovations in fuels management, climate-informed reforestation, forest carbon accounting, opportunities for investments on forests and other natural and working lands

to increase carbon sequestration, forest resilience, multi-benefit forest uses, and status of natural-resource-dependent communities.

California Natural Resources Agency

The California Natural Resources Agency oversees and supports more than 26 distinct departments, conservancies, and commissions. Our Agency Executive Team leads efforts to steward California's natural environment and to advance Governor Newsom's key priorities.

Over 19,000 Californians work within our Agency all across the state to meet our mission "to restore, protect and manage the state's natural, historical and cultural resources for current and future generations using creative approaches and solutions based on science, collaboration, and respect for all the communities and interests involved."

Plans and Strategies

California Water Action Plan

Developed by the Natural Resources Agency in conjunction with the California Environmental protection Agency (CalEPA) and California Department of Food and Agriculture), the California Water Action Plan set forth 10 priority actions that guide the state's effort to create more resilient, reliable water systems and to restore critical ecosystems.

Programs

Timber Regulation and Forest Restoration Program

Assembly Bill 1492 (2012) established the multi-agency, multi-departmental Timber regulation and Forest Restoration Program and created a new funding mechanism for forest restoration and state forest practice regulation via a one percent tax on all lumber and wood products sold at the retail level in the state. The Natural Resources Agency, through the Deputy Secretary of Forest Resources Management, has the lead role in managing this program. The program funds, coordinates, and provides policy leadership for state forest practice regulation on nonfederal lands across the review team agencies (Department of Forestry and Fire Protection, Department of Fish and Wildlife, and Department of Conservation under the Natural Resources Agency; the Regional Water Quality Control Boards under the California Environmental Protection Agency). It also funds forest and fisheries restoration programs. A current major effort is the development of ecological performance measures for forest practices.

Climate Resilience Program

The Natural Resources Agency leads and coordinates the administration's climate adaptation policy and its natural resources climate policy. The initiatives the agency is taking to reduce the future effects of climate change and make the state resilient to current and ongoing impacts of climate change include the Safeguarding California and Climate Change Adaptation Policy, AB 2800 Climate-Safe Infrastructure Working Group, Climate Change Mitigation on Natural and Working Lands, and SB 859 Wood Products Working Group.

[Grant Programs](#)

The Natural Resources Agency directly administers multiple grant programs focused on natural resources, including wildland and urban forests. Specific programs include:

- [Environmental Enhancement and Mitigation](#)
- [Urban Greening](#)
- [Green Infrastructure](#)
- [Recreational Trails and Greenways](#)

Also, various constituent units of the Natural Resources Agency, including departments, conservancies, and boards, administer their own grant programs, many of which can be applied to wildland and urban forests.

Collaborative Partnerships and Organizations

[Forest Management Task Force](#)

On behalf of the governor, the Natural Resources Agency leads the Governor's Forest Management Task Force. Originally established as the Tree Mortality Task Force the Forest Management Task Force now plays a broader and ongoing role related to the health of California's forests. The Task Force was organized to protect the environmental quality, public health, and economic benefits that healthy forests provide to California. The Task Force aims to increase the rate of forest treatments and expand state wood product markets through innovation, assistance, and investment. Advancing forest health project capacity, readiness, and completion statewide aligns with the California Forest Carbon Plan, the goal of which is to establish healthy and resilient forests that can withstand and adapt to wildfire, drought, and a changing climate.

[California Multi-Agency Information and Analysis Network \(CalMAIN\) \(new since 2010\)](#)

Still in development, CalMAIN is a collaborative effort among the Natural Resources Agency, State Water Resources Control Board, the Department of Fish and Wildlife, Department of Water Resources, Department of Conservation, and CAL FIRE. Its intent is to inform natural resource-based management decisions in oak woodland and forested watersheds, including timber harvest, cannabis, water rights and flow assessment, forest health initiatives, cumulative effects of landscape management, and other closely related focus areas.

[California Board of Forestry and Fire Protection](#)

The Board of Forestry and Fire Protection (the Board) is a government-appointed body within the Department of Forestry and Fire Protection (the Department). It is responsible for developing the general forest policy of the state, determining the guidance policies of the Department, and representing the state's interest in federal forestland in California. Together, the Board and the Department work to carry out the California Legislature's mandate to protect and enhance the state's unique forest and wildland resources. The Board is charged with protecting all wildland forest resources in California that are not under federal jurisdiction. These resources include major commercial and non-commercial stands of timber, areas reserved for

parks and recreation, woodlands, brush-range watersheds, and all private and state lands that contribute to California's forest resource wealth.

Plans and Strategies

[Strategic Fire Plan \(updated in 2018\)](#)

This plan forms the basis for assessing California's complex and dynamic natural and man-made environment, and identifies a variety of actions to minimize the negative effects of wildland fire. It establishes "A vision for a natural environment that is more fire resilient; buildings and infrastructure that are more fire resistant; and a society that is more aware of and responsive to the benefits and threats of wildland fire; all achieved through local, state, federal, tribal, and private partnerships." Community protection includes promoting the safety of the public and emergency responders as well as protection of property and other improvements. In addition to the State fire plan, each CAL FIRE unit and many counties have a specific fire plan to address each area's special conditions and concerns.

[Vegetation Treatment Program Environmental Impact Report](#)

The California Vegetation Treatment Program (CalVTP), developed by the Board, is a critical component of the state's multi-faceted strategy to address California's wildfire crisis. The CalVTP defines the vegetation treatment activities and associated environmental protections to reduce the risk of loss of lives and property, reduce fire suppression costs, restore ecosystems, and protect natural resources as well as other assets at risk from wildfire. The CalVTP supports the use of prescribed burning, mechanical treatments, hand crews, herbicides, and prescribed herbivory as tools to reduce hazardous vegetation around communities in the Wildland-Urban Interface (WUI), to construct fuel breaks, and to restore healthy ecological fire regimes.

The Department of Forestry and Fire Protection has the primary responsibility for implementing proposed CalVTP vegetation treatments, though many local, regional, and state agencies could also employ the CalVTP to implement vegetation treatments if their projects are within the scope of the CalVTP (see Final PEIR, Chapter 2, Program Description). The CalVTP will allow CAL FIRE, along with other agency partners, to expand their vegetation treatment activities to treat up to approximately 250,000 acres per year, contributing to the target of 500,000 annual acres of treatment on non-federal lands as expressed in the governor's Executive Order B-52-18.

Committees

[Effectiveness Monitoring Committee \(EMC\)](#)

The Board of Forestry and Fire Protection (Board) established the EMC in 2014 to create an independent committee to assess the efficacy of the Forest Practice Rules and other related laws and regulations. As an advisory body to the Board, the EMC helps implement an effectiveness monitoring program that provides an active feedback loop to policymakers, managers, agencies, and the public by soliciting robust scientific research that addresses the effectiveness of these laws at meeting resource objectives and ecological performance measures related to AB 1492.

The EMC includes representatives from academia, industry, state agencies, consulting firms, and non-profit agencies. This diverse membership reflects representatives from a wide range of California's natural resources, including water quality, geology, wildlife, and forestry. EMC meetings occur quarterly and are open to the public. Meetings are held in Redding, Ukiah, and Sacramento and are available through the webinar.

[Joint Institute for Wood Products Innovation](#)

Established under the Board in 2018 pursuant to Executive Order B-52-18, the Joint Institute for Wood Products Innovation is dedicated to providing California forest product information, research, and analysis to increase economic drivers for healthy forests. Institute work focuses on long-term ecological and economic sustainability; education and outreach; increased forest resilience, long-term carbon storage, and local economies; and industry retention and development in California.

[Range Management Advisory Committee](#)

The Range Management Advisory Committee was statutorily created to advise the Board of Forestry and Fire Protection, the Resources Agency, the California Environmental Protection Agency, and the California Department of Food and Agriculture on rangeland resources. The Committee recently completed a [new strategic](#) plan and [set of objectives](#).

Programs and Rules

[Anadromous Salmonid Protection Rules, 2009 \(Title 14 Sections 916.9 et seq., 936.9 et seq., and 956.9 et seq.\)](#)

The Anadromous Salmonid Protection (ASP) rules were approved by the State Board of Forestry and Fire Protection (Board) during their September 2009 meeting held in Sacramento. The ASP rules are intended to protect, maintain, and improve riparian habitats for state and federally listed anadromous salmonid species.

[Assembly Bill 1504 California Forest Ecosystem and Harvested Wood Product Carbon Inventory Annual Reports](#)

The Fire and Resource Assessment Program (FRAP) develops an annual Forest Ecosystem and Harvested Wood Product Carbon inventory (Forest Carbon Inventory) through collaboration with the U.S. Department of Agriculture (USDA) Forest Service (FS) Forest Inventory and Analysis Program (FIA), USDA FS Pacific Northwest Research Station (PNW) and the University of Montana Bureau of Business and Economic Research (BBER). This annual Forest Carbon Inventory report assists the Board of Forestry and Fire Protection in assessing whether the goal of sequestering 5 million metric ton CO₂e of forest carbon as established under Assembly Bill 32 (Global Warming Solutions Act, 2006) and 1504 are being met. This report also informs the goals identified in the California Forest Carbon Plan (released in 2018 – see below under Plans and Strategies).

[Forest Practice Rules Implementation and Effectiveness Monitoring Program \(FORPRIEM\)](#)

FORPRIEM provides sound science based data on the adequacy of the implementation and effectiveness of Forest Practice Rules specifically designed to protect water quality and riparian habitat. It uses information collected during Timber Harvest Plan (THP) Work Completion Inspections and Erosion Control Maintenance Inspections. FORPRIEM collects information on randomly located road segments, WLPZ segments, and watercourse crossings and uses a random 10 percent sample of THPs throughout the state. It is a continuation of monitoring that was previously completed under the MCR monitoring program, with data collection beginning in the fall of 2007. Data collected in this Program will complement data obtained in the IMMP monitoring program.

[Hillslope Monitoring Program](#)

The Hillslope Monitoring Program has been evaluating the implementation and effectiveness of California forest practices since 1996. The purpose of the Hillslope Monitoring Program is to determine if California's Forest Practice Rules are adequately protecting beneficial uses of water associated with commercial timber operations on nonfederal lands in California. Specific

objectives of the Hillslope Monitoring Program are: 1) implementation monitoring to determine if the Forest Practice Rules (FPRs) related to water quality are properly implemented, and 2) effectiveness monitoring to determine if the FPRs affecting water quality are effective in meeting their intent when properly implemented.

[Interagency Mitigation Monitoring Program \(IMMP\)](#)

This program built on what has been learned in CALFIRE/Board's Hill Slope Monitoring Program, Modified Completion Report, and other monitoring efforts. The IMMP will emphasize data collection and evaluation of high risk plans and the effectiveness of practices implemented at high risk locations within a plan to protect water quality and aquatic habitats. Effectiveness here refers to determining if prescribed measures applied during the plan operations resulted in the intended conditions (MOU Monitoring Workgroup 2005). The Program results will be reported to the Board. The primary monitoring related objectives of the IMMP are: determine how often practices designed to reduce impacts to water quality at high risk locations within a plan are properly implemented (including but not limited to mitigation measures developed by the Registered Professional Forester (RPF) and/or an interagency team); determine how often these practices, when properly implemented, are effective in protecting water quality on non-federal timberlands in California; provide a feedback loop to RPFs, CALFIRE Forest Practice Inspectors, state and federal agency personnel, the public, and others regarding what forestry-related practices at high risk sites require improvement to protect water quality.

[California Department of Forestry and Fire Protection \(CAL FIRE\)](#)

The Department of Forestry and Fire Protection protects the people of California from fires, responds to emergencies, and protects and enhances forest, range, and watershed values providing social, economic, and environmental benefits to rural and urban citizens.

Plans and Strategies

[California Forest Carbon Plan \(new since 2010\)](#)

CAL FIRE led multiple state, federal, and local agencies in the development of the 2018 California Forest Carbon Plan. California's forests, wildlands and adjacent communities have seen significant impacts associated with climate change over the past eight years. This plan lays out recommended actions to achieve healthy and resilient forests that sequester carbon, based on what we know today about our forests and how climate change will evolve in California.

[California Forest and Rangelands: 2010 Strategy Report](#)

Published by the Fire and Resources Assessment Program (FRAP), this was the first edition, for which this document is an update.

[Fire Prevention Planning](#)

Planning incorporates concepts of the National Fire Plan, the California Strategic Fire Plan and individual CAL FIRE Unit Fire Plans, as well as Community Wildfire Protection Plans (CWPPs). Fire Plans outline the fire situation within each CAL FIRE Unit. CWPPs do the same for communities. Each identifies prevention measures to reduce risks, informs and involves the local community or communities in the area, and provides a framework to diminish the potential loss due to wildfire. Planning includes other state, federal and local government agencies as well as Fire Safe Councils. CAL FIRE staff access a variety of

tools in the planning processes including California fire history statistics, fire weather, fire mapping, and Geographic Information Systems (GIS).

[Unit Fire Plans](#)

Individual CAL FIRE Unit Fire Management Plans document assessments of the fire situation within each of CAL FIRE's 21 Units and six contract counties. The plans include stakeholder contributions and priorities, and identify strategic areas for pre-fire planning and fuel treatment as defined by the people who live and work with the local fire problem.

Programs and Rules

[Air Program](#)

In support of its ground forces, the CAL FIRE emergency response air program includes Grumman S-2T 1200 gallon airtankers, UH-1H Super Huey helicopters, and OV-10A airtactical aircraft. 13 air attack and nine helitack bases are located statewide. The airtactical planes fly overhead directing the airtankers and helicopters to critical areas of the fire for retardant and water drops. The retardant used to slow or retard the spread of a fire is a slurry mix consisting of a chemical salt compound, water, clay or a gum-thickening agent, and a coloring agent. While both airtankers and helicopters are equipped to carry fire retardant or water, the helicopters can also transport firefighters, equipment and injured personnel. All CAL FIRE Aircraft are strategically located throughout the state at airbases and helicopter bases. During high fire activity, CAL FIRE may move aircraft to better provide statewide air support.

[California Forest Improvement Program \(CFIP\)](#)

The purpose of the California Forest Improvement Program (CFIP) is to encourage private and public investment in, and improved management of, California forest lands and resources. This focus is to ensure adequate high quality timber supplies, related employment and other economic benefits, and the protection, maintenance, and enhancement of a productive and stable forest resource system for the benefit of present and future generations.

The program scope includes the improvement of all forest resources including fish and wildlife habitat, and soil and water quality. Cost-share assistance is provided to private and public ownerships containing 20 to 5,000 acres of forest land.

[Civil Cost Recovery Program](#)

Wildland fires cost California taxpayers millions of dollars every year. If the California Department of Forestry and Fire Protection's (CAL FIRE) investigation reveals a fire was caused by a violation of law or negligence, the person responsible can be charged criminally, civilly, or both.

[Conservation Camp Program](#)

CAL FIRE is currently authorized to operate 31 Conservation Camps statewide that house several thousand inmates and wards. These camps are operated in conjunction with the California Department of Corrections and Rehabilitation (CDCR). Through these cooperative efforts CAL FIRE is authorized to operate 196 fire crews year-round. These crews are available to respond to all types of emergencies including wildfires, floods, search and rescue, and earthquakes. When not responding to emergencies, the crews are busy with conservation and

community service work projects for state, federal, and local government agencies. Fire crews perform several million hours of emergency response each year, and more on work projects.

Cooperative Fire Protection Program

In a state as large and populated as California, no one emergency response agency can do it all. That is why cooperative efforts via contracts and agreements between state, federal and local agencies are essential in response to emergencies like wildland and structure fires, floods, earthquakes, hazardous material spills, and medical aids. CAL FIRE provides fire protection services to many California citizens through the administration of 146 cooperative fire protection agreements in 35 of the State's 58 counties, 28 cities, 30 fire districts and 23 other special districts and service areas.

California Forest Legacy Program (CFLP)

The purpose of the both the federal (FLP) and the State Forest Legacy Programs (CFLP) is to protect environmentally important forestland threatened with conversion to non-forest uses, such as subdivision for residential or commercial development. To help maintain the integrity and traditional uses of private forestlands, the CFLP promotes the use of permanent conservation easements. These easements provide an approach with which the federal government, in cooperation with state and local agencies, private organizations, and individuals can preserve the rich heritage of private forests. Recent funding (since the previous edition of Strategies) for the CFLP has come in large part from the California Climate Investments (CCI), a statewide program that puts billions of Cap-and-Trade dollars toward achieving the state's climate change goals while also strengthening the economy and improving public health and the environment—particularly in disadvantaged communities.

Cultural Resources Management Program

The purpose of the CAL FIRE Cultural Resources Management Program is to identify and manage archaeological, historical, and tribal cultural resources located within project areas under CAL FIRE jurisdiction and to develop methods to protect these resources from project-related impacts. This is accomplished through regulations, policies and procedures requiring cultural resource surveys of project areas, evaluation of potential impacts, and the incorporation of protection measures before project approval. This program provides cultural resource surveys, technical assistance, project review, and training to CAL FIRE staff and other resource professionals. The legal mandates that require CAL FIRE to protect archaeological, historical and tribal cultural resources are found in the California Environmental Quality Act, the Forest Practice Rules, California Executive Order W-26-92, and the California Register of Historic Resources.

California Forest Stewardship Program

The California Forest Stewardship Program is designed to encourage good stewardship of private forestland. The program provides technical and financial assistance to influence positive changes to forestland management, assists communities in solving common watershed problems, and helps landowners.

Demonstration State Forests

The California Department of Forestry and Fire Protection (CAL FIRE) operates nine Demonstration State Forests totaling approximately 72,000 acres. The forests represent the most common forest types in the state. The State Forests grow approximately 75 million board feet of timber annually and harvest an average of 20 million board feet each year, enough to

build 12,500 single-family homes. Revenue from these harvests fund the management of the State Forests. In addition, the forests provide research and demonstration opportunities for natural resource management, while providing public recreation opportunities, fish and wildlife habitat, and watershed protection. Common activities on State Forests include: experimental timber harvesting techniques, watershed restoration, mushroom collecting, hunting, firewood gathering, cone collecting for seed, a variety of university research projects, horseback riding, camping, mountain biking, and hiking.

[Fire and Resource Assessment Program \(FRAP\)](#)

The California Department of Forestry and Fire Protection's Fire and Resource Assessment Program (FRAP) provides a variety of products including the Forest and Range Assessment, a detailed report on California's forests and rangelands. FRAP provides extensive technical and public information for statewide fire threat, fire hazard, watersheds, socio-economic conditions, environmental indicators, and forest-related climate change.

[Fire Prevention Grants Program \(new since 2010\)](#)

CAL FIRE's Fire Prevention Grant Program seeks to fund local projects that address the risk of wildfire and reduce wildfire potential to communities in, and adjacent to, forested areas. Qualified activities include hazardous fuel reduction, fire prevention planning and fire prevention education with an emphasis on improving public health and safety. This grant program is part of California Climate Investments (CCI), a statewide program that puts billions of cap-and-trade dollars to work reducing greenhouse gas emissions, strengthening the economy, and improving public health and the environment— particularly in disadvantaged communities.

[Forest Health Grant Program \(new since 2010\)](#)

The Forest Health Grant Program awards Greenhouse Gas Reduction Funds (GGRF) allocated by the legislature for California Climate Investments (CCI) to implement projects that seek to: proactively restore forest health and conserve working forests; protect upper watersheds where the state's water supply originates; promote the long-term storage of carbon in forest trees and soils; minimize the loss of forest carbon from large, intense wildfires; and, more generally, further the goals of the [California Global Warming Solutions Act of 2006](#) (AB 32).

[Forest Health Research Grant Program \(new since 2010\)](#)

The purpose of the CAL FIRE CCI Forest Health Research Program ("Research Program") is to identify outstanding scientific and management questions, prioritize research needs, and fund sound scientific studies that support knowledge and decision making for forest landowners, resource agencies, and fire management organizations in California, and to further the goals of the California Forest Carbon Plan, California Climate Investments, and AB 32 Global Warming Solutions Act. The Research Program awards grants, primarily through a competitive selection process, for projects focused on topics identified as priorities for study in the grant guidelines (subject to change annually). Research should be focused on and relevant to California ecosystems and their management. Past priority topics have included: implementation, effectiveness and impacts of significantly increased pace and scale of fuel reduction, prescribed fire and other forest health treatments; utilization of forest residues and forest products related to fuel reduction and forest health treatments; wildfire mechanics, spread and associated impacts in wildland-urban interface landscapes; wildfire impacts, recovery and resilience in an altered future climate; and natural, historical and contemporary range of variation in fire regimes and wildfire-related greenhouse gas emissions. Projects are awarded in four categories: General, State Forests, Graduate Student, and Synthesis/Tool Development. Funding allocations to these categories is subject to annual revision.

Forest Pest Management Program

Forest pests (insects and diseases) annually destroy 10 times the volume of timber lost due to forest fires. CAL FIRE's forest pest specialists help protect the state's forest resources from native and introduced pests, conduct pest surveys, provide technical assistance to private forest landowners, and promote forest health on all forest lands.

Forest Practice Act and Rules

CAL FIRE enforces the laws that regulate timber harvesting on privately-owned and non-federal public lands in California. These laws are found in the Forest Practice Act and Rules, which have been in place since 1973 to ensure that forest management is done in a manner that will preserve and protect our fish, wildlife, forests and streams. The Board of Forestry and Fire Protection is responsible for promulgating the Forest Practice Rules.

Forest Practice Program

A part of CAL FIRE's Resource Management, the Forest Practice Program ensures that private landowners abide by the Forest Practice Act and Rules when commercially harvesting trees. It is comprised of the Timber Harvesting Program, the Geographical Information Systems Group, and the California Timber Regulation and Environmental Evaluation System (CalTREES).

Reforestation Program

The L.A. Moran Reforestation Center is the core of CAL FIRE's reforestation program. Established in 1921, the Reforestation Center has produced millions of tree seedlings for the state's reforestation efforts. As the demand for reforestation seedlings has expanded dramatically over the last decade due to California's drought, tree mortality, and devastating wildfires, the Reforestation Center's seed bank and propagation nursery have responded to provide a reliable source of quality tree seedlings and seed stock to private landowners and public forests in California. CAL FIRE Forestry Assistance staff assist landowners in proper reforestation of their lands using the CAL-FIRE-produced seedlings.

Research and Development Program

The CAL FIRE Research and Development (R&D) Program receives, reviews and makes recommendations on proposals to improve the equipment and technologies used in preparation and planning, incident response, and command and control. These proposals can include ground, aerial, mobile, portable, or stationary equipment used by CAL FIRE. The R&D Program works collaboratively with other government and non-government organizations to foster the continued betterment of the Department's capabilities.

Resource Management Program

California is rich in natural resources. Of the 85 million acres classified as wildlands, nearly 17 million are commercial forest land, half privately-owned and half government-owned. This forest land grows 3.8 billion board feet yearly. Approximately 2 billion board feet of timber is harvested per year, with a value of over \$1 billion. In addition to timber, the state's wildlands also provide valuable watershed, wildlife habitat, and recreation resources. Maintaining the sustainability of all these natural resources is the goal of the CAL FIRE Resource Management Program.

State Fire Marshal State Fire Training Program

The SFT Program is a collaborative effort of the California Fire Service that work together to design and deliver courses that provide fire service personnel at all levels with the knowledge and skills to do their jobs professionally and safely. Fire departments and individuals donate thousands of hours annually to support curriculum development, issuance of instructor credentials, and certification of personnel.

[Urban and Community Forestry Program \(U&CF\)](#)

Under the authority of the Urban Forestry Act (PRC 4799.06 - 4799.12), the Urban & Community Forestry Program (UCF) works to expand and improve the management of trees and related vegetation in communities throughout California. The program's mission is to lead the effort to advance the development of sustainable urban and community forests in California. The program also administers State and Federal grants throughout California communities to advance urban forestry efforts.

[Vegetation Management Program \(VMP\)](#)

The Vegetation Management Program (VMP) is a cost-sharing program that focuses on the use of prescribed fire, and some mechanical means, for addressing wildland fire fuel hazards and other resource management issues on State Responsibility Area (SRA) lands. The use of prescribed fire mimics natural processes, restores fire to its historic role in wildland ecosystems, and provides significant fire hazard reduction benefits that enhance public and firefighter safety.

[Volunteers in Prevention Program](#)

The objectives of the VIP Program are to involve and utilize citizens and public service groups in non-salaried positions to reduce human-caused. Today, there are more than a thousand VIP's statewide, in all 21 of CAL FIRE's Operational Units, averaging over 50,000 hours of volunteer service to CAL FIRE annually. In exchange CAL FIRE pays meals and mileage for their expenses. VIP teams provide fire prevention teaching in grades K-3rd. In addition, VIPs educate thousands of children and their parents about fire prevention by participating in fairs, displays, and parades each year. Volunteers are trained to make preliminary wildland homeowner property inspections for fire safety as required by Public Resources Code 4291, and to discuss with homeowners ways to make their homes fire safe. VIPs also are trained to assist CAL FIRE's efforts during wildland fires, earthquakes, floods and other emergencies by providing information to the media and public.

[Wildland Urban Interface Building Code Standards](#)

The broad objective of the Wildland-Urban Interface Fire Area Building Standards is to establish minimum standards for materials and material assemblies and provide a reasonable level of exterior wildfire exposure protection for buildings in Wildland-Urban Interface Fire Areas. The use of ignition resistant materials and design to resist the intrusion of flame or burning embers projected by a vegetation fire (wildfire exposure) will prove to be the most prudent effort California has made to try and mitigate the losses resulting from our repeating cycle of interface fire disasters.

[California Department of Fish and Wildlife \(CDFW\)](#)

The Department of Fish and Game maintains native fish, wildlife, plant species and natural communities for their intrinsic and ecological value and their benefits to people. This includes habitat protection and maintenance in a sufficient amount and quality to ensure the survival of all species and natural communities. The department is also responsible for the diversified use of fish and wildlife including recreational, commercial, scientific and educational uses.

Programs

[Timberland Conservation Program](#)

The Timberland Conservation Program (TCP) helps to conserve natural communities on timberland. The TCP is committed to maintaining forest ecological values in managed forests, including during the environmental review of timber harvesting plans (THPs). When TCP scientists review THPs, they focus on potential significant impacts to wildlife, fish, native plants, and water bodies.

[California Endangered Species Act \(CESA\)](#)

The California Endangered Species Act (CESA) states that all native species of fishes, amphibians, reptiles, birds, mammals, invertebrates, and plants, and their habitats, threatened with extinction and those experiencing a significant decline which, if not halted, would lead to a threatened or endangered designation, will be protected or preserved. The Department will work with all interested persons, agencies and organizations to protect and preserve such sensitive resources and their habitats.

[Habitat Conservation and Mitigation Banking](#)

A conservation or mitigation bank is privately or publicly owned land managed for its natural resource values. In exchange for permanently protecting the land, the bank operator is allowed to sell habitat credits to developers who need to satisfy legal requirements for compensating environmental impacts of development projects.

[Lake and Streambed Alteration Program](#)

The Department of Fish and Game (DFG) is responsible for conserving, protecting, and managing California's fish, wildlife, and native plant resources. To meet this responsibility, the Fish and Game Code (Section 1602) requires an entity to notify DFG of any proposed activity that may substantially modify a river, stream, or lake.

[Natural Communities Conservation Planning \(NCCP\) Act and Program](#)

The primary objective of the NCCP program is to conserve natural communities at the ecosystem level while accommodating compatible land use. The program seeks to anticipate and prevent the controversies and gridlock caused by species' listings by focusing on the long-term stability of wildlife and plant communities and including key interests in the process.

[State Wildlife Action Plan \(SWAP\) and Companion Plans \(newly revised 2015-2016\)](#)

The State Wildlife Action Plan and Companion Plans examine the health of wildlife and prescribe actions to conserve wildlife and vital habitat before they become more rare and more costly to protect. The plans also promote wildlife conservation while furthering responsible development and addressing the needs of a growing human population.

[California Department of Conservation](#)

With a team of scientists and other dedicated professionals, the Department of Conservation (DOC) administers a variety of programs vital to California's public safety, environment and economy. The services DOC provides are designed to balance today's needs with tomorrow's obligations by fostering the wise use and conservation of energy, land and mineral resources.

Programs

[Division of Land Resource Protection \(DLRP\)](#)

The DLRP works with landowners, local governments, and researchers to conserve productive farm and agricultural lands, open space, and other important natural resources. They maintain the [Farmland Mapping and Monitoring Program](#) (FMMP), which tracks land use changes in agriculture and development. Other significant DLRP programs include:

[Grant Programs](#)

DLRP implements several grant programs, including:

- [Watershed Coordinator Grant Program](#)
- [Sustainable Agricultural Lands Conservation Program](#)
- [Transformative Climate Communities Program](#)
- [Working Lands and Riparian Corridors Program](#)
- [Resource Conservation District Financial Assistance Program](#)

[Regional Forest and Fire Capacity Program](#)

The Regional Forest and Fire Capacity program seeks to increase regional capacity to prioritize, develop, and implement projects that improve forest health and fire resilience, facilitate greenhouse gas emissions reductions, and increase carbon sequestration in forests throughout California. Block grants support regional planning and implementation of landscape-level forest health projects consistent with the California Forest Carbon Plan and Executive Order B-52-18. This is a California Natural Resources Agency program administered by the Department of Conservation. The Regional Forest and Fire Capacity program seeks to export the Sierra Nevada Conservancy's Watershed Improvement Program across the state.

[California Land Conservation Act of 1965 \(Williamson Act\) Program](#)

The California Land Conservation Act of 1965—commonly referred to as the Williamson Act—enables local governments to contract with private landowners for the purpose of restricting specific parcels of land to agricultural or related open space use. In return, landowners receive property tax assessments that are much lower than normal because they are based upon farming and open space uses as opposed to full market value. Local governments receive an annual subvention of forgone property tax revenues from the state via the Open Space Subvention Act of 1971. The payment of Williamson Act subventions was suspended in 2009 due to budget cuts. DLRPP maintains the Williamson Act contract lands database, and publishes reports on trends and changes in farm and ranch landowner program participation.

[California Geological Survey \(CGS\)](#)

The CGS provides scientific products and services about the state's geology, seismology and mineral resources that affect the health, safety, and business interests of the people of California. CGS is one of the several state entities that participates on timber harvesting plan review teams.

Other Organizations

State Agencies, Departments, Boards, Programs, Plans and Strategies

[California Environmental Quality Act \(CEQA\)](#)

CEQA, or the California Environmental Quality Act, is a statute that requires state and local agencies to identify the significant environmental impacts of their actions and to avoid or mitigate those impacts, if feasible.

[California Clean Air Act](#)

The California Clean Air Act (CCAA) of 1988 requires nonattainment areas to achieve and maintain the state ambient air quality standards by the earliest practicable date. The CCAA required air districts must develop plans for attaining the state ozone, carbon monoxide, sulfur dioxide, and nitrogen dioxide standards.

[California Department of Pesticide Regulation: Endangered Species Project](#)

In California, DPR has been studying endangered species protection issues with federal funding since 1988. DPR activities include mapping sites occupied by federally listed species, evaluating pesticide exposure risks to inhabited sites, classifying risk and developing protection strategies to minimize risk as needed.

[California Tahoe Conservancy](#)

The California Tahoe Conservancy is an independent State agency within the Natural Resources Agency of the State of California. It was established in its present form by State law in 1984 (Chapter 1239, Statutes of 1984). Its jurisdiction extends only to the California side of the Lake Tahoe Basin. The Conservancy is not a regulatory agency. It was established to develop and implement programs through acquisitions and site improvements to improve water quality in Lake Tahoe, preserve the scenic beauty and recreational opportunities of the region, provide public access, preserve wildlife habitat areas, and manage and restore lands to protect the natural environment.

[California Department of Water Resources \(DWR\)](#)

DWR manages California's water resources, systems, and infrastructure, including the State Water Project (SWP), in a responsible, sustainable way. Its responsibilities and duties include:

- Preventing and responding to floods, droughts, and catastrophic events
- Informing and educating the public on water issues
- Developing scientific solutions
- Restoring habitats
- Planning for future water needs, climate change impacts, and flood protection
- Constructing and maintaining facilities
- Generating power
- Ensuring public safety
- Providing recreational opportunities

[Bay-Delta Program](#)

The Sacramento-San Joaquin Delta plays a major role in California's prosperity by supplying drinking water to almost 27 million residents and fueling a \$32 billion agricultural industry. It also serves as important habitat to more than 750 animal and plant species,

including more than 40 aquatic species. Our Bay-Delta Office ensures water supply reliability and environmental needs are equally met for California.

[California Water Plan](#)

The California Water Plan is the State's strategic plan for sustainably managing and developing water resources for current and future generations. Required by Water Code Section 10005(a), it presents the status and trends of California's water-dependent natural resources; water supplies; and agricultural, urban, and environmental water demands for a range of plausible future scenarios.

[California Environmental Protection Agency \(Cal/EPA\)](#)

The California Environmental Protection Agency is charged with developing, implementing and enforcing the state's environmental protection laws that ensure clean air, clean water, clean soil, safe pesticides and waste recycling and reduction. Their departments are at the forefront of environmental science, using cutting-edge research to shape the state's environmental laws. There are five departments and several Regional Water Control Boards in CAL-EPA including Air Resource Board; Department of Pesticide Regulation; Department of Toxic Substances Control; Office of Environmental Health Hazard Assessment; State Water Resources Control Board; and nine Regional Water Control Boards.

[Air Resources Board \(CARB\)](#)

CARB's mission is to promote and protect public health, welfare, and ecological resources through effective reduction of air pollutants while recognizing and considering effects on the economy. CARB is the lead agency for climate change programs and oversees all air pollution control efforts in California to attain and maintain health-based air quality standards.

[State Water Resource Control Board \(SWRCB\)](#)

The State Water Resources Control Board's mission is to preserve, enhance, and restore the quality of California's water resources and drinking water for the protection of the environment, public health, and all beneficial uses, and to ensure proper water resource allocation and efficient use, for the benefit of present and future generations.

[Regional Water Quality Control Boards and Basin Plans \(SWRCB\)](#)

There are nine Regional Water Quality Control Boards (Regional Boards). Regional Boards develop "basin plans" for their hydrologic areas, govern requirements/issue waste discharge permits, take enforcement action against violators, and monitor water quality. The task of protecting and enforcing the many uses of water, including the needs of industry, agriculture, municipal districts, and the environment is an ongoing challenge for the State Board and Regional Boards. Regional Board staff serve as part of the interagency review teams for timber harvesting plans.

[Surface Water Ambient Monitoring Program \(SWAMP\)](#)

The SWAMP mission is to provide resource managers, decision makers, and the public with timely, high-quality information to evaluate the condition of all waters throughout California. SWAMP accomplishes this through carefully designed, externally reviewed monitoring programs, and by assisting other entities state-wide in the generation of comparable data that can be brought together in integrated assessments that provide answers to current management questions. Though a recent contract with the California Natural Resources Agency, SWAMP has increased its monitoring of water quality on forested watersheds.

[Total Maximum Daily Loads \(TMDL\) Plans](#)

Section 303(d) of the federal Clean Water Act and 40 CFR §130.7 require states to identify waterbodies that do not meet water quality standards and are not supporting their beneficial uses. These waters are placed on the Section 303(d) List of Water Quality Limited Segments (List), also known as the 303(d) List of Impaired Waterbodies. The List identifies the pollutant or stressor causing impairment and establishes a schedule for developing a control plan to address the impairment. Placement on this list generally triggers development of a pollution control plan called a Total Maximum Daily Load (TMDL) for each waterbody and associated pollutant/stressor on the list. The Clean Water Act gives the State Water Resources Control Board and the US Environmental Protection Agency (EPA) the authority to establish TMDLs under Section 303(d).

[California Department of Food and Agriculture \(CDFA\)](#)

The California Department of Food and Agriculture protects and promotes California's agriculture. California's farmers and ranchers produce a safe, secure supply of food, fiber, and shelter. These commodities are marketed fairly for all Californians and produced with responsible environmental stewardship.

[Border Protection Stations](#)

Border Protection Stations are the first line of defense for protecting our environment and resources from invasive plants and exotic pests. The California Department of Food and Agriculture (CDFA) has 16 agricultural inspection stations along shared borders with Nevada, Oregon and Arizona. Each year, inspectors intercept thousands of lots of prohibited plant material that potentially threaten the food supply, our trees and forests, and the environment.

[Plant Health and Pest Prevention Services \(PHPPS \)](#)

The Plant Health and Pest Prevention Services Division objective is to protect California's (1) Food supply from the devastating impact of exotic pests; (2) Environment and natural resources from direct pest impacts and increased pesticide use; (3) Public from pests that pose human health threats; (4) Position in the global economy.

[California Emergency Management Agency: California Local Hazard Mitigation Planning](#)

The local hazard mitigation planning process analyzes a community's risk from natural hazards, coordinates available resources, and implements actions to reduce or eliminate risks.

[California Wildlife Conservation Board \(CWCB\)](#)

The primary responsibilities of WCB are to select, authorize and allocate funds for the purchase of land and waters suitable for recreation purposes and the preservation, protection and restoration of wildlife habitat. WCB approves and funds projects that set aside lands within the State for such purposes, through acquisition or other means, to meet these objectives. WCB can also authorize the construction of facilities for recreational purposes on property in which it has a proprietary interest.

[Oak Woodlands Conservation Program](#)

The CWCB Oak Woodlands Conservation Program offers landowners, conservation organizations, cities and counties, an opportunity to obtain funding for projects designed to conserve and restore California's oak woodlands. While the Program is statewide in nature, it provides opportunities to address oak woodland issues on a regional priority basis. The

Program provides a mechanism to bring ranchers and conservationists together in a manner that allows both to achieve that which is so valued—sustainable ranch and farming operations and healthy oak woodlands.

Forest Conservation Program

The CWCB Forest Conservation Program was created through Proposition 84 (2006) and expanded by Proposition 68 (2018). The program promotes the ecological integrity and economic stability of California's diverse native forests for all their public benefits through forest conservation, preservation, and restoration of productive managed forest lands, forest reserve areas, redwood forests and other forest types, including the conservation of water resources and natural habitat for native fish and wildlife and plants found on these lands. It also provides for the protection, restoration, and improvement of upper watershed lands in the Sierra Nevada and Cascade Mountains, including forest lands, meadows, wetlands, chaparral, and riparian habitat to protect and improve water supply and water quality, improve forest health, reduce wildfire danger, mitigate the effects of wildfires on water quality and supply, increase flood protection, or to protect or restore riparian or aquatic resources. The Program supports planning, acquisition, and restoration projects throughout California's forests, consistent with the Board's Strategic Plan.

The Delta Stewardship Council

The Delta Stewardship Council was created in legislation to achieve the state mandated coequal goals for the Delta. "'Coequal goals' means the two goals of providing a more reliable water supply for California and protecting, restoring, and enhancing the Delta ecosystem. The coequal goals shall be achieved in a manner that protects and enhances the unique cultural, recreational, natural resource, and agricultural values of the Delta as an evolving place." (CA Water Code Section 85054)

Sierra Nevada Conservancy

The Sierra Nevada Conservancy (SNC) is a state agency created by bi-partisan legislation and signed into law by Governor Schwarzenegger in 2004. The SNC was created with the understanding that the environmental, economic and social well-being of the Sierra Nevada and its communities are closely linked and that the Region would benefit from an organization providing a strategic direction. The SNC Region, made up of all or part of 22 counties and over 25 million acres, is one of the most significant natural and biologically diverse regions in the world.

Watershed Improvement Program

The SNC's Watershed Improvement Program is a large-scale restoration program designed to restore the health of California's primary watershed and create resilient Sierra Nevada communities. It is recognized by state policy and planning documents as a model program for addressing the growing challenges facing the Sierra Nevada Region in a changing climate.

Strategic Growth Council

The Strategic Growth Council was formed in September 2008. The Council is a cabinet level committee that is tasked with coordinating the activities of state agencies to improve air and water quality; protect natural resource and agriculture lands; increase the availability of affordable housing; improve infrastructure systems; promote public health; assist state and local entities in the planning of sustainable communities and meeting AB 32 goals

[University of California Cooperative Extension \(UCCE\)](#)

The continuing education arm of the University of California provides innovative learning programs to adult learners in California, across the U.S. and throughout the world. By offering accessible and relevant courses, UC Extension provides knowledge and connections for people to achieve their personal and professional goals

Plans and Strategies

[Assembly Bill \(AB\) 32 Scoping Plan: Natural and Working Lands](#)

To meet California's ambitious goals of reducing greenhouse gas emissions needed to avoid the most catastrophic impacts of climate change, the State must increase its efforts to conserve, restore, and manage California's forests, rangelands, farms, urban green spaces, wetlands, and soils.

[California Desert Renewable Energy Conservation Plan \(DRECP\)](#)

To protect the area and streamline the permitting process for renewable energy projects in the California desert, the California Energy Commission, the California Department of Fish and Wildlife, the U.S. Bureau of Land Management (BLM), and the U.S. Fish and Wildlife Service developed the Desert Renewable Energy Conservation Plan (DRECP) that identifies areas in the desert appropriate for the utility-scale development of wind, solar, and geothermal energy projects. The comprehensive plan also provides for the long-term conservation and management of covered species and preserves the natural resources, recreational areas, and scenic values.

[California Emergency Management Agency \(Cal EMA\) California State Hazard Mitigation Plan](#)

The 2018 California State Hazard Mitigation Plan (SHMP) represents the state's primary hazard mitigation guidance document, and is composed of comprehensive and valuable input provided by State Hazard Mitigation Team members and stakeholders. The 2018 SHMP builds upon the state's commitment to reduce or eliminate potential risks and impacts of natural and human-caused disasters to help communities with their mitigation and disaster resiliency efforts. The plan includes: an updated statewide risk assessment, disaster history, and statistics; recent mitigation progress, success stories, and best practices; updated state hazard mitigation goals, objectives, and strategies; and updated climate mitigation progress and adaptation strategies.

[California Outdoor Recreation Plan \(CORP\) \(updated in 2018\)](#)

The California Outdoor Recreation Plan is the statewide master plan for parks, outdoor recreation, and open space for California. It provides policy guidance to all outdoor recreation providers, including federal, state, local, and special district agencies that provide outdoor recreational lands, facilities and services throughout California. The CORP is also the primary tool for prioritizing Land and Water Conservation Fund grant allocations to local governments.

[California Partners in Flight Conservation Plan](#)

The CalPIF mission is to promote the conservation of resident and migratory landbirds and their habitats in California through research, monitoring, education, and collaboration among public and private landowners and managers, government agencies, non-government organizations, and individuals and other bird conservation efforts. The California chapter of Partners in Flight (CalPIF) was established in 1992.

[California State Coastal Conservancy Strategic Plan](#)

Every five years, the Conservancy develops a Strategic Plan that identifies specific goals to guide the Conservancy's work. The Strategic Plan provides an overall vision and metrics to measure the effectiveness of the Conservancy's work. It outlines the Conservancy's priorities in the context of California's coastal management program and communicates to its partners – local governments, state and federal agencies, private landowners, nonprofit conservation organizations, and private conservation funders– and to the general public the future role of the Conservancy in protecting coastal resources.

[California State Parks Off-Highway Motor Vehicle Recreation Division Strategic Plan](#)

This Strategic Plan provides guidance to the OHMVR Division to manage state vehicular recreation areas through a statewide financial assistance program that provides off-highway vehicle-related activities including law enforcement, operations and management, education, environmental protection, and repair and restoration on local and federal lands.

[County General Plans](#)

A general plan is each local government's blueprint for meeting the community's long-term vision for the future. The General Plan Guidelines serve as a resource to help jurisdictions draft and update their general plans. Technical advisory documents provide supplemental information about general plan topics in response to local needs around the state or changes in State law.

[Metropolitan Planning Organization \(MPO\) Plans](#)

Metropolitan Planning Organization (MPO) is a federally required transportation planning body comprised of elected and appointed officials representing local, state and federal governments or agencies having interest or responsibility in transportation planning and programming. An MPO is responsible for the development of a Long-Range Transportation Plan (LRTP), the Transportation Improvement Program (TIP), and a Unified Planning Work Program (UPWP) for its metropolitan planning area. The adoption of these documents is a prerequisite for the receipt of both federal transit and federal highway funding.

[National Cohesive Wildland Fire Management Strategy](#)

This is a collaborative process to seek national, all-lands solutions to wildland fire issues, focusing on three goals:

- Restore and maintain resilient landscapes
- Create fire-adapted communities
- Safe and effective wildfire response.

[National Fire Plan](#)

The National Fire Plan (NFP) was developed in August 2000, following a landmark wildland fire season, with the intent of actively responding to severe wildland fires and their impacts to communities while ensuring sufficient firefighting capacity for the future. The NFP addresses five key points: Firefighting, Rehabilitation, Hazardous Fuels Reduction, Community Assistance, and Accountability.

[National Fish Habitat Action Plan](#)

A coalition-based effort intended to conserve fish and aquatic communities by focusing partnerships of state and federal agencies, conservation organizations, foundations and others on fish habitat issues

Federal Agencies, Departments, Plans and Programs

[United States Department of Agriculture \(USDA\)](#)

This federal agency provides leadership on food, agriculture, natural resources, rural development, nutrition, and related issues based on public policy, the best available science, and effective management.

[USDA-Farm Service Agency \(FSA\): Conservation Reserve Enhancement Program](#)

The Conservation Reserve Enhancement Program (CREP) is a voluntary land retirement program that helps agricultural producers protect environmentally sensitive land, decrease erosion, restore wildlife habitat, and safeguard ground and surface water.

[USDA-Natural Resources Conservation Service \(NRCS\): Conservation Stewardship Program](#)

The Conservation Stewardship Program (CSP) is a voluntary conservation program that encourages producers to address resource concerns in a comprehensive manner by undertaking additional conservation activities and improving, maintaining, and managing existing conservation activities.

[USDA-NRCS: Emergency Watershed Protection](#)

The purpose of the Emergency Watershed Protection (EWP) program is to undertake emergency measures, including the purchase of flood plain easements, for runoff retardation and soil erosion prevention to safeguard lives and property from floods, drought, and the products of erosion on any watershed whenever fire, flood or any other natural occurrence is causing or has caused a sudden impairment of the watershed.

[USDA-NRCS: Environmental Quality Incentives Program](#)

The Environmental Quality Incentives Program (EQIP) is a voluntary program that provides assistance to farmers and ranchers who face threats to soil, water, air, and related natural resources on their land. Through EQIP, the Natural Resources Conservation Service (NRCS) provides assistance to agricultural producers in a manner that will promote agricultural production and environmental quality as compatible goals, optimize environmental benefits, and help farmers and ranchers meet Federal, State, Tribal, and local environmental requirements.

[USDA-NRCS: Healthy Forests Reserve Program](#)

The purpose of the Healthy Forests Reserve Program (HFRP) is to assist landowners, on a voluntary basis, in restoring, enhancing and protecting forestland resources on private lands through easements, 30-year contracts and 10-year cost-share agreements. The objectives of the program are to promote the recovery of endangered and threatened species under the Endangered Species Act, improve plant and animal biodiversity and enhance carbon sequestration.

[USDA-NRCS: Wetlands Reserve Program](#)

The Wetlands Reserve Program (WRP) is a voluntary program offering landowners the opportunity to protect, restore, and enhance wetlands on their property. The NRCS provides technical and financial support to help landowners with their wetland restoration efforts. The NRCS goal is to achieve the greatest wetland functions and values, along with optimum wildlife habitat, on every acre enrolled in the program. This program offers

landowners an opportunity to establish long- term conservation and wildlife practices and protection.

[USDA-NRCS: Wildlife Habitat Incentives Program](#)

The Wildlife Habitat Incentive Program (WHIP) was a voluntary program for conservation-minded landowners who want to develop and improve wildlife habitat on agricultural land, nonindustrial private forest land, and Indian land.

[USDA-NRCS and Forest Service \(FS\): Joint Chiefs Landscape Restoration Partnership](#)

The two USDA agencies Service are working together to improve the health of forests where public forests and grasslands connect to privately-owned lands. Through the Partnership, the two USDA agencies are restoring landscapes, reducing wildfire threats to communities and landowners, protecting water quality and enhancing wildlife habitat. The partnership began in 2014, and each year the agencies select new three-year projects.

[USDA-FS: Collaborative Forest Landscape Restoration Program](#)

Congress established the Collaborative Forest Landscape Restoration Program (CFLRP) with [Title IV of the Omnibus Public Land Management Act of 2009](#) (PDF, 40 KB) and reauthorized it in the [Agriculture Improvement Act of 2018](#) Section 8629 (the Farm Bill). The purpose of the CFLRP is to encourage the collaborative, science-based ecosystem restoration of priority forest landscapes.

[USDA-FS: Forest Service Global Change Research Strategy, 2009 – 2019](#)

The Forest Service Global Change Research Strategy helps identify best management practices for urban and rural forests, woodlands, and grasslands to sustain ecosystem health and a range of ecosystem services (“adaptation”), while also increasing carbon sequestration “mitigation”)—all under changing climate conditions. The fundamental research focus of the forest Service Global Change Research Strategy is to increase understanding of forest, woodland, and grassland ecosystems so that they can be managed in a way that sustains and provides ecosystem services for future generations.

[USDA-FS: National Urban and Community Forestry Program](#)

Urban and Community Forestry (UCF) is a cooperative program of the US Forest Service that focuses on the stewardship of urban natural resources. With 80 percent of the nation's population in urban areas, there are strong environmental, social, and economic cases to be made for the conservation of green spaces to guide growth and revitalize city centers and older suburbs.

[USDA-FS: Northwest Forest Plan](#)

The Northwest Forest Plan is an integrated, comprehensive design for ecosystem management, intergovernmental and public collaboration, and rural community economic assistance for federal forests in western Oregon, Washington, and northern California.

[USDA-FS: Region 5 Best Management Practices \(BMP\) Evaluation Program](#)

Effectiveness monitoring is completed through annual BMP monitoring of randomly selected, recently completed projects and concurrent monitoring in which sites are selected based on management interest in specific ongoing projects. Effectiveness monitoring is designed to evaluate how well the Forest and Region implement BMPs and how effectively the BMPs control water pollution from National Forest lands.

[USDA-FS: Forest Health Protection \(FHP\)](#)

This program has specialists in forest entomology and pathology, invasive plants, pesticide use, survey and monitoring, suppression and control, technology development and other forest health-related services that assist with protecting and improving the health of rural, wildland and urban forests.

[USDA-FS: Revised Land Management Plans for Sequoia, Sierra, and Inyo National Forests](#)

Land and resource management plans (Forest Plans) provide direction managing the lands and resources within a national forest. These plans are periodically updated with public involvement. These documents describe the recently (2016 and 2019) revised land management plans for the Sequoia, Sierra and Inyo National Forests.

[USDA-FS: Sierra Nevada Forest Plan Amendment](#)

Amended the Land and Resource Management Plans for the eleven National Forests in the Sierra Nevada range to improve protection of old forests, wildlife habitats, watersheds and communities in the Sierra Nevada mountains and Modoc Plateau.

[USDA-FS: Southern California National Forests Land Management Plan](#)

The revised land and resource management plans (forest plans) for the southern California national forests describe the strategic direction at the broad program-level for managing the land and its resources over the next 10 to 15 years. The strategic direction was developed by an interdisciplinary planning team working with forest staff using extensive public involvement and the best science available. The revised forest plans have a focus that is different from the old forest plans. The revised forest plans are outcome based and are focused on the condition of the land after project completion rather than the products removed from the land. Each forest plan is directed toward the realization of the desired conditions using strategies that are consistent with the concept of adaptive management and sustainable resource use.

[U.S. Department of Homeland Security \(DHS\)](#)

Secures the nation from the many threats we face. This requires the dedication of more than 240,000 employees in jobs that range from aviation and border security to emergency response, from cybersecurity analyst to chemical facility inspector. Keeps the nation safe.

[DHS: Federal Emergency Management Agency \(FEMA\)](#)

FEMA's mission is to support people and first responders to ensure that as a nation we work together to build, sustain, and improve our capability to prepare for, protect against, respond to, recover from, and mitigate all hazards.

[U.S. Department of the Interior \(DOI\)](#)

The Department of the Interior conserves and manages the Nation's natural resources and cultural heritage for the benefit and enjoyment of the American people, provides scientific and other information about natural resources and natural hazards to address societal challenges and create opportunities for the American people, and honors the Nation's trust responsibilities or special commitments to American Indians, Alaska Natives, and affiliated island communities to help them prosper.

[DOI-BLM: Resource Management Plans \(RMPs\)](#)

Resource Management Plans are Bureau of Land Management land use planning tools that include issues on outdoor recreation activities, threatened and endangered species habitat, geothermal development, wilderness conservation, fire protection and land access. There are specific [California Desert Conservation Area Plans](#) that are RMPs.

These plans aim to allow use of public land while not diminishing environmental, cultural and aesthetic value of the desert and to give management direction to conflict resolution.

[DOI-USFWS: Wildlife and Sport Fish Restoration Program](#)

The U.S. Fish & Wildlife Service, Wildlife and Sport Fish Restoration Program (WSFR) works with states, insular areas and the District of Columbia to conserve, protect, and enhance fish, wildlife, their habitats, and the hunting, sport fishing and recreational boating opportunities they provide.

[DOI-USFWS: Endangered Species Program](#)

The FWS Recovery Program works with partners to take measures to prevent the extinction of species, and prepares, coordinates, and implements recovery plans. Recovery plans provide a roadmap with detailed site-specific management actions for private, Federal, and State cooperation in conserving listed species and their ecosystems. A recovery plan is a non-regulatory document. It may apply to one species or an ecosystem. The FWS also offers Safe Harbor Agreements for private landowners through the Endangered Species Program which provides opportunities for private landowners to participate in conserving and recovering imperiled species.

[DOI-USFWS: Habitat Conservation Plans \(HCPs\)](#)

HCPs are planning documents required as part of an application for an incidental take permit. They describe the anticipated effects of the proposed taking; how those impacts will be minimized, or mitigated; and how the HCP is to be funded. HCPs can apply to both listed and nonlisted species, including those that are candidates or have been proposed for listing. Conserving species before they are in danger of extinction or are likely to become so can also provide early benefits and prevent the need for listing.

[DOI-USFWS: Partners for Fish and Wildlife Service Program](#)

The Partners for fish and Wildlife Service Program objective is to efficiently achieve voluntary habitat restoration on private lands, through financial and technical assistance, for the benefit of Federal Trust Species. The 2008 Farm Bill established a tax deduction for expenditures paid or incurred for the purpose of achieving site-specific management actions recommended in recovery plans for species listed as threatened or endangered under the Endangered Species Act (ESA).

[DOI-NPS: Land and Water Conservation Fund](#)

The Land and Water Conservation Fund (LWCF) Federal program supports the protection of federal public lands and waters – including national parks, forests, wildlife refuges, and recreation areas – and voluntary conservation on private land. LWCF investments secure public access, improve recreational opportunities, and preserve ecosystem benefits for local communities.

[DOI-NPS: Federal Lands to Parks Program](#)

The Federal Lands to Parks Program helps communities to acquire, reuse and protect surplus federal properties for local parks and recreation. States, counties, and communities may acquire federal land and buildings no longer needed by the federal government at no cost on condition they are protected for public parks and recreation.

[DOI-NPS: The Rivers, Trails, and Conservation Assistance Program](#)

This program supports community-led natural resource conservation and outdoor recreation projects across the nation by partnering with community groups, nonprofits,

tribes, and state and local governments to design trails and parks, conserve and improve access to rivers, protect special places, and create recreation opportunities.

[Verified Carbon Standard \(VCS\) Program](#)

The VCS Program provides a robust, global standard and program for approval of credible voluntary offsets. VCS offsets must be real (have happened), additional (beyond business-as-usual activities), measurable, permanent (not temporarily displace emissions), independently verified and unique (not used more than once to offset emissions).

Other Organizations and Programs

[National Audubon Society](#)

Audubon's mission is to conserve and restore natural ecosystems, focusing on birds, other wildlife, and their habitats for the benefit of humanity and the earth's biological diversity. The national network of community-based nature centers and chapters, scientific and educational programs, and advocacy on behalf of areas sustaining important bird populations, engage millions of people of all ages and backgrounds in positive conservation experiences.

[CalFlora](#)

Calflora is a nonprofit organization dedicated to providing information about California plant biodiversity for use in education, research and conservation. CalFlora is structured as a digital library database and was conceived as a collaborative research project to collect and re-distribute information about California's wild plants, including habitat descriptions, photographs, observations, nomenclature, and distribution maps.

[California Association of Resource Conservation Districts \(CARCD\)](#)

CARCD works in support of thriving local communities, landscapes, and economies. They work with Resource Conservation Districts, state and federal agencies, policy makers, and individuals to ensure the resilience and health of California's water, soil, wildlife habitat, and other natural resources — today and for generations to come.

[California Fire Safe Council](#)

The Council's mission is to be California's leader in community wildfire risk reduction and resiliency. It brings together community leaders, governmental agencies, and corporations to provide education to the residents of California on the dangers of wildfires and how they could be prevented. The Council has developed and maintains an online, "one-stop shop," Grant Clearinghouse mainly for four primary federal agencies: USDA Forest Service and three Department of the Interior agencies, the Bureau of Land Management, National Park Service, and Fish & Wildlife Service. These agencies provide master grants to CFSC to conduct, select, manage and monitor subgrants to local community groups such as Fire Safe Councils, homeowner associations, local government, fire departments, and other entities working on wildfire prevention activities such as defensible space, community fire planning, and education. The Council also provides technical assistance to its grantees and local Fire Safe Councils.

[California Inter-Agency Noxious Weed Coordinating Committee \(CINWCC\)](#)

The California Interagency Noxious Weed Coordinating Committee (CINWCC) was formed in 1995 when 14 federal, state, and county agencies came together under a Memorandum

of Understanding to coordinate the management of noxious weeds. The committee's mission is to facilitate, promote, and coordinate the establishment of an Integrated Pest Management partnership between public and private land managers toward the eradication and control of noxious weeds on federal and state lands and on private lands adjacent to public lands.

[California Invasive Plants Council \(CAL-IPC\)](#)

Cal-IPC's mission is to protect California's environment and economy from invasive plants. Formed in 1992 to address one of California's top environmental threats, the organization works closely with agencies, industry and other nonprofit organizations. Their active membership includes public and private land managers, ecological consultants and researchers, planners, volunteer stewards, and concerned citizens.

[California Native Plant Society \(CNPS\)](#)

Formed in 1965, (CNPS) is dedicated to conserving California native plants and their natural habitats, while increasing the understanding, enjoyment, and horticultural use of native plants. We work closely with decision-makers, scientists, and local planners to advocate for well-informed and environmental friendly policies, regulations, and land management practices.

[California Urban Forestry Advisory Committee \(CUFAC\)](#)

The California Urban Forestry Advisory Committee was established to advise the Director of the CAL FIRE on the State's Urban Forestry Program.

[Climate Action Reserve \(CAR\)](#)

The Climate Action Reserve is a national offsets program working to ensure integrity, transparency and financial value in the U.S. carbon market. It does this by establishing regulatory-quality standards for the development, quantification and verification of greenhouse gas (GHG) emissions reduction projects in North America; issuing carbon offset credits known as Climate Reserve Tonnes (CRT) generated from such projects; and tracking the transaction of credits over time in a transparent, publicly-accessible system.

[Conservation Lands Network \(CLN\)](#)

The Conservation Lands Network is a regional conservation strategy for the Bay Area. Launched in 2011, it provides science-based for conservation efforts throughout the ten-county Bay Area. The CLN supports strategic investments in land acquisition and stewardship by focusing conservation in areas that represent the region's biodiversity and support ecological.

[Emerald Necklace Greenway \(Los Angeles\)](#)

Mission is to create an "Emerald Necklace" natural infrastructure network of green spaces (parks and trails) in underserved areas throughout the Los Angeles Basin, connecting the Mountains to the Sea and protecting urban communities from the threats of climate change. Inspired by the 1929 Olmstead Bartholomew plan for an expansive greenway for Los Angeles which, due to changing government priorities during the 1930's, was never realized.

[Forest Stewardship Coordinating Committee \(FSCC\)](#)

The California Forest Stewardship Coordinating Committee is an advisory body for the California Stewardship Program, which is administered by the California Department of Forestry & Fire Protection through the Forest Legacy Program. The committee meets quarterly to learn about forestry issues throughout the state and to make recommendations. The Joint Forestry Committee

of the California Association of Resource Conservation Districts (CARCD) has taken on the responsibility of the State Forest Stewardship Coordinating Committee. This committee has been active in forest stewardship issues, such as pre-fire fuels management and forest healthy, and represents such diverse interests as the Forest Service, Natural Resources Conservation Service, and UC Cooperative Extension; consulting foresters; forest products industry; forest landowners, land trusts, conservation and environmental organizations; and local government.

[Forest Taxation Reform Act](#)

This 1976 tax reform act made numerous revisions to the assessment and collection of taxes for timber and timberlands. Its primary features include: creation of the Timberland Production Zone (TPZ), where only timber production along with certain compatible uses are allowed, annual property taxes are reduced, and timber taxation is shifted from an annual *ad valorem* tax to a timber yield tax applied at the time of commercial harvest.

[Greenbelt Alliance](#)

The Alliance's mission is to educate, advocate, and collaborate to ensure the Bay Area's lands and communities are resilient to a changing climate. They envision a Bay Area of healthy, thriving, resilient communities made up of lands and people that are safe during climate disasters and recover quickly from wildfire, floods, and drought, where everyone is living with nature in new and powerful ways for generations to come.

[GreenInfo Network](#)

GreenInfo Network is a non-profit organization that assists others in the use of Geographic Information Systems (GIS) and related information technologies on a consultant basis. The organization works for 80-100 public interest clients annually, in California and elsewhere in the U.S. Since their founding in 1996, GreenInfo Network has assisted over 1,000 public interest groups and agencies with mapping, data, analysis and other information technology projects.

[Healthy Forests Restoration Act](#)

The Healthy Forests Restoration Act of 2003 (P.L. 108-148) contains a variety of provisions to expedite hazardous-fuel reduction and forest-restoration projects on specific types of Federal land that are at risk of wildland fire or insect and disease epidemics. The act helps rural communities, States, Tribes, and landowners restore healthy forest and range land conditions on State, Tribal, and private lands.

[Invasive Species Council of California \(ISCC\)](#)

Established in 2009, the ISCC is an inter-agency council that helps to coordinate and ensure complementary, cost-efficient, environmentally sound and effective state activities regarding invasive species. The Council is chaired by the Secretary of the California Department of Food and Agriculture and vice-chaired by the Secretary of the California Natural Resources Agency.

[Joint Fire Science Program](#)

The Joint Fire Science Program (JFSP) was created by Congress in 1998 as an interagency research, development, and applications partnership between the U.S. Department of the Interior and the U.S. Department of Agriculture. Funding priorities and policies are set by the JFSP Governing Board, which includes representatives from the Bureau of Land Management, National Park Service, U.S. Fish and Wildlife Service, Bureau of Indian Affairs, U.S. Geological Survey, and five representatives from the Forest Service.

[Land Trusts](#)

More than 200 statewide, regional and local land trusts are actively working in California to conserve areas from development, for open space and compatible working land uses. The [California Council of Land Trusts](#) is dedicated to building programs that are responsive to the needs of the land trust community and relevant to the current challenges and opportunities facing land and resource conservation.

[Local Area Formation Commissions \(LAFCOs\)](#)

LAFCOs are state-mandated, quasi-judicial countywide Commissions, whose purview is to oversee boundary changes of cities and special districts, the formation of new agencies, including the incorporation of new cities and districts, and the consolidation or reorganization of special districts and or cities.

[National Environmental Policy Act \(NEPA\)](#)

The National Environmental Policy Act (NEPA) [42 U.S.C. 4321 et seq.] was signed into law on January 1, 1970. The Act establishes national environmental policy and goals for the protection, maintenance, and enhancement of the environment and it provides a process for implementing these goals within the federal agencies. The Act also establishes the Council on Environmental Quality (CEQ).

[The Nature Conservancy \(TNC\)](#)

The mission of the Nature Conservancy conserving the lands and waters on which all life depends. Their priorities are addressing climate change, protecting land and water, providing food and water sustainability, and building healthy cities. The [California office of TNC](#) is directly involved in climate change science and policy, investment in forest restoration and fire-resistant communities, and bird habitat management in the Sacramento-San-Joaquin Delta.

[North Coast Resource Partnership \(NCRP\)](#)

Working with Northern California Tribes, counties, and diverse stakeholders, the NCRP is focused on achieving outcomes on the ground for North Coast communities and watersheds. Partnership embraces integrated goals related to water quality and supply, ecosystem function, economic vitality, collaboration, climate adaptation and energy independence, and community health and safety. Their strategies include using the best available data and science, relying on local knowledge and expertise, actively collaborating with diverse stakeholders, engaging in multi-objective, integrated planning and adaptive management, leveraging funding, accounting for benefits to economically disadvantaged communities, and ensuring project effectiveness by monitoring and economic valuation.

[Northern Sierra Partnership \(NSP\)](#)

The Northern Sierra Partnership is a partnership of five organizations with experience completing significant land protection, restoration, policy development, and community enhancement projects in the northern Sierra. Together they target locally supported conservation and planning efforts. NSP's objective is to protect wetlands, lakes, and streams; connect and manage healthy forests that will reduce the risks of catastrophic wildfire; enhance the well-being of local communities and economies through sustainable land use programs; develop proactive, science-based approaches to adapt to climate change and to mitigate its expected impacts; and build an enduring culture of land and water conservation in the northern Sierra.

[Planning and Conservation League \(PCL\)](#)

The mission of the Planning and Conservation League is to protect and restore California's natural environment, and to promote and defend the health and safety of Californians through legislative and administrative action. The PCL Foundation, the research arm of PCL, engages in public policy research that educates and empowers local communities to participate in decision-making processes that will help protect California's environment.

[Sierra Club](#)

The Sierra Club is the oldest and largest grassroots environmental organization in the United States. It was founded in 1892 in San Francisco, California by the well-known conservationist and preservationist John Muir, who became its first president. The Sierra Club has hundreds of thousands of members in chapters located throughout the US. [Sierra Club California](#) is very active in environmental policy and education in the state.

[Sierra Nevada Alliance](#)

The organization is an Alliance of conservation groups that are based or work in the Sierra Nevada region. There are over 40 member groups that span the entire 400-mile mountain range. The Alliance works to elevate and support Sierra ecosystems and communities by empowering and collaborating with their partners.

[Sustainable Agriculture Research and Education Program \(SARE\)](#)

Since 1988, the Sustainable Agriculture Research and Education (SARE) program has helped advance farming systems that are profitable, environmentally sound, and good for communities through a nationwide research and education grants program.

[Together Bay Area](#)

This coalition's mission is to be the champion and regional voice of the resilient lands that are integral to a thriving Bay Area and all people who live here. Their vision is a Bay Area that is home to healthy lands, healthy people, and healthy communities that addresses the impacts of climate change through collaboration. It seeks a just and equitable society where we people in relationship with the land that sustains us now and will sustain future generations.

[Tree City USA](#)

The Arbor Day Foundation, in cooperation with the USDA Forest Service, the National Association of State Foresters, and Urban and Community Forestry, administers the Tree City USA program. The program seeks to support efforts to create and maintain urban forests across the country, for their numerous benefits. Arbor Day Foundation has a set of four criteria by which they rate communities as to their commitment to urban forestry. [California](#) currently has 146 cities, towns and other communities that qualify as Tree Cities.

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